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ESSAYS ON APPLIED MICROECONOMICS

BY

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DISSERTATION

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ABSTRACT

This dissertation includes three chapters that use applied microeconomics theory to assess three different economic questions. The first chapter analyzes the labor market effects of import competition from high-wage countries on emerging economies. Though abundant evidence shows that import competition from low-wage countries decreases manufacturing employment and wages of high-wage countries, less is known about the reverse: the impact of import competition from high-wage countries on emerging economies. This paper uses a natural experiment to examine the effects of import competition from the United States on workers and firms in Colombia. We exploit industry variation in import exposure and regional variation in import access in the wake of a free trade agreement that increased import competition in Colombia but left its exports unaffected. Using administrative employer-employee data to identify proxies for productivity and skills, we find that a 10 percent increase in import competition from the United States decreases employment in Colombia by 6.4 percent in affected industries and states. The impacts are driven largely by the exit and shrinking of less-productive firms. Less-skilled workers experience the greatest impacts, with effects on employment lasting for at least four years. Import competition induces workers to shift from affected to unaffected industries and states, and decreases the wage of workers employed in less-productive firms.

The second chapter includes an analysis of how school shootings affect housing prices in the United States. We find that house prices within a school district decline by 7.8 percent in the three year period after the episode, along with a decline in number of transactions. The drop in property prices is stronger among houses with more bedrooms, a measure that serves as a proxy for properties most likely to have school-age children in the household. We also find evidence of decrease in school enrollment and in the number of

teachers in the aftermath of the shooting. The analysis suggests that it is the deterioration in school quality that results in lower willingness to pay.

The third chapter compares the economic returns of workers' skills with the returns to schooling degrees. Using longitudinal data of all college graduates in Colombia, we estimate labor market returns to postsecondary degrees and to various skills –including literacy, numeracy, foreign language, field-specific, and non-cognitive skills. Graduates of longer programs, of private institutions, and of schools with higher reputation earn higher wages. Even after controlling for all the characteristics of the degree, a one standard deviation increase in each skill predicts an average wage increase of two percent. Returns to skills vary along the wage distribution, with tenure, with the field of specialization and the type of job obtained immediately after graduation.

Para Myriam y la memoria de Augusto...

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CHAPTER 1

ENTERING THE MAJOR LEAGUES: THE EFFECT OF IMPORT COMPETITION FROM THE UNITED STATES ON WORKERS AND FIRMS IN AN EMERGING ECONOMY

Leonardo Bonilla and Juan S. Muñoz

1.1 Introduction

Emerging economies are plagued by a glut of persistently low-productivity firms, which can decrease total factor productivity up to 60 percent compared to developed countries (Hsieh & Klenow, 2009, 2014; Busso, Madrigal, & Pagés, 2013). A less dynamic exit of firms can therefore affect economic development and explain cross-country differences in employment growth (Eslava, Haltiwanger, & Pinzón, 2019; Eslava, Haltiwanger, Kugler, & Kugler, 2004). Lack of market competition may induce slow firm exit, and enable less productive firms to coexist with more productive ones. Trade barriers and protectionist policies in emerging countries, for instance, are one way in which competition is hindered, benefiting unproductive firms and reducing productivity and employment (Eslava, Haltiwanger, Kugler, & Kugler, 2013). Import competition from low-wage countries such as China has also been shown to reduce manufacturing employment and wages in low- and high-wage countries.[1] However, it is still not clear if import competition from high-wage countries induces firm exit and employment growth within emerging economies.

Estimating the effects of increased competition on firms and workers in

[1]For effects among high-income countries, see Autor, Dorn, and Hanson (2013), Autor, Dorn, Hanson, and Song (2014), Bloom, Handley, Kurman, and Luck (2019) ,Bernard, Jensen, and Schott (2006) and Pierce and Schott (2016). For the effects among low-income countries, see Dix-Carneiro and Kovak (2017), Jenkins, Peters, and Moreira (2008), Moreira (2007).

emerging economies has proved challenging, especially because of difficulties in separating the effects of imports from the effects of exports. Trade is widely believed to increase competition and induce firm exit (Melitz, 2003; Melitz & Ottaviano, 2008), usually by highlighting the positive effects of exports, but the role of imports is ambiguous. Several developing countries have adopted measures to increase trade with developed countries in an effort to increase productivity and employment. Multiple free trade agreements have been signed between emerging countries and the United States to induce a more dynamic trade. At the same time, tensions over free trade between countries of differing economic development levels have emerged, as evidenced by the U.S.-China tariff wars beginning in 2018.

In this paper, we analyze how an increase in import competition from the United States affects the distribution of firms and employment in Colombia. As opposed to previous work (such as (Autor et al., 2013)), we focus on a setting in which import competition from a high-wage country affects the local labor market effects of a lower-wage country, and use a novel identification strategy. To the best of our knowledge we are the first to analyze the effect of imports from a developed country on an emerging economy. Previous literature has struggled to analyze such a phenomenon due to: 1) empirical difficulties in the identification of an import competition shock that does not affect exports; and 2) data restrictions that do not allow longitudinal tracking of firms and workers. This paper overcomes these two difficulties.

First, we use exogenous variation induced by a free trade agreement implemented between the United States and Colombia, and regional variation in access to imports, to evaluate the effects of U.S. import competition. The agreement, which took effect in 2010, affected imports but not exports. Tariffs charged by Colombia fell, but tariffs imposed by the United States were unaffected because they were already low (or zero) due to previous trade agreements.[2] We combine the variation induced by the free trade agreement at the industry level with state variation across states with and without cus-

[2]The free trade agreement mainly reduced Colombian tariffs on manufacturing and service goods. It left in place protections for certain agriculture products for longer periods of time, with protections lasting five years for most products, but continuing for up to 20 years for some products, such as rice.

toms ports. Transportation within Colombia, which lacks train and river transportation options, relies on an underdeveloped road system to trade across states. Political neglect and civil conflict hindered historical construction of roads within the country (Duranton, 2015; Bushnell, 1993), and, as a result, 90 percent of incoming products from the United States stay within eight states that have customs ports. We design a triple-difference model by combining variation across industries that reduced tariffs, variation between states with and without customs ports, and time variation before and after the implementation of the free trade agreement. We find that the agreement increases import competition from the United States in around 15 to 18 percent, but, as expected, does not affect exports. We additionally show that imports increase exclusively among states with customs ports, which gives strong empirical support to our triple-difference estimates.

Second, we take advantage of administrative matched, formal-sector employer-employee data that allow us to identify which firms employ which workers, for what extent of time, and at what wage level.[3] These data include longitudinal earnings records for all workers that contribute to health or pensions in a given month. The data include firm, municipality, and four-digit industry identifiers, and are ideal to study the effects of import competition from the United States on workers and firms in Colombia. We further combine these records with administrative trade data on Colombian imports and exports at the industry and state levels, and with information on tariffs charged by Colombia and the United States. Our final data track individuals and firms from 2008 to 2014 (four years after the tariff reductions in 2010), and are merged with four-digit industry- and state-level imports and exports.

The data allow us to study the role of employers and employees by es-

[3]The data, accessed through the Colombian central bank, extend through 2016. However, due to confounding reasons involving currency devaluations and oil prices (explained in greater detail in Section 1.3.1), we exclude data from 2015 and 2016 in our main analysis. We note, however, that results are unchanged when we do include these data. In addition, we cannot distinguish if the flows out of the firms go to informality or unemployment since the data only include formal workers. We observe steady (or even decreasing) informality rates for agriculture, manufacturing, mining, and services during the analyzed period (2007-2014). Thus, we do not have strong evidence to suspect that our results are driven by informality, and even if this were the case, we still consider the results to reflect interesting dynamics across good-quality jobs.

timating high-dimensional worker and firm-fixed effects models, and identifying low- and high-paid workers and low- and high-paying firms. We follow Abowd, Kramarz, and Margolis (1999) (here onward called AKM) –and further Card, Cardoso, Heining, and Kline (2018) and Card, Heining, and Kline (2013)– to estimate the *ex-ante* distribution of worker- and firm-specific wage premiums using the years previous to the tariff reduction. The firm-specific wage premiums quantify the amount each employer pays to their workers irrespective of the type of workers hired. This wage premium can be interpreted as a combination of firm-specific productivity and rent-sharing elasticities (Card et al., 2018). We consider it to be a good proxy for firm productivity since these two measures strongly correlate (Card et al., 2018; Alvarez, Benguria, Engbom, & Moser, 2018). The worker-specific wage premium quantifies the amount earned by a given worker irrespective of where he/she works. We interpret this as a measure of worker skill. From here onward we refer to the firm-specific wage premium as productivity, and the worker-specific wage premium as skill. To the best of our knowledge we are the first to estimate the heterogeneous effects of import competition on wage premiums across the distribution of specific firms and workers.

Our empirical analysis yields three main results. First, we find that a 10 percent increase in import competition from the United States decreases Colombian employment in around six percent, a magnitude that is similar to that of Chinese imports on employment in the United States (Autor et al., 2013). We additionally find that the decrease in employment is explained by a decrease in both extensive margin (the number of firms) and intensive margin (the average firm size). These results are driven by low-skilled workers who are more likely to lose their jobs, whereas more skilled workers remain unaffected.

Second, the decrease in employment is explained by low-productivity firms, which are more likely to shrink or exit the market. We do not observe any effect on high-productivity firms, suggesting that missing production from low-productivity firms is not reallocated to more-productive firms within Colombia. Instead, consumers likely substitute to imports from the United States. These results contrast to that of Chinese imports on firms in the United States, where losses in employment were mainly driven by high-wage

multinational firms (Bloom et al., 2019).

Third, we do not find an effect of import competition on wages, but we do find evidence that import competition induces reallocation of workers from affected to unaffected industries or states (i.e., from industries that reduced tariffs, and from states with customs ports, to industries that did not reduce tariffs, and to states without customs ports). However, for workers originally employed in low-productivity firms (that were most likely to close or shrink), exclusively, we observe a wage reduction of around 0.75 percent. We interpret this as workers in affected industries and states shifting into other positions, and moving down the job ladder by accepting lower-paid offers after a job loss. (Haltiwanger, Hyatt, Kahn, & McEntarfer, 2017).

We address two potential threats to identification that could undermine our results. First, firm- and worker-level regressions would be subject to selection bias if the sample were not constant in time, and some particular types of workers or firms could be more likely to lose their jobs, or to exit the market. To deal with this we follow Autor et al. (2014) and restrict the sample to a panel of incumbent workers and firms. We track them over time by keeping the sample constant. Second, our empirical strategy relies on the assumption of no differential trends between treated (workers and firms in industries that reduced tariffs, and in states with customs ports) an untreated (all the rest) units before the tariff reduction. We estimate event study models around the implementation of the free trade agreement using imports, employment, number of firms, and firm size as dependent variables. We do not find evidence of any pre-trends for any of these outcomes

.

In addition, we conduct a large number of robustness checks that highlight the stability of our results. The estimates include a considerable amount of fixed effects, and the main results may change depending on the specification chosen. We show, however, that our results do not change when considering alternative structures of fixed effects, or when using a difference-in-difference model that accounts only for the changes in tariffs, and excludes variation across states with and without customs ports. We also control for the value of imported inputs, which, if omitted, could confound our estimations; our results are robust to this. Finally, we include the mining sector and two

additional years in the sample.[4] The results again remain unchanged.

This paper contributes to a large literature quantifying the effects of import competition on local labor markets. Most of this literature has focused on the effect of low-price imports from China (an emerging economy) on industries in high-income countries, such as the United States or countries within Europe.[5] Other analysis that does focus on the effect of Chinese imports on emerging economies are largely descriptive, and they do not provide any causal estimates (Jenkins et al., 2008; Moreira, 2007; Wood & Mayer, 2011). In addition, Dix-Carneiro and Kovak (2017), Dix-Carneiro (2014), and Attanasio, Goldberg, and Pavcnik (2004) study how unilateral liberalization decreases employment and earnings in Brazil and Colombia. However, these papers explore incoming import competition from all types of countries including, for example, China. We contribute directly to this literature in two ways. First, we analyze patterns by low- and high-productivity firms and low- and high-skilled workers using the wage premiums estimated from the AKM model. Second, we analyze the effect of imports coming from the United States, and complement the results by understanding the importance of firm exit.

We also contribute to the literature that studies the effects of trade on firm productivity and exit. Melitz (2003) and Melitz and Ottaviano (2008) show theoretically that an increase in trade increases the productivity of firms by motivating low-productivity firms to exit the market and high-productivity firms to export. Furthermore, Pavcnik (2002), Forlani (2017), Halpern, Koren, and Szeidl (2015), and Olper, Curzi, and Raimondi (2017) empirically evaluate this result finding strong evidence for Chilean, Irish, Hungarian, Italian, and French firms. Fieler, Eslava, and Xu (2018), Medina (2018), and Bas and Strauss-Kahn (2015) additionally find that an increase in trade motivates quality upgrading, whereas Egger and Kreickemeier (2009) theo-

[4]The mining sector is excluded because of confounding issues with oil prices, whereas the years 2015 and 2016 are excluded because of a great devaluation of the Colombian peso.

[5]The effect on the United States see: Autor et al. (2013), Autor, Dorn, and Hanson (2015), Pierce and Schott (2016), Autor et al. (2014), Feenstra and Hanson (1999), Bloom et al. (2019), Bernard et al. (2006).For the effect on Europe, see: Bloom, Draca, and Van Reenen (2016), Branstetter, Kovak, Mauro, and Venancio (2019), Hummels, Jørgensen, Munch, and Xiang (2014)

retically analyze the effect of trade on profits and unemployment. Goldberg, Khandelwal, Pavcnik, and Topalova (2010) analyze the effects of imported intermediate inputs on firms' production. Our results contribute to this literature by showing how imports from the United States motivate firm exit and shrinkage in emerging countries, where the dispersion of firm productivity is larger (Hsieh & Klenow, 2009) compared to developed countries where unproductive firms exit faster (Eslava et al., 2019).

Finally, we contribute to the literature that explains the importance of firm heterogeneity by estimating employer- and worker-specific fixed effects. This literature (Card et al., 2013, 2018; Abowd, McKinney, & Zhao, 2018) studies the relevance of employers on determining wages and how sorting between firms explain wage variation.[6] We contribute to this literature by estimating heterogeneous effects of import competition along the distribution of firm- and worker-specific wage premiums. We find that the responses to import competition are heterogeneous across firms with different wage structures, and workers earning different wages. Ignoring this heterogeneity can have negative policy implications because of potential erroneous conclusions about who is really affected by import competition.

The rest of the paper is organized as follows: Section 1.2 presents some conceptual considerations relating the heterogeneous effects of increased import competition and its relationship with the AKM model. Section 1.3 describes the background as well as the data we use. Section 1.4 details the empirical strategy that identifies the casual effect of import competition from the United States on workers' and firms' outcomes. Section 1.5 presents the results. Section 1.6 provides some robustness checks. Section 3.6 concludes.

1.2 Conceptual Framework

Melitz (2003) shows how trade induces high-productivity firms to export, and, simultaneously, forces low-productivity firms to exit the market. We

[6]Some recent papers have highlighted the importance of firm-wage premiums on economic phenomena like the effects of job displacement (Lachowska, Mas, & Woodbury, 2018) and the gender pay gap (Card, Cardoso, & Kline, 2016)

analyze a setting in which tariffs were reduced in one country (Colombia) and not in the other (United States), leading to an increase in import competition but not in exports. Applying the predictions of (Melitz, 2003) to our setting, we expect import competition to heterogeneously affect low-and high-productivity firms and low- and high-skilled workers. We present a basic framework here (for a more detailed version, see Appendix A.1).

1.2.1 The Heterogeneous Effects of Increased Import Competition

Consider a continuum of J firms indexed by j that combine labor (L) and capital goods (X) to produce output Y. Labor can be either skilled (L_S) or unskilled (L_U), whereas capital can be foreign (X_F) or national (X_N). Each firm is indexed by a level of productivity φ_j that is randomly drawn from a distribution G. The production function takes the form of:

$$Y_j = \varphi_j f\left(L(L_S, L_U), X(X_F, X_N)\right),$$

where $f(\cdot)$ is a production function with usual properties. $\pi_j(\varphi) = \varphi_j P_j(\tau) Y_j - c_j(\tau)$ stands for the firms' profits, where $c_j(\tau)$ is a cost function of firm j, $P_j(\tau)$ is the price of goods, and τ is an ad valorem tariff rate charged to foreign products. Firms have market power and act as monosopnies in a monopolistic competition framework.

Denote $\varphi^* = \inf\{\varphi : \pi(\varphi) \geq 0\}$ as the threshold that determines firm exit. Any firm with $\varphi < \varphi^*$ will exit the market, and any firm with $\varphi \geq \varphi^*$ will stay. Formally, we can express this as:

$$\varphi^* = \frac{c_j(\tau)}{P_j(\tau) Y_j}.$$

The threshold φ^* depends on the tariffs charged to foreign products in two ways: 1) a decrease in τ decreases φ^* by decreasing the cost of inputs; 2) a decrease in τ increases φ^* by reducing the demand for local goods and, consequently, decreasing the price $P_j(\tau)$. This framework suggests that a decrease in Colombian tariffs will induce low-productivity firms to exit the market if the decrease in demand for local goods is larger than the decrease

in costs. It is then expected that import competition affects the likelihood of exiting of low- and high-productivity firms differently.

In addition, the effects of decreasing τ on employment are also heterogeneous between skilled and unskilled workers. In this setting, workers can lose employment in two ways. First, if import competition induces the exit of low-productivity firms, then we will expect that workers in those firms switch to unemployment, move to other firms, or leave the labor force. If firms' productivity is correlated with workers' skills, then the effect of import competition on employment should be heterogeneous by low- and high-skilled workers. Second, skilled and unskilled employment can be also affected heterogeneously if there is a differential substitution between each type of labor and imported goods. Consider the cost function as follows:

$$c_j(\tau) = W_{Uj}L_{Uj}(\tau) + W_{Sj}L_{Sj}(\tau) + X_N Q_N(\tau) + X_F Q_F(\tau),$$

where W_{ij} is the wage obtained by individual $i \in \{S, U\}$ and Q_N and Q_F are prices of national and foreign goods, respectively. A decrease in tariffs will increase the demand for foreign goods, and potentially decrease the demand for skilled and/or unskilled labor. The magnitudes of $L'_S(\tau)$ and $L'_U(\tau)$ depend on the degree of substitution of workers for foreign goods. For instance, we expect to see a larger effect on unskilled employment if unskilled workers have a higher degree of substitution with products from the United States.

1.2.2 Firms and Worker-Specific Wage Premiums

Given this setting, a fundamental task is to account for the heterogeneous effects of import competition across firms' levels of productivity and workers' skills. Both measures are difficult to empirically quantify, but, following Card et al. (2018), we suggest a framework that derives close proxies of these measures and can be computed using matched employer-employee data. Firms have market power and workers have heterogeneous preferences over firms. As we show in Appendix A.1, the equilibrium wages for skilled and

unskilled workers can be expressed as:

$$\ln W_{ij} = \ln \lambda + \underbrace{\ln(\varphi_j f_L(L_j, X_j))}_{FWP} + \underbrace{\ln(L'(L_i))}_{WWP} - \ln(M_{ij}(\gamma_{ij})), \tag{1.1}$$

where M_{ij} is a wage markdown caused by the firms' monopsonic power, and γ_{ij} is the wage elasticity of labor supply of worker i to firm j.

Equation (1.1) has two important components that proxy for firms' productivity and workers' skills. First, the firm-specific wage premium (FWP) is a common measure across all individuals working in firm j and proxies that firms' productivity. It is composed of the firm's level of productivity and the marginal product of labor in that firm. In other words, it is composed of the firms' productivity and a measure of rent sharing from the firm to its workers. Second, the worker-specific wage premium (WWP) captures the common component among all workers of skill type i across firms, and accounts for the productivity of individuals with a given level of skills. It can be therefore considered as a proxy for the worker's level of skill in production. These measures can be estimated using the AKM model on matched employer-employee data (Section 1.4 provides greater detail), and correspond to proxies of firms' productivity and workers' skills that help account for heterogeneous effects of import competition.

1.3 Data and Background

1.3.1 Background

The Free Trade Agreement:- Trade between the United States and Colombia grew remarkably after the beginning of the 1990s, when both countries took measures to increase trade.[7] In 1991 the United States, under the Andean Trade Preference Act (ATPA), eliminated tariffs on a large number of

[7] Appendix Figure A.1 presents the evolution of trade between both countries by industries. Colombian exports to the United States are mostly composed of mining products, whereas imports are composed primarily of manufactured goods. Imports from the United States are mainly manufacturing goods, while exports correspond primarily to mining and secondly to manufacturing.

Colombian products.[8] At the same time, Colombia liberalized markets by decreasing the tariffs charged to the United States to around 15 percent.[9]

Later, in 2006, both countries started negotiations on the implementation of a free trade agreement, that was approved in 2007 by the Colombian government. Four years later, in 2011, the U.S. senate approved it, and was legally implemented in May 2012. However, two years before the implementation date, the Colombian government, under Decree 4114 of 2010 (implemented on November 5th, 2010), unilaterally implemented the tariff cuts originally stipulated in the free trade agreement aiming to boost employment, reduce costs, and increase production. Therefore, Colombia effectively adopted the tariff reductions two years before it was officially implemented by the United States.

The free trade agreement renewed the existing tariff exemptions that had been granted to Colombian products under the ATPA. In return, Colombia reduced tariffs on incoming products from the United States. Tariffs were eliminated for most manufacturing, services, and mining products. Some other goods, most of them agricultural products, remained protected for additional years (in most cases for five years, but for some products such as rice, the tariffs were set to continue for an additional 20 years), allowing local producers to adapt progressively to the incoming competition.[10]

Figure 1.1 presents the evolution of the tariffs charged by Colombia to the United States (Panel 1.1a), and the evolution of tariffs charged by the United States to Colombia (Panel 1.1b). Panel 1.1a shows that tariffs on mining goods decreased with the free trade agreement, whereas tariffs for manufacturing and service goods decreased after 2010. Such decrease is explained by Colombian decree 4111 of 2010. A big portion of agriculture goods, on

[8] ATPA was established to promote Colombia's export industries, as well as to help the fight against drug production. It was continuously renewed after 2002, when it was called the Andean Trade Promotion and Drug Eradication Act (ATPDEA).

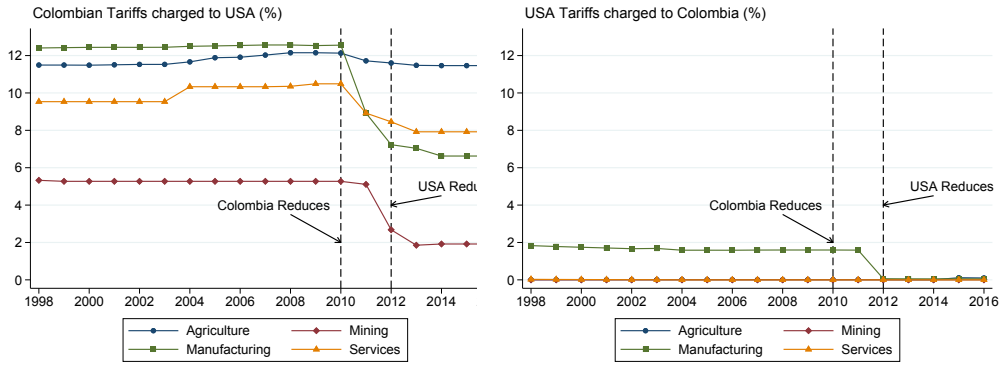
[9] A more detailed discussion about Colombian liberalization can be found in Eslava et al. (2004).

[10] The main protected products were rice, chicken, milk, cheese, butter, corn, meats, motorcycles (between 1500 and 3000 cc.) paper, ink, iron and steel products, glass, and plastics. The agreement additionally regulated competition, customs, environmental rights, intellectual property, and investment procedures.

the contrary, remained protected for some additional years. Panel 1.1b show that tariff cuts for Colombian products entering the United States were much lower, and basically renewed the already low tariff rates imposed years before. Such cuts, nonetheless, were implemented with the agreement in 2012.

Figure 1.1: Tariffs Charged by Country

(a) Colombia (b) United States



Notes: These graphs present the average tariffs charged by Colombia and the United States by two-digit industry codes. These industries correspond to agriculture, manufactures, mining, and services. The left panel presents the historical tariffs that Colombia charged on products from the United States. The right panel plots the historical tariffs charged by the United States on incoming imports from Colombia.

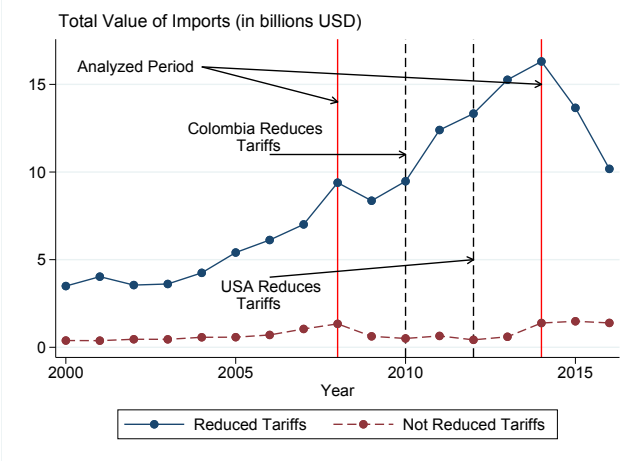
Imports from the United States increased strongly between 2010 and 2014 among industries that reduced tariffs, passing from 9 to 16 billion dollars.[11] In figure 1.2 we present the dollar value of imports by industries that did and did not reduce tariffs. The solid line depicts a continuous increase in imports among the industries that experienced tariffs cuts, exclusively. After 2014, however, the value of imports decreased presumably because of a 30% appreciation of the U.S. dollar with respect to the Colombian peso or a strong decrease in international oil prices.[12]

Regional Trade in Colombia:- Connectivity and trade among Colombian regions are very limited because train and river transportation is practically

[11] Appendix Table A.13 presents a list of the goods most frequently imported from the United States to Colombia, and those that are seldom imported from the United States to Colombia, before and after the free trade agreement.

[12] Appendix Figure A.2 shows the evolution of the exchange rate and oil prices for the analyzed period. The decrease in oil prices affected the dollar value of exports, whereas the peso devaluation (of around 30 percent) increased the price of importing goods from the United States.

Figure 1.2: Imports by Industries With and Without Tariff Cuts

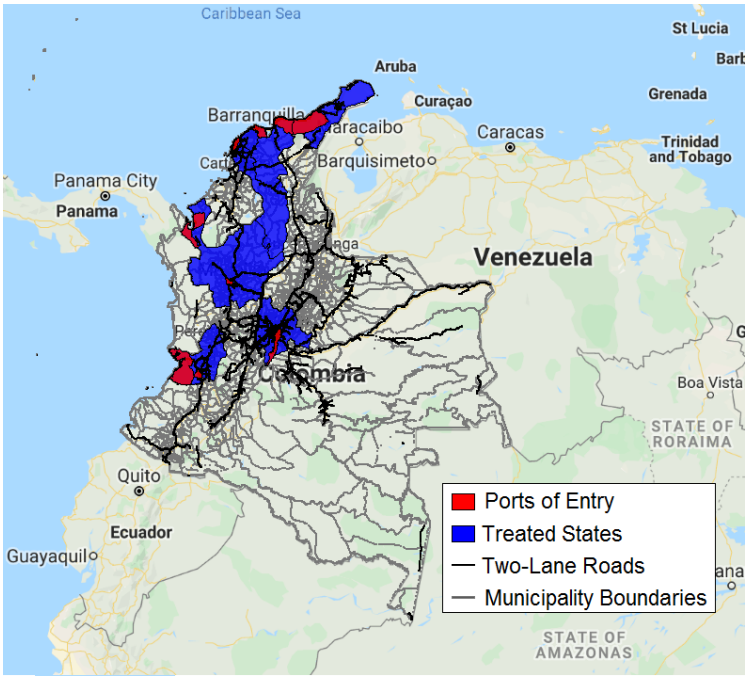


Notes: This graph plots the value of imports in billions USD, by industries were tariffs were and were not reduced. The solid connected line depicts the value of imports among industries that experienced tariff cuts. The dashed connected line presents the value of imports in industries were tariffs did not change. The dashed vertical lines depict the period 2008 to 2014, which is the period we analyze herein. The solid vertical lines plot the years in which Colombia and the United States decreased tariffs.

nonexistent, and road infrastructure is very scarce (Duranton, 2015). The Colombian government historically neglected road construction due to geographic limitations and civil conflict in many parts of the country (Bushnell, 1993). As a result, incoming products from the United States affect mainly states with customs ports, and do not reach isolated areas. In Figure 1.3 we present a map of the different customs ports that receive incoming goods from the United States, and the states around these ports. The map also shows the main roads built as of 2010. All of the roads shown are two-lane highways (i.e., one lane going in each direction). Red areas highlight municipalities with customs ports, and blue areas highlight the surrounding states. The three red areas within Colombian mainland correspond to Bogotá, Medellín, and Cali, which are the three biggest Colombian cities with customs in international airports. Around the coastline the red areas depict the maritime customs of Buenaventura, Barranquilla, Santa Marta, Cartagena, and Riohacha.

Incoming products from the United States stay primarily in states with customs ports. In figure 1.4 we present the value (in millions of U.S. dollars)

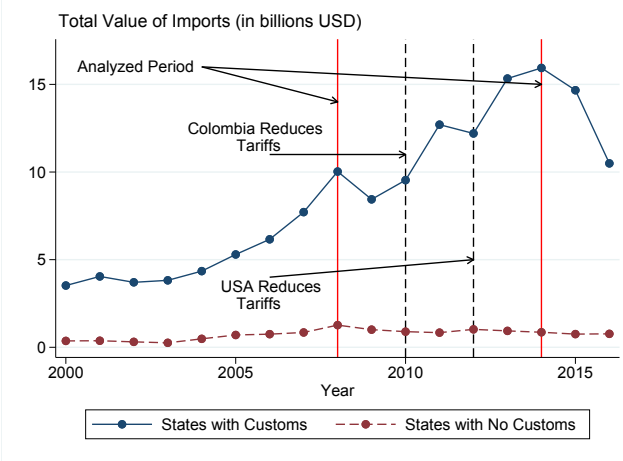
Figure 1.3: Regional Trade



Notes: This map depicts the Colombian territory with all the municipality boundaries. The areas depicted in red correspond to municipalities with custom ports for products imported from the United States. They correspond to: Barranquilla, Bogotá, Buenaventura, Cali, Cartagena, Medellín, Riohacha, and Santa Marta. In blue we depict the states where such municipalities are located. These states correspond to: Atlántico, Bolívar, Cauca, Cundinamarca, La Guajira, Magdalena, and Valle del Cauca.

of imported goods by states with and without customs ports. Around 80 percent of imports from the United States stay in the state where the customs port is located, and 90 percent stay within states that have customs ports. The 10 percent difference stems from goods that move across states that both have customs ports. The majority of the incoming goods stay in the coasts, or reach the heartland by plane. A total of eight states (out of 33) are mainly affected by import competition, while imports in the remaining states are limited. Local producers outside states with customs ports rarely buy products from the United States. In general, they produce products locally and use local inputs. Import competition from the United States thus affects specific regions rather than the entirety of the country.

Figure 1.4: Imports in States With and Without Customs Ports



Notes: This graph plots the value of imports in billions USD, by states with and without customs ports. The solid connected line depicts the value of imports among states with customs ports. The dashed connected line presents the value of imports in states with no customs ports. The dashed vertical lines depict the period 2008 to 2014, which is the period we analyze herein. The solid vertical lines plot the years in which Colombia and the United States decreased tariffs.

1.3.2 Data

We use a rich and unique Colombian data set from three different sources. First, we use yearly matched employer-employee earnings records from 2008 to 2014.[13] This is a confidential, administrative data set that includes all formal-sector workers who contribute to health or pension in any given month. The data include firm identifiers, four-digit industry codes, and the municipality where the payment occurred. A primary feature of the data is that we are able to follow every worker and firm. A limitation, however, is that the data set contains only formal-sector workers, who correspond to around 60 percent of Colombian workers.[14] We complement these data using head-

[13]The data include the years 2015 and 2016, but we excluded the use of data from these years because of the strong devaluation (depicted in Appendix Figure A.2a).

[14]Even though informality is a big issue in Latin America, Appendix Figure A.3 shows that informality by sector in Colombia did not increase during the period we analyze. In fact, during the relevant time frame, we observe a slight decrease in informal employment in the informal sector. This provides evidence that our results are not mainly driven by people moving to informal work. Even if the results were driven by workers shifting from formal to informal employment, we would still consider formal employment to be a first-best option compared to either informal employment or unemployment.

counts of workers per state and industry from the 2005 census.[15]

Second, we use highly detailed administrative records on Colombian imports and exports from 2008 to 2014. These data, recorded at the 10-digit level, are broken up by state where the good was bought (in the case of imports) or sold (in the case of exports).[16] These data constitute administrative information from the customs authorities that is sent to the Colombian central bank. We complement this information with data records on every single imported and exported product by each firm in 2008.

Finally, we use tariff information from the United Nation's Trade Analysis Information System, the World Trade Organization, and the U.S. International Trade Commission. We use multiple tariff data due to differences in the level of industry code aggregation, and because individually no single source covers our entire period of study.[17]

We merge all the data sets using four-digit ISIC codes (845 industries) and state identifiers (33 states), and create three different estimating samples. First, we collapse the data at the industry-state-year level and create a yearly panel of industries by states (N = 142,212).[18] The agreement could have motivated the creation or destruction of industries in some states, so we replace by zeros those industries-states-years that reported having at least one worker between 2008 and 2014. So, for instance, if Bogotá reported production of shoes in 2008 exclusively, then we replace further years with zeros, and vice versa. This sample is static and does not change in time, so there are not issues of potential selection bias.

Second, we create a panel of incumbent (before the free trade agreement) workers in 2010. Individual-level samples change over time, raising concerns about non-random selectivity if particular types of workers leave or enter the

[15]The 2005 census provided the number of workers for every state and industry. Such information is available at: `http://systema59.dane.gov.co/cgibin/RpWebEngine.exe/PortalAction?BASE=CG2005BASICO`.

[16]Colombia has 33 different states that are very heterogeneous.

[17]These different data sources work with different industry codes. Therefore, we homogenize and merge them using four-digit industry codes.

[18]We drop industry by state cells that do not report any person employed from 2008 to 2014. So from a perfectly balanced panel of 195,195, we keep 142,212 observations.

sample non-randomly. Therefore, following Autor et al. (2014), we restrict the sample to workers observed before the tariff reduction, and track them up to four years after. We limit the sample to workers between 25 and 64 years old working in 2010 (N = 6,615,624). Third, we create a panel of incumbent firms (N = 165,724), and again track them up to four years after the tariff reduction. This sample also suffers from potential selection bias, so it is also restricted to those observed in 2010. Following (Card et al., 2013), we keep firms with more than five employees so that the AKM can be estimated. Appendix Table A.1 presents descriptive statistics across all samples.[19]

1.4 Empirical Strategy

Our empirical analysis aims to find the causal effect of imports competition from the United States on Colombian labor markets. The direct correlation between import competition and employment estimated by OLS is unlikely to reflect causal effects due to endogeneity concerns. Reverse causality (e.g., industries with more employment may be more subject to trade effects) and potential unobserved confounders (e.g., unobserved productivity in certain industries or states may correlate positively with trade and labor market outcomes) can upwardly bias the coefficients. For this reason, we implement the strategy detailed herein.

1.4.1 Measuring Import Competition

We first introduce a measure for import competition as following:

$$\log(\text{IC})_{sjt} = \log\left(1 + \frac{\text{USD Value of Imports}_{sjt}}{L_{sj}^{2005}}\right). \tag{1.2}$$

This measure accounts for the per capita import competition from the United States faced by every industry j in state s, and year t in Colombia. We add a one in the logarithm to include all those sectors and regions that have imports equal to zero. It is important to include the zeros since some industries and regions may have not imported before the free trade agreement took effect,

[19] We also use a fully balanced panel with a total of N = 195,195 observations (33 states, seven years, and 845 industries), and the results remained unchanged.

and, as a result of the reduction in tariffs, they may have started importing goods. We normalize by the number of workers in 2005 using data from the Colombian 2005 census at the state and industry levels.[20]

The purpose of this measure is to account for the degree of import competition from the United States across industries and states. We normalize the measure by the size of the workforce since smaller industries or states are more subject to competition. This measure varies by state, industry, and year, and it captures the degree of competition faced before and after the free trade agreement.

Similarly, we define the following measure to evaluate whether the free trade agreement affects exports:

$$\log(\text{EW})_{sjt} = \log\left(1 + \frac{\text{USD Value of Exports}_{sjt}}{L_{sj}^{2005}}\right).$$

This measure replaces the value of exports, and accounts for the degree of exports per worker in each industry, state, and year.

1.4.2 Industry and State Shocks

To address the endogeneity concerns we introduce some shocks that use the exogenous decrease in tariffs and state disparities in access to imports. Consider, first, the following imports and exports penetration shocks:

$$\begin{aligned} IMP_{jt}^{dd} &= 1(\text{COL Reduction})_j \times \text{Post}_t \\ EXP_{jt}^{dd} &= 1(\text{USA Reduction})_j \times \text{Post}_t, \end{aligned} \tag{1.3}$$

where $1(\text{COL Reduction})_j$ and $1(\text{USA Reduction})_j$ are categorical variables that take the value of one if industry j experienced a tariff reduction in Colombia and in the United States, respectively.[21] Post_t is a dummy indicator that takes the value of one after 2010, and zero otherwise. We denote

[20] We divide by workers in 2005 since the free trade agreement may have affected the number of workers in later years.

[21] We define a reduction in tariffs by comparing tariffs in 2014 with tariffs in 2010. A tariff reduction implies that tariffs in 2014 are smaller than tariffs in 2010.

these shocks using the superscript "dd" to indicate that the shocks correspond to a double difference.

The shocks in (1.3) do not take into account regional variation. However, as shown in Section 1.3.1, Colombia has large regional variation in trade flows, with eight states serving as the destination for almost 90 percent of the imported goods from the United States. To account for this, we define the following alternative and more reliable trade shocks:

$$IMP^{ddd}_{sjt} = IMP^{dd}_{jt} \times 1(\text{Customs in State})_s$$
$$EXP^{ddd}_{sjt} = EXP^{dd}_{jt} \times 1(\text{Customs in State})_s,$$

where $1(\text{Customs in State})_s$ is a categorical variable that takes the value of one if state s has a customs port that receives imported goods from the United States. We use the superscript "ddd" to denote that these shocks come from a triple difference.

These measures rely on temporal, industry, and regional variation. We expect that, after the implementation of the agreement, industries with lower tariffs and regions with access to import competition from the United States are more affected than protected industries in isolated regions. It is expected, therefore, that imported goods have a greater effect on manufacturing goods in states with customs ports, such as Bogotá, than non-tradable goods in isolated states, such as the Amazon.

1.4.3 Triple-Difference Model (First Stage)

Consider now the following differences-in-differences model:

$$Y_{sjt} = \alpha_1 IMP^{dd}_{jt} + \alpha_2 EXP^{dd}_{jt} + \mu_s + \mu_j + \mu_t + \varepsilon_{sjt}, \tag{1.4}$$

where Y_{sjt} corresponds to an outcome in state s, industry j and year t. Such outcomes include the measure of import competition from the United States and the measure of export per worker. This model includes state (μ_s), industry (μ_j), and year (μ_t) fixed effects to account for within variation.

Equation 1.4 pools the effect of the decrease in tariffs across states with and without customs ports. However, as suggested in Section 1.3.1, import competition from the United States affects exclusively those states with customs ports. For this reason, we estimate the following model that accounts for these differences:

$$Y_{sjt} = \alpha_1 IMP_{sjt}^{ddd} + \alpha_2 EXP_{sjt}^{ddd} + \mu_{sj} + \mu_{st} + \mu_{jt} + \varepsilon_{sjt}. \tag{1.5}$$

This model includes state-by-industry (μ_{sj}), state-by-year (μ_{st}), and industry-by-year (μ_{jt}) fixed effects and is identified using within industry-by-state variation. The fixed effects account for potential confounding effects and control for potential pre-existing differences. As treated units, the model uses industries that experienced a tariff reduction, and that are located in states with customs ports; as control units, the model uses industries that did not receive tariff reductions, or that are located in states that do not have customs ports. The model treats industries-by-states as separate units and therefore isolates the effect of the tariff reduction by states with and without customs ports.

1.4.4 The Effects of Import Competition (Second Stage)

The results of the triple-difference model quantify the effect of the free trade agreement on import competition. We take advantage of these results and estimate a second stage that uses variation induced by the free trade agreement to identify the effects of import competition on worker and firm outcomes. The richness of the data allows to estimate aggregated (i.e., at the industry-state-year level) and individual level (i.e., at the worker and firm level) models.

Aggregated Model:- We collapse the data at the state-industry-year level, and use the variation in import competition to identify aggregated effects by industries and states. These estimations do not suffer from sample selection since the units of observation remain constant. Using equation 1.5 as a first stage, we can find the effect of import competition from the United States

on aggregated measures by estimating:

$$Y_{sjt} = \gamma_1^{ddd} \log(\widehat{\text{IC}})_{sjt} + \gamma_2^{ddd} \log(\text{EW})_{sjt} + \mu_{sj} + \mu_{ts} + \mu_{jt} + e_{sjt}, \qquad (1.6)$$

where Y_{sjt} correspond to an aggregated outcome (employment, number of firms, and average firm size), $\log(\widehat{\text{IC}})_{sjt}$ stands for the predicted import competition, and $\log(\text{EW})_{sjt}$ is the measure of exports per worker.[22] We also include state-by-industry (μ_{sj}), year-by-state (μ_{ts}), and year-by-industry (μ_{tj}) fixed effects. Excluding such fixed effects will bias the coefficients because the instrument may be no longer exogenous. The parameter of interest is γ_1^{ddd}. Standard errors are clustered at the industry-by-state level.

Individual Level Model:- We additionally estimate regressions at the worker and firm level. To deal with selection issues, we restrict the sample to incumbent observations observed before the implementation of the free trade agreement (i.e., in 2010). Our individual level model takes the form:

$$Y_{isj,t} = \delta_1 \log(\widehat{\text{IC}})_{sj,t} + \delta_2 \log(\text{EW})_{sj,\tau} + \delta_3 X_i + \mu_s + \mu_j + u_{ijs,t}, \qquad (1.7)$$

where $Y_{isj,t}$ corresponds to an outcome, that varies for workers or firms i, state s, and industry j. We measure the outcome year-by-year, up to four years after the tariff reduction, and estimate separate regressions for each year. The measure on import competition from the United States, $\log(\text{IC})_{sj,t}$, corresponds to that in equation (1.2) but included separately for each year. The term $\log(\text{EW})_{sj,t}$ corresponds to the measure exports per worker.[23] Note that this model includes state (μ_s) and industry (μ_j) fixed effects to control for pre-existing differences, and resemble a differences-in-differences specification with repeated cross sections. We additionally control for a set of baseline characteristics X_i.[24] Standard errors are again clustered at the state–industry level, and the parameter of interest is δ_1.

[22] In the first stages we drop the term EXP_{sjt}^{ddd} because it adds noise to the estimations. We also control for the value of exports on the second stage, but excluding it does not affect any of the results.

[23] Again these results are not sensitive to excluding this control.

[24] For workers we include age, age-squared, gender, earnings averaged from 2008 to 2010, worker wage premiums averaged from 2008 to 2010, and number of days worked averaged from 2008-2010. For firms we include wages, firm wage premiums, and the number of days worked. Each of these measures is averaged across all workers within the firm, and then averaged from 2008 to 2010.

This specification is of course subject to reverse causality or omitted variable bias, similar to the aggregated level estimations. To deal with this we instrument using the interaction of the dummy that takes the value of one for industries that decreased tariffs and a dummy for states that have customs ports. Formally, this instrument is defined as:

$$\text{IC}_{sj} = 1(\text{Col Reduction})_j \times 1(\text{Customs in State})_s.$$

1.4.5 Firm and Worker Wage Premiums

Individual level estimations are useful to estimate heterogeneous effects across workers and firms. As we showed in Section 1.2, the effects of import competition are expected to be heterogeneous across firms' levels of productivity and workers' skills. Even though in our data we are not able to directly measure firm productivity and worker skills, we are still able to find proxies by extracting information from wages. In particular, we follow Abowd et al. (1999) and Card et al. (2013) and estimate the following high-dimensional firm and worker fixed-effect model:

$$\ln W_{ijt} = \boldsymbol{\alpha_i} + \boldsymbol{\psi_{J(i,j)}} + X'_{ijt}\beta + \varepsilon_{ijt}, \tag{1.8}$$

where W_{imt} corresponds to income of individual i, in firm j, in period of time t. Equation (1.8) is the sample counterpart of equation (1.1). The components α_i and $\psi_{J(i,j)}$ correspond to the worker- and firm-specific wage premiums, respectively.[25]

The firm wage premium is the component of the wage that is common to all the workers of the firms. Even though it is not an exact measure of productivity, for the purposes of this paper, it can nevertheless serve as a good proxy for productivity. As shown in Section 1.2, and following Card et al. (2018), the firm wage premium can be a function of the firm's productivity. Furthermore, it has been shown empirically that firm productivity and firm wage premiums are highly correlated (Card et al., 2016; Alvarez et al.,

[25] We additionally control for year fixed effects.

2018).[26]

The individual wage premium, on the other hand, measures the workers' compensation irrespective of the firm where he/she works. If wages are equivalent to the marginal product of labor, then the worker wage premium is a measure of worker productivity that is highly correlated with the level of skills. As in the case of the firm wage premiums, we can interpret the worker wage premium as a proxy for skills, but, if we do not accept such an interpretation, then the worker wage premiums still rank workers according to earnings irrespective of the employer. Such ranking reflects discrepancies among workers that should be reflected in heterogeneous effects of import competition.

We classify incumbent firms and workers based on their wage premiums in the period 2008-2010. To do a more reasonable comparison, and to be in line with our estimation strategy, we classify the samples by quartiles within industries and states.[27] We then follow this set of incumbents after the tariff reduction, to identify the effect of the policy on firms and workers of different ex ante ranks.

We estimate heterogeneous effects in three main ways. First, we break down the sample of firms into quartiles of the distribution of firm-specific wage premiums. Such partition distinguishes between low- and high-fixed effect firms, which proxy low- and high-productivity firms. Second, we break down the sample of workers into quartiles of the distribution of firm-specific wage premiums. This allows us to see what happens to workers who were originally employed in low- and high-productivity firms. Lastly, we divide the sample of workers by quartiles of the distribution of worker-specific wage premiums. With this we are able to distinguish among low- and high-fixed effect workers, which proxy for low- and high- skilled workers.

[26] In any case, even if this interpretation of productivity were not accepted, the firm wage premiums still rank firms between lower- and higher-paying firms, and the effects of imports among them are expected to be different.

[27] To do this, we run a regression of the worker and firm wage premiums on state and industry fixed effects, and take the residuals. We examine the results by quartiles of the distribution of these residuals.

1.4.6 Macroeconomic Confounders

Variation in oil prices and the exchange rate (Colombian pesos per dollar) can cause spurious correlations that should be avoided. As we show in Appendix Figure A.2, oil prices and exchange rates fluctuate strongly, especially after 2015. Therefore, we exclude the years 2015 and 2016 from our analysis. As a consequence, our results are valid in the short term only, and we are unable to say much about longer-term effects of import competition. Figure A.2b shows, additionally, a strong decrease in the price of oil in 2009. This variation certainly affects the trade between Colombia and the United States, so we drop the mining sector from our analysis. In Section 1.6 we show that including 2015 and 2016, or including the mining sector, does not alter our main findings, even though those results can be biased due to confounded measures.

1.5 Results

1.5.1 Effect of the Free Trade Agreement on Import Competition (First Stage)

We begin by presenting the results of the free trade agreement on import competition and exports per worker. Columns (1) to (3) in panel A) of Table 1.1 present the results of estimating the differences-in-differences model in equation (1.4).[28] The results suggest that imports increased among the sectors that reduce tariffs, and that the result is robust to alternative fixed effects used. In columns (4) and (5), we present the results separately between states without and with customs ports, respectively, and find that the increase in imports is entirely driven by a strong effect among states with customs ports. In states without customs ports we do not see any effects on imports, presumably because road connectivity hindered the access of these products. We do not observe any effect regarding exports.

These results motivate the estimation the triple-difference model. The re-

[28] We control for the tariff reductions in the United States, but do not display the point estimates.

sults are displayed in columns (6) to (8) of Table 1.1. We observe a strong increase in imports from the United States of around 15 to 18 percent among industries that reduced the tariffs in states with customs ports. The results are robust to the alternative structure of fixed effects. As expected, imports from the United States increase remarkably among non-protected industries in states with customs ports. We again do not observe any effects in exports among industries in which the United States reduced its tariffs.

Table 1.1: Effect of the Free-Trade Agreement on Imports and Exports

	Differences-in-Differences			Het. Effects by States		Triple-Differences		
				No Customs	Customs			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A) Log(Import Competition)								
1(COL reduction)**Post*	2.434***	0.077***	0.077***	0.027	0.175***			
	(0.100)	(0.023)	(0.023)	(0.029)	(0.037)			
1(COL reduction)**Post**1(Customs)						0.172***	0.175***	0.150***
						(0.037)	(0.037)	(0.045)
Ind. F	587.714	11.577	11.574			22.035	22.857	10.878
B) Log(Exports per Worker)								
1(USA reduction)**Post*	0.054	-0.060	-0.060	-0.040	-0.096			
	(0.175)	(0.047)	(0.047)	(0.043)	(0.110)			
1(USA reduction)**Post**1(Customs)						-0.095	-0.096	-0.101
						(0.109)	(0.109)	(0.121)
Ind. F	.097	1.591	1.591			.76	.776	.698
Observations	142,212	142,212	142,212	100,051	42,161	142,212	142,212	142,212
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Industry FE		Yes	Yes	Yes	Yes	No	No	No
State FE			Yes	Yes	Yes	No	No	No
Industry*State FE						Yes	Yes	Yes
State*Year FE							Yes	Yes
Industry*Year FE								Yes

Note: This table presents the estimation of equation 1.5 on imports and exports. Panel A uses the import competition measure, defined as equation 1.2, as dependent variable, and control for the reduction in U.S tariffs. Panel B uses as dependent variable log exports per worker defined as equation 1.2, but changing the value of imports for the value of exports, and controls for the reduction in Colombian tariffs. Columns (1) to (5) present the results of a difference-in-difference model. Columns (6) to (8) presents the results of a triple-difference model as in equation 1.5. 1(COL Reduction) and 1(USA Reduction) are categorical variables that take the value of one if Colombia and the USA reduced tariffs in a given industry, respectively. 1(Customs) is a categorical variable that takes the value of one if the observation is within a state with customs port. *Post* is a categorical variable that takes the value of one for observations after 2010. *** p¡0.01, ** p¡0.05, * p¡0.1

In general, Table 1.1 suggests a strong impact of the free trade agreement on import competition. No effect on exports was expected because tariffs in the United States remained low under the agreement. In fact, the F-stats for exports are very low, and the export shocks are not even significant. Thus, in the rest of the specifications we focus only on imports, and we use exports as a control to account for potential gains from the free trade agreement.[29]

1.5.2 The Effects of Import Competition from the United States (Second Stage)

Aggregated-Level Results

Table 1.2 presents the results of the estimation of the second stage results detailed in equation (1.6). We present the reduced form, OLS, and IV estimates. Columns (1) to (3) suggest that import competition from the United States decreases employment with an elasticity of employment to import competition of around -0.64. Recall that the increase in import competition estimated in table 1.1 is of around 15 to 18 percent, so this point estimate implies a reduction in employment of around 9.6 to 11.5 percent. [30] This magnitude is similar to the United States' elasticity of employment to imports from China found in Autor et al. (2013) (around -0.6), and resembles many other results estimating the effect of import competition on employment around the world.

The decrease in employment can be driven by an extensive margin (firms exit) or an intensive margin (firms downsize). Thus, we analyze the effect of import competition from the United States on the number of firms and the average firm size. Columns (4) to (6) show a strong and robust negative effect on the number of firms, equivalent to an overall decrease of 8.4 to 10 percent.[31] Columns (7) and (9), on the contrary, show a less precise

[29]When we use alternative specifications with and without exports as controls, the results do not change.

[30]Even though we reject the null only at the 10 percent level, we provide in Section 6 additional evidence that this effect is robust to many other specifications in which it is more precisely estimated. We do this for simplicity in the presentation of the paper.

[31]This result is very robust to alternative specifications, as suggested in appendix table 1.4.

Table 1.2: Imports from the United States and the Decline in Employment

	log (Employment)			log (Number of Firms)			log (Firm Size)		
	RF	OLS	IV	RF	OLS	IV	RF	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(COL reduction)*$Post$	-0.080*			-0.064***			-0.023		
1(Customs)	(0.041)			(0.016)			(0.037)		
log(Import Competition)		-0.007	-0.640*		-0.003	-0.559***		-0.006	-0.105
		(0.006)	(0.331)		(0.002)	(0.193)		(0.005)	(0.243)
Observations	142,212	142,212	142,212	142,212	142,212	142,212	142,212	142,212	142,212
F-first stage			11.352			11.352			11.352
Industry*State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry*Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table presents the estimation of equation 1.6. Columns (1)-(3) use log of employment as outcome. Columns (4)-(6) use the log number of firms as outcome. Columns (7)-(9) use the log of the average firm size as outcome. Columns (1), (4), and (7) present reduced form estimates. Columns (2), (5), and (8) presents OLS estimations. Columns (3), (6), and (9) presents two-stage least squares results. All specifications control for industry–state, state–year, and industry–year fixed effects. Standard errors clustered at the industry*state level. *** p¡0.01, ** p¡0.05, * p¡0.1

and smaller, although still negative, effect on the average firm size.[32] Even though the point estimate is not significant, alternative specifications (shown in Section 1.6) yield significant and meaningful results on this outcome, so we cannot rule out that import competition does not affect firm size. This estimate implies an overall reduction in firm size of 1.6 to 1.9 percent.

Such results suggest that import competition from the United States decreases employment among affected industries and states in Colombia. The decrease in employment is mediated by the extensive and the intensive margin, even though the weight of the extensive margin is larger implying a bigger importance for firm exit rather than for firms shrinking.

Individual-Level Results by Wage Premiums of Firms and Workers

Our main interest is to distinguish the effect between low- and high-productivity firms and low- and high-skilled workers. So, as detailed in Section 1.4.5, we estimate heterogeneous effects across the within industry-state distributions

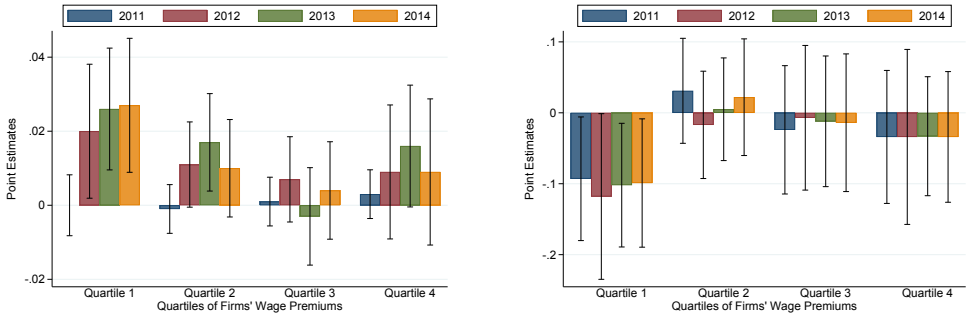
[32] We measure firm size by counting the number of workers registered in each firm by each year.

of firm and worker wage premiums.[33] As a result, we have three sets of results: 1) effects on firms by firm wage premiums; 2) effects on workers by firm wage premiums; and 3) effects on workers by worker wage premiums. The first two show heterogeneous effects by types of firms, whereas the third shows effects that emerge according to types of workers.

Effects on Firms by Firm-Specific Wage Premiums: In Figure 1.5 we present the estimation of equation 1.7 on two outcomes at the firm level: Panel 1.5a uses an indicator variable for whether the firm exits or not; Panel 1.5b uses the log of firm size. We estimate heterogeneous effects using the quartiles of the distribution of firm-specific wage premiums, and track the firms yearly. Quartile 1 corresponds to those firms in the first quartile of the within industry and state distribution of firm wage premiums.

Figure 1.5: Effects on Firms' Outcomes by Quartiles of Firm-Specific Wage Premiums

(a) 1(Firm Exit) (b) Firm Size



Notes: These graphs present estimates of the parameter δ_1 in equation 1.7 at the firm level by quartiles of the distribution of firm-specific wage premiums. Panel 1.5a uses an indicator that takes the value of one if the firm is not observed in year τ as outcome. Appendix table A.3 presents the underlying regressions. Panel 1.5b use the log of firm size as dependent variable, and appendix table A.4 presents the regression results. All quartiles are defined in the distribution of firm-specific wage premiums within industry and state. Panel A presents estimates of a linear probability model. 90% confidence intervals displayed.

We observe in Panel 1.5a that a 10 percent increase in import competition from the United States increases the probability of exiting among firms in the first quartile by around 0.003 percentage points. We also see a positive effect among firms in the second quartile, although the magnitude (0.002 percent-

[33]The results of the first-stage estimation are presented in Appendix Table A.2. We show a very robust first stage with a positive increase in imports.

age points) and the precision are smaller. Firms in the fourth quartile also show positive point estimates, but these are even less precise and smaller. These results suggest that import competition motivates an exodus of firms, especially low-productivity firms. We test for this by grouping below and above the median, and we find that firms below the median are significantly more likely to exit compared to firms above the median.

We also test for effects in firm size among the firms that do not exit. We find that firm size decreases among firms in the first quartile, as shown in Panel 1.5b. As opposed to the effects of Chinese import competition in the United States (Bloom et al., 2019), we do not observe that more productive firms increase in size. Such result implies that missing production generated by firms exiting and shrinking is not appropriated by high-productivity local firms. Instead consumers seem to substitute their previous consumption of locally produced goods with imported goods from the United States.

In general, these results suggest two effects. First, low-productivity firms are more likely to be driven out of the market, inducing a decrease in the supply of locally produced goods. Second, low-productivity firms are likely to shrink, whereas high-productivity firms do not grow. This provides some evidence that local consumers (either individual consumers or firms) decrease consumption, or stop buying locally produced goods and begin buying imported substitutes or other goods from the United States. If this were not the case, then we should observe increases in the size of high-productivity firms induced by the decrease in the supply of locally produced goods. This contrasts with the effects of Chinese competition in the United States, where import competition affects mainly high-paying and bigger firms.

Firm dispersion in productivity is very large in developing countries. Such distribution, especially in Latin America, is usually bimodal, with a predominance of very low- and very high- productive firms Busso et al. (2013); Eslava et al. (2019). The results herein suggest that an increase in import competition from the United States induces the exit of firms, especially among low-productivity firms, and that remaining low-productivity firms decrease in size.

Effects on Workers by Firm Wage Premiums: Results this far leave open the question regarding what happens with the workers who were originally working in firms with different levels of productivity, especially those who lose their job because the firm exit or shrink. To answer this we estimate equation 1.7 on the sample of incumbent workers, and estimate heterogeneous effects across the firms where these workers were initially employed. Quartile 1 refers to workers who were initially working in a firm in the first quartile of the distribution of firm-specific wage premiums. Figure 1.6 presents the results using as outcome a dummy for whether the worker works (panel 1.6a), a dummy for whether working in an industry that did not reduce tariffs (panel 1.6b), an indicator variable for whether the worker works in a state without customs ports (panel 1.6c), and the worker's log daily wage (panel 1.6d). We again follow the same format as in Figure 1.5 separating the results by year and quartiles.

The results in Panel 1.6a present a small and positive effect on the probability of working among workers in the first quartile. Surprisingly, we do not observe that workers in low-productivity firms transition into unemployment (or informality), but instead they are more likely to remain employed than workers in unaffected industries or states. This can occur if workers from low-productivity firms in affected industries or states are reallocated into low-productivity firms in unaffected industries and/or states.

We find direct evidence of this by presenting, in panels 1.6b and 1.6c, the effect of import competition on a dummy for working in an industry that did not reduce tariffs and a dummy for working in a state with no customs ports, respectively. We see that workers in the first and second quartiles are more likely to move to industries that did not reduce tariffs, whereas there is a generalized movement across all quartiles to states without customs ports. The point estimates in panels 1.6b, nonetheless, are much larger in magnitude, implying a bigger reallocation across industries than across states. Such reallocation exists more prominently among workers in low-productivity firms.

This reallocation of workers to unaffected industries and states does not explain entirely the positive point estimates in panel 1.6a. In fact, workers can reallocate and that does not necessarily increase the probability of being

Figure 1.6: Effects on Workers' Outcomes by Quartiles of Firm-Specific Wage Premiums of Initial Firms

Note: These graphs present estimates of the parameter δ_1 in equation 1.7 by quartiles of the distribution of firm-specific wage premiums of incumbent firms. Quartile 1 corresponds to workers working in a firm in the first quartile of the within state-industry distribution of firm-specific wage premiums. Panel 1.6a uses an indicator that takes the value of one if the workers is observed working in year τ as outcome. Appendix table A.5 presents the underlying regressions. Panel 1.6b uses an indicator variable for whether the worker is an industry that did not reduce tariffs as dependent variable, and appendix table A.7 presents the regression results. Panel 1.6c uses in indicator variable for whether the worker works in a state without a customs port. Appendix Table A.8 presents the point estimates. Panel 1.6d uses the log of workers' daily wage, and appendix table A.6 presents the point estimates. Panels A, B, and C present estimates of a linear probability model. 90% confidence intervals displayed.

employed. A possibility is that workers in unaffected industries or states are displaced, decreasing therefore their probability of being employed. This result is equivalent to a situation of job displacement in times of recession as shown in (Haltiwanger et al., 2017).

Theoretically, reallocation of workers mitigates wage losses after a labor demand shock. We observe in panel 1.6d that wages are not affected for workers in the top three quartiles, but workers in the first quartile experience a wage decrease of around -0.05%. Two main reasons can explain this result: 1) workers who switch jobs accept lower-paid offers; 2) workers who

stay experience wage losses. Unfortunately, we cannot rule out either of these explanations. However, the evidence in panels 1.6a, 1.6b, and 1.6c shed some light about workers in low-productivity firm relocating and displacing other workers in low-productivity firms by accepting lower paid offers.

In general, two results can be derived here. First, workers in low-productivity firms move into unaffected industries and states, and they can potentially displace former workers by accepting lower-paid job offers. Second, workers in all the other quartiles shift to unaffected industries and/or states and mitigate wage loses. Those in the second quartile are more likely to move into other industries, whereas workers in high-productivity firms move geographically, across states.

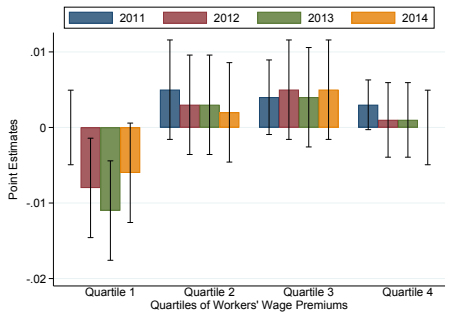
Effects on Workers by Worker Wage Premiums: The effects of import competition from the United States can differ depending on the levels of skills of the workers. Thus, we analyze what happens to workers with different levels of worker-specific wage premiums, which can proxy for skills. We analyze the same outcomes included in Figure 1.6 but now separate the results by quartiles of the distribution of worker-specific wage premiums. These results are presented in Figure 1.7 that follows the same format as Figure 1.6.

The results in panel 1.7a show a precise and negative effect on the probability of being (formally) employed among workers in the first quartile of the workers wage premium distribution. The workers in this quartile do not shift into unaffected industries (panel 1.7b) or into unaffected states (panel 1.7c), and we do not see any effect in wages among the ones who remained employed (panel 1.7d). Such results imply that the losses in employment are mainly driven by low-skilled workers who transition to informality, unemployment, or leave the labor force. The results in employment in Table 1.2 are thus mainly driven by low-skilled workers.

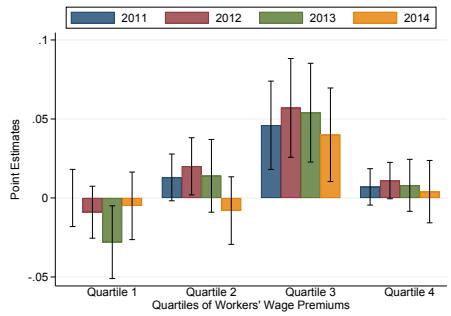
These results contrast with those among workers in other quartiles. We do not observe any effect on the likelihood of working, but we do observe positive point estimates on the probability of shifting into industries that did not reduce tariffs (for workers in the third quartile) and into states without custom ports (for all quartiles except the first). We do not see any effects

Figure 1.7: Effect on Worker Outcomes by Worker-Specific Wage Premiums

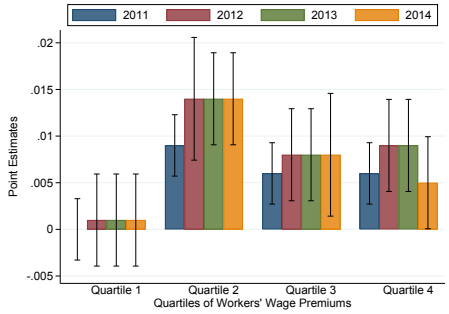
(a) 1(Employed)



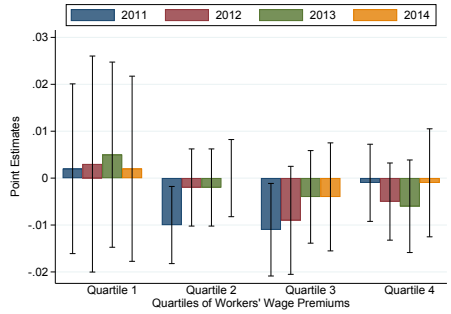
(b) 1(Employed in Unaffected Industry)



(c) 1(Employed in Unaffected State)



(d) log Daily Wage



Notes: These graphs present estimates of the parameter δ_1 in equation 1.7 by quartiles of the distribution of worker-specific wage premiums. Panel 1.7a uses an indicator that takes the value of one if the workers is observed working in year τ as outcome. Appendix Table A.9 presents the underlying regressions. Panel 1.7b uses an indicator variable for whether the worker is an industry that did not reduce tariffs as dependent variable, and Appendix Table A.11 presents the regression results. Panel 1.7c uses in indicator variable for whether the worker works in a state without custom ports. Appendix Table A.12 presents the point estimates. Panel 1.7d uses the log of workers' daily wage, and Appendix Table A.10 presents the point estimates. All quartiles are defined in the distribution of worker-specific wage premiums within industry and state. Panels A, B, and C present estimates of a linear probability model. 90% confidence intervals displayed.

on wages, suggesting that reallocation could have mitigated potential wage losses.

In general, we observe four main sets of results. First, the free trade agreement between Colombia and the United States decreases the amount of import competition but leaves exports relatively unaffected. Second, import competition reduces employment by decreasing the number of firms, and the average size of remaining ones. Such effects are mainly driven by low-skilled workers who are more likely to lose their (formal) jobs. Third, low-productivity firms are more likely to exit the market and shrink in size. We interpret this as evidence of firm selection due to import competition. Fourth, import competition from the United States induces worker movement from

affected industries (i.e., those in which tariffs on related goods fell) and states (i.e., those with customs ports) to unaffected industries (i.e., those in which related tariffs did not fall) and states (i.e., those without customs ports). Such reallocation of labor mitigates or increases wage loses, depending on the case, and can potentially induce job displacement of workers who were originally employed in unaffected industries or states.

1.6 Robustness Checks

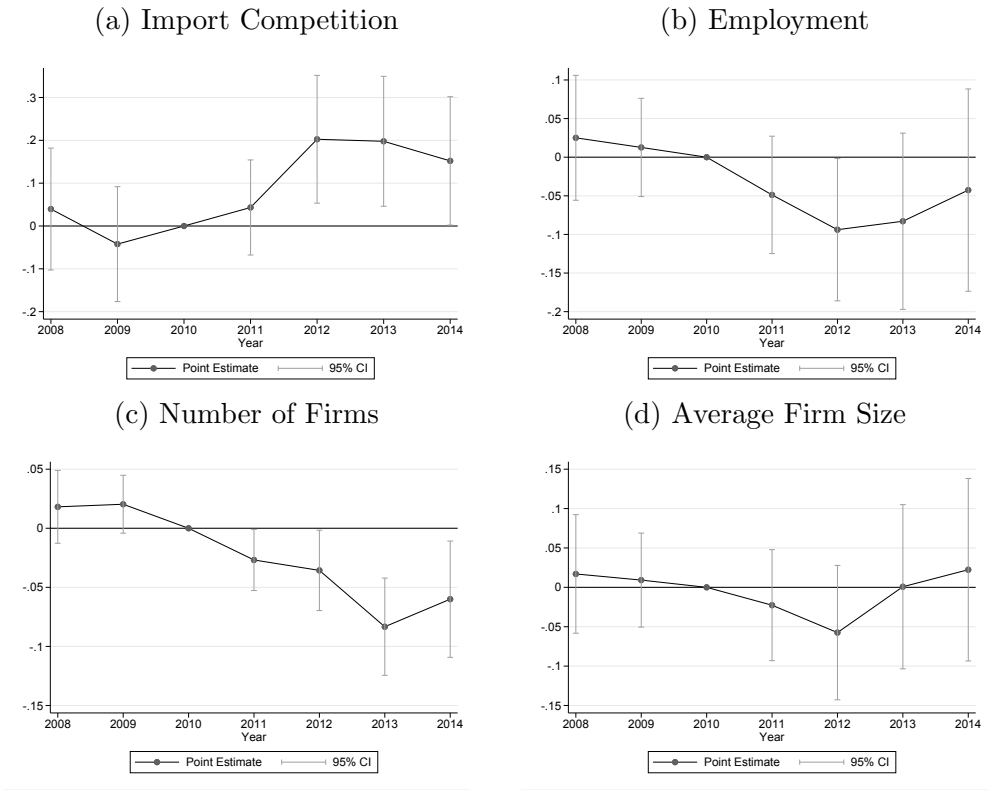
1.6.1 Pre-trends

The biggest threat to identification comes by potential pre-trends between our treated and non-treated units. We test for this estimating event study models that identify potential pre-trends between the treated and control groups. Figure 1.8a plots the coefficients of an event study model, and normalize with respect to 2010, which is when Colombian tariffs were reduced. Panel 1.8a uses the measures of import competition from the United States as outcome, and resembles the first stage of the model. Panels 1.8b, 1.8c, and 1.8d use log employment, log number of firms, and log average firm size as dependent variables, respectively. These results resemble the reduced form estimates of Table 1.2. All the models are estimated in the sample aggregated at the industry-state-year level.

We do not observe any pre-trends prior to 2010, suggesting that our main results may not be driven by differentiated trends among industries that experienced tariff reductions nor states with and without customs ports. We do observe an increase in imports after 2011 among industries where tariffs were reduced and states with custom ports, as the first stage suggested. We also see a decrease in employment, number of firms, and the average firm size after 2010. The effect fades on time, except for the number of firms where we do not see any process of mean reversion.

These results give strong evidence that our estimations are not driven by differential pre-trends, and suggest that we are not violating the parallel trends assumption that is necessary for the proper identification of the double

Figure 1.8: Event Studies to Check Existence of Pre-Trends



Notes: This graph presents the point estimates of an event study model using equation 1.5. Panel 1.8a uses import competition as dependent variable, panel 1.8b uses log Employment, panel 1.8c use log of the number of firms, and panel 1.8d uses log of average firm size. We interact IMP_{sjt}^{ddd} and EXP_{sjt}^{ddd} with yearly dummies, and exclude 2010. We present the coefficients on the interactions of IMP_{sjt}^{ddd} and the yearly dummies. Standard errors are clustered at the state*industry level. 95% confidence intervals are shown.

and triple-difference model. Instead, it gives evidence of the exogeneity of our instrument and the validity of the exclusion restriction.

1.6.2 Differences-in-Differences

Our results are additionally robust to just using a differences-in-differences strategy. In particular, we use the tariff reduction of the free trade agreement before and after 2010 to design a differences-in-differences estimator, and ignore differences among states with and without customs ports. For such we use an import shock as in equation 1.3 and first identify the effect of that shock on the measure of import competition. This estimation is the

first stage of the model, and the results are presented in columns (1) to (3) of table 1.1. Using this as first stage, we instrument the measure of import competition and estimate the following model:

$$Y_{sjt} = \gamma_1^{dd} \log(\widehat{\text{IC}})_{sjt} + \gamma_2^{dd} \log(\text{EW})_{sjt} + \mu_j + \mu_s + \mu_t + e_{sjt}, \tag{1.9}$$

where we control by the measure of exports, and include industry (μ_j), state (μ_s), and year (μ_t) fixed effects. Standard errors are clustered at the industry–state level. The results are presented in table 1.3. We present reduced forms, OLS, and IV results. Columns (1) to (3) use log employment as outcome, columns (4) to (6) use the log number of firms, and columns (7) to (9) use the log of average firm size.

Table 1.3: Imports from the U.S. and Decline on Employment: *D-i-D* Model

	log (Employment)			log (Number of Firms)			log (Firm Size)		
	RF	OLS	IV	RF	OLS	IV	RF	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(COL reduction)**Post*	-0.268***			-0.135***			-0.130***		
	(0.022)			(0.009)			(0.018)		
log(Import Competition)		0.044***	-2.020***		0.011***	-0.952***		0.031***	-1.065***
		(0.008)	(0.379)		(0.004)	(0.175)		(0.006)	(0.218)
Observations	142,212	142,212	142,212	142,212	142,212	142,212	142,212	142,212	142,212
F-first stage			33.315			33.315			33.315
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table presents the estimation of equation 1.9. Columns (1)-(3) use log of employment as outcome. Columns (4)-(6) use the log number of firms as outcome. Columns (7)-(9) use the log of the average firm size as outcome. Columns (1), (4), and (7) present reduced form estimates. Columns (2), (5), and (8) presents OLS estimations. Columns (3), (6), and (9) presents two-stage least squares results. All specifications control for industry, state, and year fixed effects. Standard errors clustered at the industry*state level. *** p¡0.01, ** p¡0.05, * p¡0.1

We observe again negative and very precise results in all three estimations. These results suggest again that the decrease in employment is mediated by firms exiting and shrinking. The magnitudes are bigger than the triple-difference model, but they mix results in states who experienced import competition with states that did not. The triple difference accounts for this issue and gives more reliable estimates.

1.6.3 Moving Fixed Effect Structure

All the estimates so far include a very conservative structure of fixed effects. Our strategy relies on within industry and state variation as the proper exogenous source of variation that identifies the model. However, there are alternative combinations of fixed effects that use different source of time variation to estimate the model. Our results are not altered when considering alternative less-demanding combinations. We present in Table 1.4 alternative models that change the structure of the fixed effects, and therefore the identifying variation.

Table 1.4: Robustness of the Effects of Import Competition from the U.S.

	Differences-in-Differences				Triple-Differences			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A) log *Employment*								
log(IC)	-0.697***	-1.874***	-2.020***	-2.057***	-2.176***	-2.258***	-0.640*	-0.639*
	(0.057)	(0.374)	(0.379)	(0.383)	(0.472)	(0.484)	(0.331)	(0.331)
B) log *Number of Firms*								
log(IC)	-0.390***	-0.875***	-0.952***	-0.972***	-1.044***	-1.227***	-0.559***	-0.559***
	(0.029)	(0.173)	(0.175)	(0.177)	(0.217)	(0.249)	(0.193)	(0.193)
C) log *Average Firm Size*								
log(IC)	-0.270***	-1.000***	-1.065***	-1.079***	-1.229***	-1.029***	-0.105	-0.105
	(0.029)	(0.215)	(0.218)	(0.220)	(0.296)	(0.265)	(0.243)	(0.243)
Observations	142,212	142,212	142,212	142,212	142,212	142,212	142,212	142,212
F-first stage	377.439	33.349	33.315	33.418	26.318	27.098	11.352	11.338
Year FE	Yes	Yes	Yes	Yes	Yes	No	No	No
Industry FE		Yes	Yes	Yes	No	No	No	No
State FE			Yes	Yes	No	No	No	No
Industry*State FE					Yes	Yes	Yes	Yes
State*Year FE						Yes	Yes	Yes
Industry*Year FE							Yes	Yes
Inputs Cost Reduction				Yes				Yes

Note: This table presents IV estimations using the log of employment as outcome in panel A), the log of number of firms in panel B), and the log of average firm size in panel C). Log(IC) stands for log of import competition. Columns (1)-(4) present the estimates of equation 1.9, but vary on the fixed effects included. Column (1) includes year fixed effects, column (2) adds industry fixed effects, and column (3) adds state fixed effects. Column (4) includes the measure of inputs costs. Columns (5)-(8) present the results of the estimation of equation 1.6. Column (5) includes year and state by industry fixed effects. Column (6) includes state by industry and state by year fixed effects. Column (7) adds state by industry fixed effects. Column (8) includes the measure of inputs costs. All specifications additionally control for the log of exports per worker. Standard errors clustered at the industry*state level. *** p¡0.01, ** p¡0.05, * p¡0.1.

Columns (1) to (3) show that the result in the difference-in-difference model

remains unchanged when different fixed effects are included. The precision of the estimates is the same, although the magnitudes change. In columns (5) to (7) we do the same using the triple-difference model. All specifications include state-by-industry fixed effects since our instrument is valid exclusively using within state and industry variation. We see that the sign does not change when including year-by-state not year-by-industry fixed effects. However, we do observe a decrease in magnitude and precision when industry-by-year fixed effects are included. This estimate is our preferred specification since it shows more conservative estimates.

1.6.4 Input Prices

An increase in imported products from the United States increases competition for local firms, but also decreases the cost of some inputs. Firms may be harmed if there is an increase in import competition, but they may also benefit if there is a decrease in the cost of inputs Goldberg et al. (2010). The effects of import competition on the number of firms, for instance, may be mediated by the direct effect of competition or by a decrease in the cost of inputs. To account for the effect on input prices and isolate the effect of import competition, we compute a measure for the cost of inputs and include it as a control.

To construct this measure, we use detailed data on all the goods imported by every firm in 2008. These data includes information on the firms' industry and state. Using this, we compute the following measure of input prices at the state (s), industry (j), and year (t) level:

$$IP_{sjt} = \frac{\sum_k \left[\text{Imports}_{ksjt} \times \left(T_{ksj}^{2008} \times \log(\text{Import Intensity})_{sj}^{2008}\right)\right]}{K_{sj}},$$

where k indexes each input and K_{sj} is the total number of inputs used per industry by state. Imports_{ksjt} correspond to the imported value of input k by industry j, state s, and year t. We multiply this value times some weights that measure the relevance of input k and the intensity in imports of industry j by state s. T_{ksj}^{2008} is a weight equivalent to the percentage imported of good k with respect to the total imported goods in industry j, state s, in 2008.

Formally, the weights are equivalent to:

$$T_{ksj}^{2008} = \frac{\text{Imports}_{ksj}^{2008}}{\sum_k \text{Imports}_{ksj}^{2008}}.$$

Import intensity is a measure of how much the output sector j in state s imported in 2008. We expect that industries that import more are more affected by input prices. Therefore, we multiply the weights for a measure of import intensity. We normalize such measure by the number of employees in 2005 and take logs. Formally such measure is equivalent to:

$$\log(\text{Import Intensity})_{sj}^{2008} = \log\left(\frac{\sum_k \text{Imports}_{ksj}^{2008}}{L_{sj}^{2005}}\right).$$ [34]

We then average across the k inputs to construct a weighted average of the prices of imported inputs for each industry, state, and year.

We include this measure as a control, and present the results in columns (4), for the differences-in-differences model, and columns (8), for the triple-difference model, of Table 1.4. For both cases we observe that the measure of input costs do not have much of an effect on the point estimate of the coefficient associated with the measure of import competition. Therefore, we are able to conclude that the effect of import competition from the United States is neither mediated nor affected by a reduction in input prices. Similar results are found in Autor et al. (2013) when analyzing the employment effects of Chinese import competition among labor markets in the United States.

1.6.5 Mining and additional Years

Finally, we present the results of our main strategy, including the mining sector and the years 2015 and 2016. Recall that we dropped the mining sector because of potential confounders due to the high volatility in the price of oil. A decrease in the price of oil can affect the mining industry, which is one of the biggest industrial sectors in Colombia. Therefore, imports from this industry could be correlated with fluctuations in oil prices, rather than with the free trade agreement; this could confound the estimates. Furthermore,

recall that data from 2015 and 2016 were also dropped because of a steep devaluation that affected Colombia in those two years. A peso devaluation (or a dollar appreciation) changes the prices of imports and, thus, could confound the timing of the free trade agreement's potential effects. Both forces could potentially bias our estimations, so we chose not to include them in our main strategy.

However, our main results do not change much when these additional observations are included. In Table 1.5 we again present the results of the difference-in-difference and triple difference models, including the reduced form, OLS, IV, and input cost-controlled specifications. The results are very close to our main estimates, but they may be somewhat biased due to the variability in the price of oil in 2009 and 2010.

The same happens when we include data from 2015 and 2016. We present these results in Table 1.6, in which we follow the same structure but include these two additional years. Again the main results do not change much, but these results do show larger point estimates that can be confounded by exchange rate fluctuations that primarily affect our measure of imports. We can evidence this by observing the decrease in the F-stat of the first stage compared to the estimations without 2015 and 2016. Nonetheless, all the estimates maintain the signs, and precision does not change dramatically.

In general, our results are robust to alternative samples, controls, and specifications. The signs, magnitudes, and precision are not affected. We also show strong evidence about the non-existence of pre-trends, that validate our identifying assumptions.

1.7 Conclusion

In this paper we explore how import competition from the United States affects firms and employment in Colombia. We additionally take advantage of a high-dimensional workers and firms fixed model (i.e., the AKM model) to find proxies for firms' productivity and workers' skills, and estimate het-

erogeneous effects of import competition among them. To the best of our knowledge we are the first to study the effects of import competition from a developed country on a developing country, to analyze the empirical effects of free-trade agreements in general, and to estimate the heterogeneous effects of import competition across the distribution of firm- and worker-specific wage premiums.

We take advantage of the implementation of a free trade agreement signed between Colom- bia and the United States, and regional variation in access to trade, to estimate the effect of increased import competition from the United States on Colombian firms and local labor markets. The free trade agreement led to increased competition from goods coming from the United States, but left exports unaffected because already low tariffs on Colombia goods exported to the United States remained intact. Furthermore, competition from U.S. goods increased exclusively in specific Colombian regions: those states with access to customs ports. Colombia's lack of infrastructure (trains and roads) hinders wider distribution of the imported products. This unique setting enables us to identify the effect of import competition from the United States on Colombian firms and formal-sector workers.

We use administrative matched employer-employee that allow us to track job spells for formal workers up to four years after the tariff reduction. These data also allow us to estimate high-dimensional firm- and worker-fixed effect models, and estimate the heterogeneous effects among low- and high-fixed effect firms and workers. These fixed effects account for firm and worker wage premiums, which can be interpreted as measures of productivity and skills, respectively. Though the data set excludes informal workers, these data offer a nearly ideal way to identify the effects of import competition on firm exit and employment.

Our results suggest that a 10 percent increase in import competition from the United States decreases employment on 6.4 percent in Colombia. The decrease in employment is mainly driven by low-productivity firms that exit or shrink. High-productivity firms do not grow, suggesting that local consumers substitute locally produced goods for imported products from the United States. We also observe that workers shift mainly from affected to

unaffected industries and regions. We do not observe effects on wages suggesting that reallocation mitigates such effects, except for the case of workers originally working in low-productivity firms who presumably reallocate and displace former workers in unaffected industries and regions by accepting lower paid jobs. Low-skilled workers seem to be the most affected since they are more likely to lose their (formal) job.

This paper provides evidence of the positive and the potentially negative effects of import competition from high-wage countries for developing countries. Spurring the exit of low-productivity firms can boost employment and development, as many have suggested. However, this process exacts a price in terms of employment among lower-skilled workers. The question regarding how long these effects last remains unanswered and should motivate future research on the topic. Negative effects on employment may be mitigated in a longer run, potentially boosting the benefits.

Table 1.5: Effects of Trade of Import Competition Including the Mining Sector

	Reduced Form		OLS		IV		Inputs Cost	
	D-D	*D-D-D*	*D-D*	*D-D-D*	*D-D*	*D-D-D*	*D-D*	*D-D-D*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A) log *Employment*								
1(COL reduction)**Post*	-0.275***							
	(0.021)							
1(COL reduction)**Post*		-0.095**						
1(Customs)		(0.040)						
log(Import Competition)			0.035***	-0.010*	-2.718***	-0.816**	-2.706***	-0.816**
			(0.009)	(0.006)	(0.636)	(0.393)	(0.616)	(0.393)
B) log *Number of Firms*								
1(COL reduction)**Post*	-0.134***							
	(0.009)							
1(COL reduction)**Post*		-0.060***						
1(Customs)		(0.015)						
log(Import Competition)			0.009**	-0.003	-1.252***	-0.585***	-1.245***	-0.585***
			(0.004)	(0.002)	(0.289)	(0.216)	(0.279)	(0.216)
C) log *Average Firm Size*								
1(COL reduction)**Post*	-0.144***							
	(0.018)							
1(COL reduction)**Post*		-0.044						
1(Customs)		(0.035)						
log(Import Competition)			0.024***	-0.008*	-1.503***	-0.269	-1.497***	-0.269
			(0.006)	(0.005)	(0.368)	(0.275)	(0.359)	(0.275)
Observations	149,611	149,611	149,611	149,611	149,611	149,611	149,611	149,611
F-first stage					20.59	9.782	21.593	
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Industry FE	Yes		Yes		Yes		Yes	
State FE	Yes		Yes		Yes		Yes	
Industry*State FE		Yes		Yes		Yes		Yes
State*Year FE		Yes		Yes		Yes		Yes
Industry*Year FE		Yes		Yes		Yes		Yes
Input Import Cost							Yes	Yes

Note: This table presents estimations using the log of employment as outcome in panel A), the log of number of firms in panel B), and the log of average firm size in panel C). All specifications include the mining sector. Column (1) and column (2) present the estimates of equations 1.4 and 1.5, respectively. The coefficients on EMP^{dd}_{sjt} and EMP^{ddd}_{sjt} are not shown. Column (3) presents the estimation of the OLS version of equation 1.9, and column (4) of equation 1.6. Columns (5) and (6) present the second stage estimates of equations 1.9 and 1.6, respectively. Columns (7) and (8) additionally control for input costs. Standard errors clustered at the industry*state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Table 1.6: Effects of Trade of Import Competition Including 2015-2016

	Reduced Form		OLS		IV		Inputs Cost	
	D-D	*D-D-D*	*D-D*	*D-D-D*	*D-D*	*D-D-D*	*D-D*	*D-D-D*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A) log *Employment*								
1(COL reduction)**Post*	-0.353***							
	(0.025)							
1(COL reduction)**Post*		-0.107**						
1(Customs)		(0.046)						
log(Import Competition)			0.044***	-0.003	-3.731***	-1.165*	-3.341***	-1.136*
			(0.008)	(0.006)	(0.914)	(0.679)	(0.746)	(0.660)
B) log *Number of Firms*								
1(COL reduction)**Post*	-0.178***							
	(0.011)							
1(COL reduction)**Post*		-0.083***						
1(Customs)		(0.018)						
log(Import Competition)			0.010***	-0.002	-1.779***	-1.023**	-1.574***	-0.977**
			(0.003)	(0.002)	(0.433)	(0.478)	(0.348)	(0.453)
C) log *Average Firm Size*								
1(COL reduction)**Post*	-0.173***							
	(0.020)							
1(COL reduction)**Post*		-0.026						
1(Customs)		(0.040)						
log(Import Competition)			0.033***	-0.002	-1.963***	-0.100	-1.786***	-0.131
			(0.006)	(0.005)	(0.498)	(0.393)	(0.416)	(0.389)
Observations	182,844	182,844	182,844	182,844	182,844	182,844	182,844	182,844
F-first stage					17.768	5.203	21.582	5.353
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Industry FE	Yes		Yes		Yes		Yes	
State FE	Yes		Yes		Yes		Yes	
Industry*State FE		Yes		Yes		Yes		Yes
State*Year FE		Yes		Yes		Yes		Yes
Industry*Year FE		Yes		Yes		Yes		Yes
Input Import Cost							Yes	Yes

Note: This table presents estimations using the log of employment as outcome in panel A), the log of number of firms in panel B), and the log of average firm size in panel C). All specifications include 2015-2016. Column (1) and column (2) present the estimates of equations 1.4 and 1.5, respectively. The coefficients on EMP^{dd}_{sjt} and EMP^{ddd}_{sjt} are not shown. Column (3) presents the estimation of the OLS version of equation 1.9, and column (4) of equation 1.6. Columns (5) and (6) present the second stage estimates of equations 1.9 and 1.6, respectively. Columns (7) and (8) additionally control for input costs. Standard errors clustered at the industry*state level. *** p¡0.01, ** p¡0.05, * p¡0.1

CHAPTER 2

DO SCHOOL SHOOTINGS ERODE PROPERTY VALUES?

Juan S. Muñoz and Ruchi Singh

2.1 Introduction

On March 24, 1998, two students from Westside Middle School near Jonesboro, Arkansas, opened fire, killing five people and wounding ten others. A year later, two students from Columbine High School in Colorado, killed 13 people and injured 23 others. More recently, a gunman at Sandy Hook Elementary School, in Newtown, Connecticut, killed 20 children, and six staff members. The United States has more mass shootings at schools and elsewhere than any other country. The number rose by more than five times in the period from 2014 to 2017, as shown in Figure 1.[1] About 15 percent of these mass shootings occurred in schools. These incidents have sparked a political debate over gun violence, zero tolerance policies and gun control.

School shootings are a type of crime that is unpredictable; they are exceptionally traumatic events, but are highly unlikely to be repeated in the same location. Our research examines the effects of such occurrences on residential housing values and sheds light on mechanisms behind the relationship between crime and house prices. This relationship has been broadly documented as negative and strong.[2] Households might avoid areas with high levels of crime because of the associated potential loss if they were to be victimized in the future. This link appears to be the logical explanation for

[1] We define mass shootings as gun-related episodes with three or more victims (excluding perpetrators); these incidents are those that do not involve gangs, drugs, or organized crime.

[2] See for instance: (Thaler, 1978); (Hellman & Naroff, 1979); (Linden & Rockoff, 2008); (J. C. Pope, 2008); (D. G. Pope & Pope, 2012); (Lynch & Rasmussen, 2001); (Gibbons, 2004); (Ihlanfeldt & Mayock, 2010) ; (Abadie & Dermisi, 2008); (Gautier, Siegmann, & Vuuren, 2009) ; and (Ratcliffe & von Hinke, 2015).

the relationship between crime and housing prices, but it is not the only one. Crime, in fact, may have some externalities that may shift housing demand due to some other understudied or unrecognized channels. In the case of school shootings, potential homebuyers might want to avoid the affected area due to the associated trauma, lower perceived school quality (due to decline in enrollment, scores etc.), or other stigmas associated with the place.

Literature on mass shootings has traditionally focused on the effects on victims. Empirical studies have shown that such horrific shootings can lead to trauma, stress and increased frequency of mental health conditions such as anxiety, fear, depression which might result in poor academic achievement and have implications for long-term outcomes ((Nader, Pynoos, Fairbanks, & Frederick, 1990)).

We extend this literature and analyze whether mass shootings in schools affect house prices in the school districts in which they occur. Our results suggest that house prices within the affected school districts fall by an average of 7.8 percent (or $15,051 on average), and its effects persists for, at least, five years. Additionally, we find that in the wake of such an incident the number of transactions decreases in the affected school district.

We focus on mass shootings that took place in schools during the period 1998 to 2014. Our analysis employs two key sources of data: 1) the Stanford Mass Shootings of America data project; and 2) individual transaction and assessment records for the school districts where shootings took place and the adjacent school districts for the period from 1991 to 2017. These data were obtained from CoreLogic, Inc., which collects real estate information nationwide. Our analysis uses micro-level transaction data from all U.S. school districts in which a mass shooting took place on a school campus. The coverage of the data makes our results externally valid.

The key challenge in estimating the effect of mass shooting on property values is identifying the counterfactual scenario, i.e., how prices would have evolved in absence of the shooting. Relying on cross-sectional variation alone might lead to biased estimates because house prices might vary across geographic administrative boundaries due to both observed and unobserved characteristics. We address this potential concern by exploiting an exogenous shock in the timing of the shootings and using a difference-in-differences strategy. To estimate the causal effects of school shootings on housing prices,

we compare prices in the affected school district with those in neighboring school districts.

Descriptive statistics at the census tract level suggest differences in observable characteristics between affected and neighboring school districts prior to the school shootings. The difference-in-difference strategy will take care of these preexisting differences, but to test the robustness of our results we use two alternative identification strategies. First, we use a boundary discontinuity approach within a difference-in-differences framework to compare houses within half a mile of the school-district boundary to better control for unobserved amenities. The effect is stronger when we restrict the analysis to properties near a school-district boundary; prices in these areas fall by 13.6 percent (or $20,337 on average), and the decline remains persistent again for five years after the shooting.

Second, we use a propensity score matching approach, within the difference-in-difference framework, to reduce preexisting differences. Given that we use repeated cross sections, we match at the census tract level using observable characteristics before the shooting. Then, we compute kernel weights using the propensity score, and use them to weight our observations. This strategy suggest a smaller but still significant decrease in prices of around 3.5 percent.

We also perform additional robustness checks. Our results are not driven by changes in the composition of houses sold. Furthermore, graphical evidence and falsification tests provide no evidence of spurious negative effects due to differential housing price trends. These tests support our findings that the declines we document in housing prices are indeed an outgrowth of the mass shooting incidents.

We then explore mechanisms that explain the decrease in housing demand in affected schools districts. We first rule out the mechanism under which housing demand decrease because homebuyers are afraid of experiencing an associated crime in the aftermath of the shooting episode. We do so by presenting difference-in-differences estimates that suggest that crime does not increase in cities were the shooting occurred.

Next, we evaluate two alternative plausible mechanisms: deterioration in valuation of school quality and place-based stigmas. To discern whether the results are driven by homebuyers valuing school quality, we examine the market for houses that have more bedrooms, which serve as a proxy for family-size households (those that are most likely to include children

who attend or will attend local schools). The decline in prices is higher for houses with more bedrooms, suggesting that the response is larger in case the family has children. Furthermore, we find a decrease in enrollment and number of teachers in schools that experienced a school shooting and also in neighboring schools that did not experience the shooting but that are within the same school district. This suggest that the quality of school district has deteriorated and that is being capitalized into house prices. Finally, we find no effect on the price of non-residential properties. These findings suggest that lower perceived quality of schools represent particular concerns for potential homebuyers.

Next we test if place-based stigmas explain the decrease in housing demand. If place-based stigmas explain the decrease in demand, then the properties closer to the area of the shooting should witness a larger decrease in prices. We find, however, that house prices seem to remain unaffected when comparing properties closer to the school where the shooting occurred with those farther away. Therefore, place based stigmas do not seem to explain the observed results.

This paper is the first analysis of the effect of school shootings on house prices. It contributes to two strands of research. First, we contribute to the literature on the capitalization of school quality into house prices. Existing research shows that housing prices respond to local school quality as measured by test scores, value added, level of capital expenditure per pupil, school report cards, popularity of school, etc.[3] These papers suggests a lower willingness to pay for housing in neighborhoods in which schools are reputed to be of poor quality.[4] In fact, in a study of school shootings, (Beland & Kim, 2016) find that school quality decreases after a school shooting. Thus, our paper adds to this literature by providing further evidence that deterioration in school quality results in decline in house prices.

Second, we contribute to the literature on the effect of crime on house prices by analyzing how crimes with almost zero probability of repetition

[3]A summary of this literature is provided by (Gibbons & Machin, 2008), (Black & Machin, 2011), (Nguyen-Hoang & Yinger, 2011) and (Machin, 2011). There is a consensus estimate of around 3–4 percent house price premium for one standard deviation increase in school average test scores.

[4]See (Black, 1999), (Agarwal, Rengarajan, Sing, & Yang, 2016) , (Andreyeva & Patrick, 2017), (Davidoff & Leigh, 2008), (Fack & Grenet, 2010), (Gibbons, Machin, & Silva, 2013), among others.

affect property values. Our work adds to the existing works of (Linden & Rockoff, 2008) and (J. C. Pope, 2008), who analyze how proximity to the home of a registered sex offender decreases house prices, and to the work of (Abadie & Dermisi, 2008), (Gautier et al., 2009), and (Ratcliffe & von Hinke, 2015) who analyze the effects of terrorism.

The remainder of the paper is organized as follows: Section 2 describes the data used in our analysis. Section 3 describes our empirical methodology. Section 4 presents the empirical results, and section 5 explores the mechanisms that explain our key results. Section 6 concludes.

2.2 Data

We combine data from two main sources. First, we use arm's length real estate transaction data for the period 1996-2017 for the school districts that were affected by mass shooting in schools, and for the neighboring districts that were unaffected.[5] We merge these data with assessment records using a unique property identifier for each property to ascertain the characteristics of the house. Both, the real estate transaction data and assessment records, come from Corelogic Inc., a national real estate company. The data contain information on transaction, price, and date of sale, along with the geographic coordinates of the house and characteristics of the house like size, age, number of bedrooms, baths, presence of garage, fireplace etc.[6]

We match the sales data to the school districts by using the latitude and longitude coordinates of the property. The school-district boundary maps are obtained from the National Center for Education Statistics (NCES). We

[5]The counties used in our analysis include Craighead, Greene, Lawrence, Jackson, Poinsett in Arkansas; Lane in Oregon; Jefferson, Park, Clear Creek, Gilpin, Boulder, Adams, Denver, Araphoe in Colorado; Rockdale, Dekalb, Gwinnett; Walton; Newton and Henry in Georgia; San Diego, Lake, Modoc, Lassen, Pulmas, Sierra, Nevada, Placer, El Dorado in California; Beltrami, Marshall, Clearwater, Pennigton, Polk in Minnesota; Orange, Almance, Durham, Chatam, Caswell, Person in North Carolina; Lancaster, Chester in Pennsylvania; Multnomah, Clackmass in Oregon; Cuyahoga in Ohio; Saginaw, Bay in Michigan, Geauga, Lake in Ohio, New haven, Fairfield, Litchfield in Connecticut; Washoe, Harney, Carson City, Churchill, Douglas, Humboldt, Lyon, Pershing, Storey in Nevada and Snohomish in Washington.

[6]We drop transactions with sales prices in top and bottom 1 percent of the distribution for each county to eliminate outliers. We also clean the data to remove outliers for any other property characteristics. We normalize the sale prices using quarterly Case-Shiller Home Price Indices for each state to September 2017.

also identify the corresponding census tract by overlaying the transaction data with Census Tract shapefile (2010 definition) obtained from the U.S. Census Bureau.

Second, we use data for mass shootings in America from the Stanford Mass Shootings of America (MSA) data project (courtesy of the Stanford Geospatial Center and Stanford Libraries). The project started in 2012 in reaction to the Sandy Hook mass shooting incident in Connecticut, and collects data from online media sources. The project defines mass shootings as those that involve three or more victims (not necessarily fatalities), excluding the shooter. The shootings do not include those that are gang-, drug- or organized crime-related. The dataset includes the time, date, and location of the shooting, along with number of victims and number of fatalities. It also indicates whether the shooting took place at a school or not. We consider all mass shootings at schools that happened after the year 1998.[7]

These data do not include socio-demographic information about homeowners, although it is very rich and descriptive about house prices and amenities. In order to describe the setting, we therefore use census information at the census tract level prior to the shootings (i.e. we use census 1990 data) merged to the affected and non-affected school districts. The first three columns of Table I present average socio-demographic characteristics for census tracts located within an affected school district (treatment), census tracts located within adjacent school districts (controls), and census tracts in the rest of the country. Column (4) presents the p-value of a difference in means between treated and control areas.

Treated and control census tracts differ across majority of characteristics. In general, control areas are wealthier, have a bigger share of white population, less unemployment, more female labor force, and a bigger share of college graduates. Treated areas, on the contrary, are much similar to the average census tract with a population of around 3,200, a median home value of $93,000, a share of white people close to 80%, and a share of college graduates close to 12%. These results suggest a big degree or preexisting differences among treated and control areas. Additionally, we also use crime data at the city level from City-Data.com and the school enrollment and data on number of teachers from National Center for Education Statistics (NCES).

[7]The date and episodes are presented in Appendix Table 1.

2.3 Empirical Strategy

2.3.1 Effect of Mass Shootings on House Prices

Individuals choose where to live based on many factors such as housing characteristics, school quality, local amenities, proximity to labor markets, etc. This individual sorting usually hinders any potential estimation of the effect of crime on housing values. It is expected that there is higher demand for areas with low crime rates but is also the case that crime is endogenously determined in certain locations. Furthermore, unobserved characteristics also play an important role by including potential confounding factors into the estimation.

School shootings, however, are isolated exogenous episodes that homeowners and buyers are not able to predict. They occur in a random fashion, and thus enable a potential estimation framework free of confounding factors such as individual sorting.

The key empirical challenge, nonetheless, is finding a valid counterfactual distribution - i.e., what would have happened if the shooting had not taken place. We, therefore, use a difference-in-differences strategy to compare house prices in the school district where shooting took place ("treated" school district) to the adjacent school districts ("control" school districts).[8] This strategy estimates the ex-post average price difference between treated and control areas by taking into account the preexisting differences across locations. We will call this our main specification.

We append 15 shooting episodes from 1998 to 2014, and analyze a time window of three years before and after the episode. We also explore longer-term effects by varying the temporal window around the incident. Figure 2 describes our strategy using a map for Fairfield County, Connecticut, where the Sandy Hook Elementary School shooting took place in 2012. The blue

[8] We use school-district boundaries instead of school-attendance zones (as treated and control units) because the attendance boundaries are not available for some of the schools in our datasets. Moreover, it is more difficult to clearly identify the control areas (which are the adjacent areas to the treated unit) for our analysis as the attendance zones overlap. The advantage of using school district-level data is that the schools within the district are subject to the same policies and regulations. For more information on the advantages of school district boundaries see (Dhar & Ross, 2012). The other key advantage of using school-district boundaries is that they do not change much. To our knowledge, for our sample, the school district boundary has changed only for one school.

triangle plots the exact location where the shooting took place. The red area marks the affected school district, whereas the green areas plot the adjacent school districts. Areas in grey are dropped. Our strategy compares the red and green areas.

The estimating equation for the effects of mass shootings on house prices is, therefore:

$$\begin{aligned}\widetilde{ln(p_{ijt})} =& \alpha + \beta_1 T_{ij} + \beta_2 (T_j * 1(\text{After shooting})_t) + \gamma X_{it} + \mu_t \\ &+ \text{Census Tract Fixed Effects} + \sum_{(k=1)}^{14} \mu_t \times 1(\text{episode})_k + \varepsilon_{ijt}\end{aligned} \tag{2.1}$$

Where $\widetilde{\ln(p_{ijt})}$ is the log of the deflated house price for property i in school district j in year t. T_j is a dummy variable that takes the value of one if the property is inside the treated school district and zero otherwise. X_{it} is a matrix of observable housing characteristics such as log of building area, log of land area, dummy for condos, fireplace, brick construction, etc. We include year-month fixed effects (μ_t) to control for time trends and census tracts fixed effects to control for the unobserved characteristics of the neighborhood. We also include episode specific time trends ($\mu_t * 1(\text{episode})_k$) to account for time-varying trends across episodes.[9] ε_{ijt} is the error term. Standard errors are clustered at the census-tract level.

It is worth noting that the treated area is the school district, and thus our strategy is based on an economic intuition. Homeowners in treated school districts are likely to be affected as their children are likely to attend the affected school, whereas homeowners in control areas are eligible to enroll their children outside the affected school district. Thus, the shooting episode affects homeowners in the entire school district and not only those living closer to the school.[10]

The difference-in-difference estimator controls for preexisting differences across treated and control areas. However, to further reduce the concern about preexisting differences we employ two alternative strategies as robust-

[9] $1(\text{episode})_k$ is a dummy that equals one if the observation corresponds to mass shooting episode k which is interacted with time to control for differing time trends across the regions.

[10] The previous literature usually defines treated areas to be within an arbitrary radius around the place of the episode. Our strategy does not rely on this, but instead defines treated areas as areas within a given school district where the shooting took place.

ness checks. First, we use an alternative identification strategy which compares the affected school districts with the adjacent ones, but restricts the sample to observations within half mile from the school-district boundary. Figure 3 describes this strategy for the shooting in Orange High School, NC, in 2006. The different colors represent the distance from the border. We use the properties marked in blue inside and outside the border as treated and control locations, respectively.[11] The estimation strategy is the same as equation (1), but includes properties which are physically closer and are, thus, likely to be similar in observed and unobserved amenities.

Second, in order to reduce preexisting differences, we additionally employ a propensity score matching approach in which we match census tracts based on observable characteristics. First, we identify the census tracts within treated and control school districts and compute a propensity score based on observable attributes prior to the shooting (we use 1990 census data). Second, we match treated and control census tracts using a kernel that computes weights that minimize the observable differences between both distributions using the propensity score. Lastly, we merge these weights into the individual data and compute a reweighting estimator for equation (1) that is balanced in the pretreatment period and further controls for preexisting unobserved heterogeneity. Columns (5) and (6) of Table I show the average value of the observed characteristics weighted by such kernel weights. Column (7) presents the p-values of a difference in means test. The weights balance quite closely the treatment and control areas.

2.3.2 Effect of Mass Shootings on Number of Sales

In addition to the effect of school shootings on sales prices, we also estimate the change in number of transactions taking place after the shooting. We aggregate the data at the census tract - year level for one to three years before and after the incident. We then estimate the following difference-in-

[11] We vary the width of the radius around the border in the section detailing robustness checks. The results are consistent across different specifications.

differences specification.

$$\ln(\text{Sales})_{jt} = \alpha + \beta_1 T_j + \beta_2 (T_j * 1(\text{After shooting})_t) + \text{Year Fixed Effects} \tag{2.2}$$
$$+ \text{Census Tract Fixed Effects} + \varepsilon_{ijt}$$

Where $\ln(\text{Sales})_{jt}$ is the number of sales in census tract j in year t, T_j is a dummy variable that takes the value of one if the census tract is in the school district where shooting took place, and $1(\text{After shooting})_t$ is a dummy that equals one for observations after the shooting. We also include year and census tract fixed effects, and ε_{ijt} represents the error term.

2.4 Results

In this section, we present the graphical evidence and our key findings.

2.4.1 Graphical Evidence

The key identifying assumption for our difference-in-differences strategy is the parallel trends assumption (i.e., house prices had similar time trends in treated and control areas before the mass shooting took place).

Figure 4 presents the evolution of house prices in treated and control areas for three years before and after the shooting.[12] Panel (a) compares the school district where shooting took place to the adjacent school districts, while Panel (b) includes only properties within a half mile of the school district boundary. The pre-trends seems to be similar across the treated and control areas in both panels.

If mass shootings in schools negatively affect housing prices in the school district, then we should expect a decrease in prices after the incident. This decrease should be, at least, larger in the affected school district as opposed to the adjacent ones. Figure 4 suggests that there is an immediate decrease in the price of properties in treated as well as in neighboring school districts. However, the decrease is larger in the affected school districts, and, in partic-

[12] Mass shootings took place at different times across the region. Thus, we normalize the time with time period zero reflecting the date of the shooting.

ular, for properties near the school district boundary. This evidence supports the fact that school shootings decrease property prices, in particular, among residents near the school district boundary. The ex-post decline in prices reflects the causal effect of school shooting on house prices.

2.4.2 Main Results

In columns (1) to (4) of Table II, we present the estimation result of equation (1), comparing house prices in affected and adjacent school districts three year before the shootings to three year after the incident. For illustrative purposes, we first present estimates including only month-year and census-tract fixed effects, but no other control variables (column 1). The result suggests an average decline of 6.6 percent in affected school districts as compared to house prices in the neighboring school districts over a three year period.

In column (2), we include house characteristics to control for observable heterogeneity in properties. Our most complete specification is shown in column (3) of Table II, where we include episode-specific time trends to account for time varying heterogeneity across shooting episodes. Our estimates suggest that housing prices decline by 7.8 percent in the school district where shooting took place.

Finally, in column (4) we weigh each observation by the inverse of the total number of transactions in each episode before the shooting.[13] We estimate an 11.1 percent decline in house prices when using the weighted regression. This estimator reinforces our results and suggests that our findings are not driven by any specific episode where higher number of transactions have taken place.[14]

Next, we analyze whether the relative decline in house prices in affected districts persist. We estimate equation (1) but compare prices three year prior to the shooting with prices one to five years after the episode. The results are presented in columns (5) to (9) of Table II, and are similar to an event-study analysis. As column (5) shows prices decrease 9.3 percent

[13]We sum the number of transactions three years before shooting and weigh each observation by the inverse of this number.

[14]In addition, we estimate in Appendix Table II the same weighted regression but excluding each episode separately. The results hold and are still robust meaning that one episode is not explaining the entire result.

during the first year after the shooting and then decline by 6.8 percent and 4.5 percent in the second and third year. We observe negative effects even four to five years after the incident. This result implies that the effect of shooting declines but persists up to almost five years in the aftermath of the shooting.

2.4.3 Effect on Number of Transactions

The decline in prices may be explained by shifts in housing supply or demand after the shooting. To understand what drives the decrease we estimate the effect of shootings on the number of sales using equation (2). The results are presented in Table III. Columns (1) to (3) present estimates for the entire school district while Columns (4) to (6) present the result for area half mile around the boundary. Columns (1) and (4) show results for the effect of shooting when we include observations three year before and after the shooting. Columns (2) and (5) include observations two years around the shooting, and columns (3) and (6) repeat the analysis for one year before and after the shooting.

The results are insignificant for the first year. The point estimates increase in magnitude in the following years, revealing a decrease in the number of transactions. After two years, for instance, the number of properties sold in the affected school districts decrease by 8.3 percent and 13.9 percent after three years. We also find suggestive evidence of decline in number of transactions in half mile area around the school district boundary but the results are not statistically significant, probably due to small sample size. Supply and demand analysis tells us that for prices and quantity (number of transactions) to decrease jointly, the decrease in the demand for housing must be larger than the possible increase in the supply of housing. If demand decreases and supply remains steady, then we may observe a new equilibrium with lower prices and quantities. However, the supply of housing can still increase but not as much as the decrease in demand, and we can also observe a decline in prices and quantities. The above results suggest, therefore, that the underlying cause of the fall in prices in the wake of a school shooting is a significant decrease in demand– at least higher than the potential increase in supply.

2.5 Robustness

2.5.1 Alternative Specifications

We now analyze alternative specifications that minimize preexisting differences between treated and control school districts. We first restrict the analysis to properties near the school district boundary, which may be more comparable in observed and unobserved characteristics. Then we use kernel weights obtained from propensity score matching to estimate a weighted estimator that balances the sample of census tracts in the pretreatment period.

Table IV summarizes the results obtained from the regression where we restrict our analysis to houses within half a mile of the school-district boundary. We keep the same structure as Table II. The results suggest that mass shootings decrease property values by 13.6 percent (column 3) over a three year period after the shooting among households who live near the school district border. This result is again robust to alternative specifications and weighing strategy.[15] The effect on properties near the boundary persists as suggested by columns (5) to (8) suggest. In fact, in the first year after the shooting we observe a 17.2 percent decrease in house prices, although such effect decreases monotonically in magnitude.

We then present the results using the propensity score weights in Table V. These estimates resemble a reweighting estimator that balances the observable characteristics between treated and control census tracts. The point estimate suggests a 3.5 percent decline and remains statistically significant.

These results suggest that the estimated effects are robust to alternative methods that further control for preexisting differences among treated and control areas.

2.5.2 Falsification Tests

To test the robustness of the previous estimates, we next need to determine whether the shooting was indeed uncorrelated with any other observable or unobservable factor. Moreover, there is a possibility that the estimated decrease in home values is a result of differential trends between affected

[15] In Appendix Table III we vary the radius around the boundary. We show that the effect is not driven by picking particular radius around the boundary.

and non-affected areas before the shooting. This would be a serious concern for our estimates because if the affected areas were experiencing a relatively slower decline in prices, our findings could be due to a spurious negative effect. To test for these concerns, we perform placebo experiments by leveraging the length of our transaction dataset. Instead of using the actual year of the mass shooting, we use false shooting dates two, three, and four years before the actual shooting. We estimate the placebo test for the entire school district and for properties within a half mile boundary of the school district, and re-estimate equation (1) using a one-year windows before and after the fake episode.[16]

The results of the placebo experiments are presented in Table VI. Neither of the six specifications reveal a negative and significant decline after the fake episode, suggesting no systematic differences between treated and control areas previous to the shooting. These results suggest that the decline in housing prices after the shooting is indeed a result of the mass shooting itself.

2.5.3 Composition of Houses

A final potential concern is the possibility that the types of houses being sold before and after the incident are different. This could happen if the potential sellers of expensive houses in affected school districts had decided to postpone placing their houses on the market in the hope that housing prices would eventually recover after some time passed. Thus, our results might not be capturing the dis-amenity due to mass shootings but instead might be reflecting the changes in composition of houses being sold by virtue of a market characterized by having more houses available at the lower end of the price range and fewer houses available at the higher end of the market. To test this we compare the observable characteristics of the houses sold before and after the incident. We use the same model as in equation (1), but instead of using house prices as the dependent variable, we include one-by-one the housing characteristics. Results are presented in Table VII. We do not find any evidence of changes in composition of houses on the market.

[16] We use one-year windows because we do not have enough observations for all the episodes to estimate three-year windows in the pre-shooting period that do not include observations after the shooting.

2.6 Mechanisms

The relationship between crime and property values may be explained by many potential channels. Perhaps the most prominent channel is that housing demand in high-crime areas is low because individuals do not want to personally experience crime. However, the probability that a school district experiences a shooting again is very low, and is no different from the probability that any other school district experiences its first school shooting.

In fact, we test for this by estimating a difference-in-differences model at the city level using yearly crime rates from 2002 to 2016 as the dependent variable. We use cities where the school shooting took place as treated units and neighboring cities as controls.[17] The results are presented in Table VIII. Each column uses the number of crimes per 100,000 as dependent variable, except for column (9) that uses a principal component index computed using the previous eight columns. The results provide suggestive evidence of decline in crime. Therefore, increase in crime (i.e. a higher probability of repetition) is not the reason for the decline in house prices.

Two other potential channels might be driving the observed decrease in demand for affected areas. First, parents might want to avoid the area because of a possible decline in perceived school quality. (Beland & Kim, 2016), in fact, find strong evidence of a decrease in school quality after a school shooting. School shootings may give strong negative signals to homebuyers about the quality of the schools within the school district and also about the future quality of those schools. Second, place based stigmas may motivate people to avoid the area in general. Both explanations may shift the demand for housing in a particular location, and may coexist in the case of school shootings. We find strong evidence for the mechanism concerning perceived school quality, but not for the mechanism concerning broader social stigma.

2.6.1 Deterioration of School Quality

If deterioration in school quality is driving the results, we would expect the effects to be stronger for families with school-age children. In our dataset, we cannot directly observe which families have children. Thus, we use the number of bedrooms in a house as a proxy for family size because families

[17] This analysis does not cover all of the cities but only cities where data was available.

are likely to have houses with more bedrooms. Our prior assumption posits that houses with more bedrooms will witness a larger decrease in prices as compared to houses with fewer bedrooms.

In Table IX, we estimate equation (1) for properties with one, two, three, and four bedrooms. In the last two columns we pooled properties with fewer and more than two bedrooms. We find that the effect becomes stronger as the number of bedrooms increases, whereas one-bedroom houses have a positive point estimate. These estimates support the fact that families with children are driving the price decrease, presumably because of a deterioration in the perceived school quality, or a desire to avoid low quality schools.

To explain the mechanism better, we test whether school quality has deteriorated after shooting. We observe a remarkable decrease in school enrollment and the number of teachers, which is in line with (Beland & Kim, 2016). In Table X, we present difference-in-difference estimates at the school level using the number of students and teachers as dependent variable. Columns (1) and (2) present the results on enrollment, whereas columns (3) and (4) present the results for the number of teachers. We present only the interaction term of the difference-in-differences model, and in columns (1) and (3) we define treated schools as those within the school district of the school where the shooting occur. In the remaining columns we estimate separately the effect for the schools where the shooting occurred and for neighboring schools that are within the affected school district but did not experience the shooting directly.

The results show an overall decrease of 16 percent in enrollment and 8 percent in the number of teachers among all the schools within the school district where the shooting occurred. When we look separately at the schools where the shooting took place we see a bigger reduction of 17.6 percent in enrollment and 12.6 percent in the number of teachers. Furthermore, in neighboring schools within the affected school district (excluding the directly affected one) we also observe a decrease of 7.5 percent in enrollment and 5.2 percent in the number of teachers, although the latter is not precisely estimated.

These results suggest that school shootings reduce enrollment and the number of teachers across the entire school district. In fact, not only schools where the shooting took place where affected, but we also find some spillover effects to neighboring schools within the same school district. This argument

reinforces the mechanism that school quality deteriorates after shooting and results in decline in house prices.

Finally, if deterioration in school quality is the key reason for decline in house prices, then we should not expect prices of commercial and industrial properties to decline. Thus, we estimate equation (1) for non-residential properties and present the results in Appendix Table IV. The coefficients of these regressions turn out to be positive and insignificant, suggesting no effect on prices of non-residential properties.

2.6.2 Place-Based Stigmas

To determine if place-based stigmas, and not valuation of school quality, explain the decrease in demand for housing we analyze what happens with housing values within the school district. In particular, the entire school district has to experience a decrease in prices if perceived school quality is what really declined. That is, if stigmas about the area led to decline in prices, then properties closer to the shooting will have a larger decrease in prices compared to the ones farther away, but if school quality concerns drive the decreases, then everyone within the district will experience a price decrease. To test for this, we limit our estimation to properties close to the shooting, and compare the effect on properties within this limited range of houses.

Following (Linden & Rockoff, 2008) and (J. C. Pope, 2008), we construct radius of 0.3, 0.5, and one mile around the school, and compare property value inside these radii to properties outside of it. For consistency, we again restrict the time frame to three years before and after the shooting. The results of these estimations are presented in Table XI. Columns (1) to (3) include properties within three miles of the shooting, whereas columns (4) to (6) include properties within five miles.

We do not find any differential effects among properties closer to and farther from the affected schools. The point estimates are not statistically different from zero. These results suggest that stigmas may not be driving the result; if they were, properties closer to the school would have been "more stigmatized" than properties farther away. Instead, the results imply a general erosion of property values within the entire school district. This

supports our contention that valuation of school quality drives the shift in demand, and not potential place based stigmas.

2.7 Conclusion

In this paper, we estimate the effect of mass shooting in schools using a difference-in-differences framework. We find that, on average, the home values in affected school districts decrease by 7.8 percent (or $15,051) one year after the shooting. The effect is stronger when we look at homes closer to the school district boundary (about 13.6 percent or $20,337).[18] We also observe some persistence in this effect for, at least, five years throughout the district. Additionally, we find a decrease in the number of transactions taking place after the shooting.

Furthermore, we find strong evidence that our results are driven by the trauma witnessed by school children and perceived deterioration in school quality. Literature has suggested that schools are a highly valued amenity among households. Our results validate these findings and suggest that potential homebuyers avoid school districts in which a shooting has taken place, and that school quality decreases (lower enrollment rates, scores etc.) after the shooting. We do not find much evidence of place based stigmas as being the mechanism for decline in house prices.

The magnitude we estimate for the entire school district (around 7.8 percent over a three-year period) is slightly larger compared to previous estimates of the effects of schooling outcomes on property values. For instance, (Black, 1999) estimates a 2.5 percent increase in housing values for a 5 percent increase in test school scores, whereas (Gibbons et al., 2013) estimate a 3 percent increase in prices for an increase of one standard deviation in average value added.

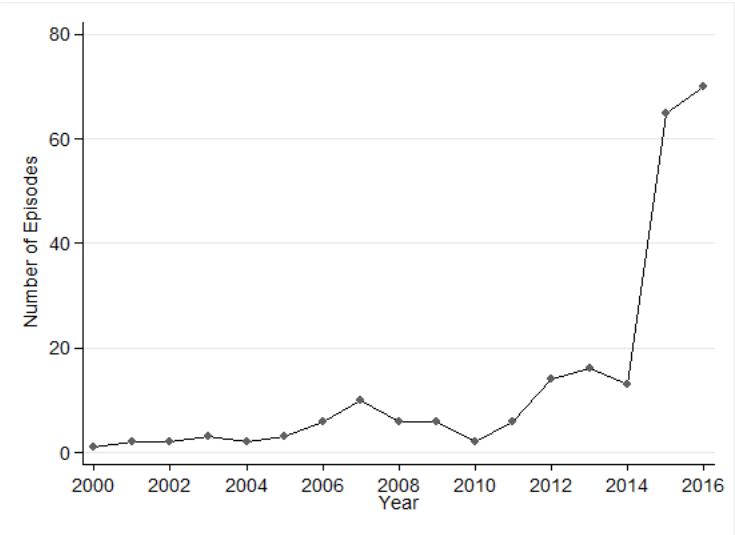
The decrease in prices we find near the school-district boundary (around 13.6 percent) is comparable to the dis-amenity found by (Linden & Rockoff, 2008) in work conducted in North Carolina that examines the effect on prices of houses in close proximity to the residence of a registered sex offender (a

[18]The average price of houses in affected school districts prior to the shooting was around $192,961. A 7.8 percent decrease then implies a decrease of approximately $15,051. The average value of a house near the school-district boundary was $149,539, so a 13.6 percent decrease is equivalent to approximately $20,337.

decline of 11.6 percent). Our results are also similar to the effects on house prices that stem from the discovery of a cancer cluster of child leukemia (a decline in values of 14 percent) ((Davis, 2004)), and the temporary, one-year effect of getting a school quality rating of "A" rather than "B" (20 percent) found by (Figlio & Lucas, 2004).

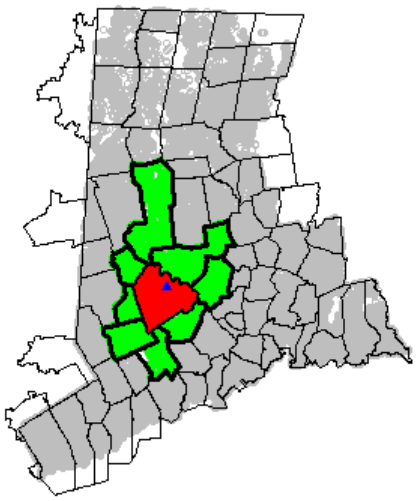
Overall, our results suggest that households have a strong preference to reside in a good school district. Incidents such as school shootings that deteriorate the school quality, might therefore lead to decline in house prices in that school district. Future research is needed to understand how to deal with locations affected by crime shocks, particularly with school- related crime.

Figure 2.1: Number of Mass Shootings 2000-2016



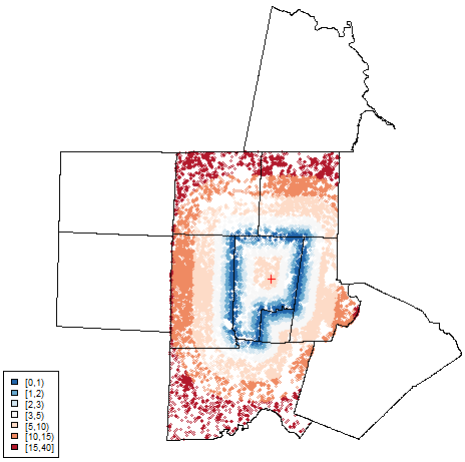
Note: Data source is the Stanford Mass Shooting in America data Project. Episodes included had more than three victims and were not drugs, gangs, or organized crime related.

Figure 2.2: Sandy Hook and Adjacent School Districts



*Note:*This map plots Sandy Hook school district in Newton, CT. The blue triangle indicates the location of the 2012 shooting. The red area indicates the affected school district whereas the green area the adjacent school districts.

Figure 2.3: Distance of Properties to Border in Orange School District, NC



Note: This map plots Orange County, NC County. The red cross indicates the location of Orange high school, where a shooting took place in 2006. The different colors show the distance to the school district boundary.

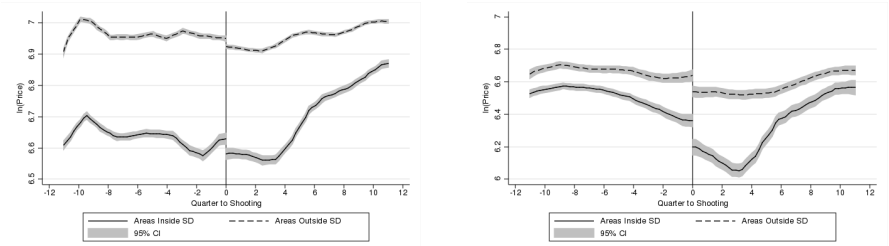
Table 2.1: Summary Statistics at the Census Tract Level for Affected and Non-affected School Districts

Covariate	Raw				Weighted		
	Treated	Controls	Rest	P-values (1) vs (2)	Treated	Controls	P-values (5) vs (6)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Population	3217.9	3337.2	3423.7	0.094	3269.7	3267.0	0.967
Median Home Value	93003.3	112291.2	101291.1	0.000	97926.3	98490.8	0.810
Median Rent	415.8	462.0	391.9	0.000	424.5	434.4	0.112
Income Per Capita	14054.6	16066.7	14447.5	0.000	14425.5	14738.0	0.167
Median Household Income	31940.4	35519.4	32079.0	0.000	33138.3	33702.3	0.306
Percentage White	80.1%	82.6%	83.2%	0.026	83.7%	85.0%	0.219
Percentage Black	15.1%	10.6%	10.9%	0.000	11.2%	10.1%	0.276
Percentage Hispanic	4.9%	6.6%	6.9%	0.000	5.3%	5.0%	0.306
Labor force	50.4%	53.2%	49.5%	0.000	51.8%	52.0%	0.435
Employment Rate	91.6%	94.5%	93.3%	0.000	93.1%	93.3%	0.404
Unemployment Rate	8.4%	5.5%	6.7%	0.000	6.9%	6.7%	0.404
Manufacturing Share	16.7%	16.7%	17.3%	0.830	16.2%	16.5%	0.497
Female Labor Force	58.9%	61.3%	56.6%	0.000	60.3%	60.6%	0.490
Self-Employment Share	6.6%	7.4%	7.7%	0.000	6.9%	6.9%	0.925
Share College Graduates	12.0%	17.1%	13.0%	0.000	12.6%	12.8%	0.477
Percentage Married	40.4%	43.2%	43.4%	0.000	41.9%	42.2%	0.408
Poverty Rate	14.2%	9.9%	12.8%	0.000	12.2%	11.3%	0.059
White Poverty Rate	7.1%	6.1%	7.8%	0.000	7.3%	7.1%	0.597
Percentage of Old Houses	39.0%	32.4%	40.1%	0.000	35.0%	35.6%	0.627

Note: This table presents summary statistics from the census 1990 at the census tract level. Columns (1)-(3) presents mean values using raw data. Columns (5) and (6) present weighted mean values that use weights from a kernel propensity score matching algorithm. Census tracts are matched using 1990 characteristics, and the sample is restricted to the common support. Columns (4) and (7) present the p-values of a difference in means test between affected and non-affected school districts for the raw and the weighted averages, respectively.

Figure 2.4: Price Evolution before and After the Shooting of Affected and Adjacent School Districts

(a) All School District (b) Half Mile from Boundary



*Note:*Both panels present the results of a kernel-weighted local polynomial regression for treated and control areas. Bottom panel includes only properties within half mile to the school district boundary, whereas the top panel includes the full school district. The plots include observations three years before and after the shooting.

Table 2.2: Average Effect of School Shootings in the School District

	Dependent Variable : ln(Price)								
	3 year window around the shooting				Persistence (Comparing to three years before episode)				
					<1 year	1-2 years	2-3 years	3-4 years	4-5 years
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(Within Affected SD)	-0.066***	-0.075***	-0.078***	-0.111***	-0.093***	-0.068***	-0.045***	-0.038***	-0.046***
*1(After Shooting)	(0.013)	(0.013)	(0.011)	(0.016)	(0.012)	(0.011)	(0.010)	(0.010)	(0.011)
Observations	797,611	797,611	797,611	797,611	484,259	495,798	492,208	478,298	470,415
R-squared	0.643	0.752	0.758	0.706	0.764	0.769	0.773	0.769	0.761
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend			Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted Regression				Yes					

Note: The dependent variable is the log of the deflated price per property. Estimations include all properties within the school district. Columns (1) to (4) include observations three years before and after the shooting. Columns (5) to (9) compares observations three years before with observations 1,1-2, 2-3, 3-4, and 4-5 years after the shooting, respectively. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. The regression in column 4 weighs the observations by the inverse of the total number of properties sold within each episode. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.3: Effect of School Shooting on the Number of Sales at the Census Tract Level

	Dependent Variable: ln(Sales)					
	Full School District			Half Mile Around Boundary		
	+3\-3 yr	+2\-2 yr	+1\-1 yr	+3\-3 yr	+2\-2 yr	+1\-1 yr
	(1)	(2)	(3)	(4)	(5)	(6)
1(Within Affected SD)	-0.139***	-0.083*	0.016	-0.101	-0.039	-0.023
*1(After Shooting)	(0.044)	(0.045)	(0.051)	(0.078)	(0.083)	(0.110)
Observations	14,331	4,804	4,744	3,049	1,055	2,924
R-squared	0.725	0.826	0.777	0.727	0.825	0.850
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes	Yes

Note: These regressions are done at the census tract-year level. The dependent variable is the log of the number of sales at the census tract and year. Columns (1) to (3) include the full school district. Columns (4) to (6) include properties one mile from boundary. Columns (1) and (4) include observations on the number of transactions three years before and after the shooting. Columns (2) and (5) include observations of the number of transactions two years before and after the episode. Columns (3) and (6) include observations of the number of transactions one year before and after the shooting. Property characteristics include the average log of area of land, the log of area of building, percentage of apartments, percentage of condos, percentage of properties with fireplace, percentage of properties constructed with brick, and the average number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.4: Effect of School Shootings in Properties Half Mile around School District Boundary

	Dependent Variable : ln(Price)								
	3 year window around the shooting				Persistence (Comparing to three years before episode)				
					¡1 year	1-2 years	2-3 years	3-4 years	4-5 years
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(Within Affected SD)	-0.144***	-0.142***	-0.136***	-0.176***	-0.172***	-0.133***	-0.107***	-0.100***	-0.118***
*1(After Shooting)	(0.039)	(0.038)	(0.036)	(0.042)	(0.044)	(0.042)	(0.031)	(0.028)	(0.030)
Observations	59,948	59,948	59,948	59,948	38,896	38,754	38,367	37,093	36,992
R-squared	0.670	0.711	0.718	0.679	0.697	0.698	0.698	0.700	0.691
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend			Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted Regression				Yes					

Note: The dependent variable is the log of the deflated price per property. Estimations include properties within 1 mile of the school district boundary. Columns (1) to (4) include observations three years before and after the shooting. Columns (5) to (9) compares observations three years before with observations 1,1-2, 2-3, 3-4, and 4-5 years after the shooting, respectively. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. The regression in column 4 weighs the observations by the inverse of the total number of properties sold within each episode. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.5: Effect of School Shootings using Propensity Score Matching Weights

	Dependent Variable: ln(Price/HPI)		
	(1)	(2)	(3)
1(Within Affected SD)*1(After Shooting)	-0.026*	-0.035**	-0.046***
	(0.015)	(0.015)	(0.014)
Constant	6.867***	2.009***	113.709***
	(0.041)	(0.081)	(42.381)
Observations	771,530	771,530	771,530
R-squared	0.640	0.731	0.738
Year*Month FE	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes
Property Characteristics		Yes	Yes
Episode Specific Trend			Yes

Note: The dependent variable is the log of the deflated price per property. Estimations include all properties within census tracts matched with a kernel propensity score algorithm using census 1990 characteristics at the census tract level. All observations are weighted by the kernel weights generated in the propensity score algorithms. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.6: Falsification Test using Placebo Episodes Prior to the Shooting

	Dependent Variable: ln(Price)					
	Full School District			Half Mile Around Boundary		
	2 years before	3 years before	4 years before	2 years before	3 years before	4 years before
	(1)	(2)	(3)	(4)	(5)	(6)
1(Within Affected SD)	0.006	-0.024	-0.007	0.025	-0.016	-0.016
	(0.027)	(0.033)	(0.036)	(0.035)	(0.038)	(0.043)
1(After X years before shooting)*1(Within SD)	-0.004	0.008	-0.010	-0.019	0.017	0.003
	(0.005)	(0.006)	(0.007)	(0.012)	(0.012)	(0.012)
Observations	223,916	198,927	158,096	19,263	17,988	15,825
R-squared	0.751	0.759	0.755	0.706	0.727	0.724
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the log of the deflated price per property. Columns (1) to (3) include the full school district. Columns (4) and (6) include properties half a mile from boundary. Columns (1) and (4) use a dummy that takes the value of one for a placebo test two prior to the shooting. Columns (2) and (5) use a dummy that takes the value of one for a placebo test three years prior to the shooting. Columns (3) and (6) use a dummy that takes the value of one for a placebo test four years prior to the shooting. All regressions include a one year window before and after the placebo shooting date. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.7: Balance Test of Housing Characteristics

	Full School District			Half Mile Around Boundary		
House Characteristic	Coefficient (1)	Std. Error (2)	P-value (3)	Coefficient (4)	Std. Error (5)	P-value (6)
1(Fireplace)	0.040	0.033	0.228	0.010	0.024	0.668
1(Brick)	-0.044	0.034	0.194	-0.020	0.023	0.391
1(Condo)	0.057	0.037	0.120	0.036	0.023	0.114
Number of Bedrooms	-0.002	0.061	0.971	-0.051	0.078	0.512
ln(land)	0.135	0.159	0.395	0.250	0.160	0.119
ln(Building)	-0.040	0.024	0.097	-0.015	0.032	0.650
Years Since Built	4.017	2.200	0.068	1.001	1.446	0.489

Note: Each row of the table presents a different housing characteristic. We estimate a linear regression using equation (1) and varying the outcome. Columns (1) to (3) include observations in the complete school district. Columns (4) to (6) include properties half mile around the boundary. Coefficients, standard errors, and p-values associated with the treatment parameter are presented.

Table 2.8: Average Effect on Crime

	Dependent Variable: Crime Rate								PCA index
	Murders	Rapes	Robberies	Assaults	Burglaries	Thefts	Auto Thefts	Arson	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1(Within Affected City)	-0.532	-5.155	-9.161	-104.704***	-85.046	261.614**	-133.462***	-18.292*	-24.523
*1(After Shooting)	(1.300)	(4.099)	(15.465)	(37.139)	(85.028)	(117.979)	(27.772)	(10.247)	(17.907)
Observations	1,514	1,490	1,491	1,476	1,474	1,474	1,475	1,450	1,474
R-squared	0.447	0.589	0.837	0.799	0.784	0.823	0.752	0.496	0.854
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Regressions are estimated at the city level using the affected city as treated and surrounding cities as controls. All dependent variables are rates per 100,000, except for the PCA index in column (9) which corresponds to a principal component index of the variables in columns (1) to (8). Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.9: Average Effect of School Shootings by House Size

	1 BR	2 BR	3 BR	4 BR	Less than 2 BR	More than 2 BR
	(1)	(2)	(3)	(4)	(5)	(6)
1(Within Affected SD)*1(After Shooting)	0.084***	-0.036***	-0.054***	-0.118***	-0.033**	-0.087***
	(0.032)	(0.013)	(0.011)	(0.015)	(0.013)	(0.012)
Observations	24,556	155,870	386,242	183,028	180,426	617,185
R-squared	0.805	0.792	0.732	0.765	0.787	0.753
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the log of the deflated price per property. All specifications include properties with different number of bedrooms. Column (5) and (6) pool together properties with less and more than two bedrooms, respectively. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.10: Average Effect on School Enrollment and Number of Teachers

	Dependent Variable: Ln(Enrollment)		Dependent Variable: Ln(Teachers)	
	(1)	(2)	(3)	(4)
1(Affected SD)*1(After Shooting)	-0.161***		-0.083*	
	(0.042)		(0.044)	
1(Affected School)*1(After Shooting)		-0.176**		-0.126*
		(0.085)		(0.069)
1(Surrounding School within SD)*1(After Shooting)		-0.075*		-0.052
		(0.042)		(0.035)
Constant	6.103***	5.537***	3.341***	2.561***
	(0.065)	(0.097)	(0.061)	(0.073)
Observations	26,115	26,115	25,255	25,255
R-squared	0.254	0.818	0.315	0.820
Year FE	Yes	Yes	Yes	Yes
Episode FE	Yes	Yes	Yes	Yes
School District FE	Yes		Yes	
School FE		Yes		Yes

Note: Regressions are done at the school level using the total number of students and teachers as dependent variable. Columns (1) and (3) present the pooled effect for all the schools within the affected school district. Columns (2) and (4) present results separately for the schools where the shooting occurred, and non-affected schools within the affected school district. Columns (1) and (3) include school district fixed effects. Columns (2) and (4) include school fixed effects. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 2.11: Average Effect of School Shootings Using as Treatment a Radius around Affected School

	Properties within 3 Miles			Properties within 5 Miles		
	0.3 Mile	0.5 Mile	1 Mile	0.3 Mile	0.5 Mile	1 Mile
	(1)	(2)	(3)	(4)	(5)	(6)
1(Within X Mi.)*1(After Shooting)	0.050	-0.002	-0.003	0.055	0.001	0.010
	(0.065)	(0.043)	(0.022)	(0.065)	(0.044)	(0.023)
Observations	49,229	49,229	49,229	109,924	109,924	109,924
R-squared	0.735	0.735	0.735	0.734	0.734	0.734
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the log of the deflated price per property. Columns (1) to (3) include only observations within a three mile radius from school shooting. Columns (4) to (6) include only observations within a five mile radius from school shooting. Columns (1) and (4) use properties within 0.3 miles around the school as treated units. Columns (2) and (6) use as treatment properties within 0.5 miles around the school. Columns (3) and (6) use as treatment properties within one mile around the school. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

CHAPTER 3

UNBUNDLING RETURNS TO SKILLS AND DEGREES: EVIDENCE FROM POST-SECONDARY EDUCATION IN COLOMBIA

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3.1 Introduction

There is a long-standing tradition in economics of estimating returns to human capital by using the years of education with which the individual enters the labor market. This paper shows that returns to education are not enough to capture all the returns to human capital. We jointly estimate the labor market returns to various types of postsecondary degrees as well as to several types of skills (academic, specific and non-cognitive). We find that for individuals who graduated from similar programs, a one standard deviation increase in the measure of skills yields an average wage premium of 2 percent. Similarly, for students of comparable skills, the market valuation of postsecondary degrees is positive and varies widely with the length and quality of the program.

We combine administrative records from a variety of sources to build a longitudinal data set that follows students from high school to college and into the labor market. Students in Colombia are evaluated after high school graduation by means of a mandatory standardized test (analogous to the SAT in the United States) on mathematics, language (Spanish), foreign language, and other subjects. We combine those data with information on students' enrollment and graduation in postsecondary programs. For all college graduates we observe a rich set of characteristics including their field of study and the length of the program. Students who are about to graduate from two- and four-year colleges undergo evaluations on mathematics, literacy and foreign language, as well as on specific tests related to their field of studies (akin

to the subject GRE). Furthermore, after graduation, Colombia's Ministry of Education surveys a subgroup of recent graduates in a follow-up survey that contains measures of non-cognitive skills. We combine all these data with records from the Colombian Social Security Administration that contain information on wages and employment characteristics.

We use an expanded Mincer earnings function to jointly estimate the labor market returns to different types of degrees and skills. We assuage concerns of an ability bias by controlling for a broad range of measures of both baseline and contemporaneous skills as well as quality measures of the different programs. A similar approach was taken recently, for instance, by Saiz and Zoido (2005), Hanushek, Schwerdt, Wiederhold, and Woessmann (2015), and Lindqvist and Vestman (2011).

We first estimate returns to finishing different types of higher education degrees. We distinguish among four types: i) public two-year programs, ii) private two-year programs, iii) public four-year programs, and iv) private four-year programs. We find that the annual returns to graduating from a four-year private school (instead of a two-year public program) are 6 percent. Following MacLeod, Riehl, Saavedra, and Urquiola (2017), we compute a measure of college reputation: for each individual the reputation of her college is calculated as the average of the high school exit exam among graduates in the same college. A one standard deviation in our measure of college reputation carries a wage premium of 3 percent. There is also a large heterogeneity in the returns by fields of study. Similarly to Kirkeboen, Leuven, and Mogstad (2016), we find that the returns to certain fields are as large as the returns to completing a four-year college.

We then estimate that, conditional on the degree, the program and its reputation, an increase in one standard deviation in skills yields a wage return of 2 percent. Returns to education are not enough to capture all the returns to human capital. Returns to skills are fairly homogeneous. The returns to purely academic skills, numeracy and literacy, are 2 and 2.5 percent, respectively. Returns to foreign language are 1.6 percent. Returns to specific skills are 2.2 percent and the wage premia to non-cognitive skills are 2.3 percent.[1]

[1]The estimates of returns to one year of education were obtained using the Integrated

Because tests scores are imperfect measures of a person's skills, returns could be biased towards zero. We provide instrumental variables estimates that use high school exams as instruments to alleviate the attenuation bias potentially caused by measurement error.

We explore different patterns of heterogeneity in the returns to skills. First, consistent with Farber and Gibbons (1996), we find that the returns to those skills that are less observable to the employer (numeracy and literacy), increase with tenure in the firm while returns to a more easily observable skill (such as foreign language) is constant. We assume that foreign language skills are easier to evaluate in an interview than skills in numeracy or literacy. Second, individuals realize what we term "returns to specialization." For example, people who work or study in more math-oriented fields and industries have a higher return to numeracy, whereas individuals who graduate from social sciences have a larger return to literacy, and individuals who work in tourism have a higher return to foreign language skills. Third, returns to numeracy are higher for people graduating in most fields of study, and working in most economic sectors. Fourth, we find that women have more returns to foreign language skills than men.

This study contributes to a large literature estimating returns to skills. Most of the previous literature provide estimates for developed countries. Our estimates for Colombia, a developing country with relatively low quality of education, fall within the range of estimates previously found in other studies. Returns to numeracy skills have been found to be on the order of 2 to 20 percent Levine and Zimmerman (1995); Murnane, Willett, and Levy (1995); Tyler (2004); De Coulon, Marcenaro-Gutierrez, and Vignoles (2008); Song, Orazem, and Wohlgemuth (2008); Joensen and Nielsen (2009); James (2013); Hanushek et al. (2015). Returns to foreign language have been found to be around 2.5 to 60 percent Bleakley and Chin (2004); Saiz and Zoido (2005); Christofides and Swidinsky (2010); Azam, Chin, and Prakash (2013); Guo and Sun (2014); Budría and Swedberg (2015); Di Paolo and Tansel (2015); Stöhr (2015). Returns to literacy have been found to be as high as 20 percent Ishikawa and Ryan (2002); De Coulon et al. (2008); Fasih, Patrinos,

Household Survey data (GEIH, according to the Spanish acronym) and are available upon request.

and Sakellariou (2013); Hanushek et al. (2015); Sanders (2016). Several papers find that non-cognitive skills are valued in the labor market as much as cognitive skills Bowles, Gintis, and Osborne (2001); Heckman, Stixrud, and Urzua (2006); Lindqvist and Vestman (2011); Dasgupta, Mani, Sharma, and Singhal (2017). Similarly to this literature, we find that field-specific and non-cognitive skills are also as valued in the labor market as academic skills are.[2] To the best of our knowledge we are the first to estimate all these measures jointly.

This paper also contributes to the literature analyzing the heterogeneity in the returns to different types of postsecondary degrees on two dimensions Hastings, Neilson, and Zimmerman (2013); Rodríguez, Urzúa, and Reyes (2015); Kirkeboen et al. (2016); Busso, Dinkelman, Martínez, and Romero (2017).[3] First, controlling for a wide range of measures of skills and for the quality of the college (as measured by its reputation), we find that the types of degrees as well as the field of study affects dramatically future income. Second, we estimate the returns to college reputation and find that this measure leads to sizable increases in wages.

The rest of the paper is organized as follows. Section 3.2 describes the Colombian education system and the data used. Section 3.3 presents the results for the returns to skills and postsecondary degrees. Section 3.4 analyzes the robustness and heterogeneity in the estimation of the returns to skills. Section 3.5 then presents the robustness and heterogeneity in the estimation to different types of postsecondary degrees. Section 3.6 concludes.

3.2 Background and Data

Education in Colombia is divided into primary school (first to fifth year), middle school (sixth to ninth year), high school (tenth and eleventh), and postsecondary education. Programs in postsecondary education are divided

[2]Appendix Table C.1 summarizes the results, methodologies and samples used in previous studies.

[3]Several studies review the evidence of returns to education in Latin America. See for instance, Psacharopoulos and Ng (1994), Behrman, Birdsall, and Székely (2007), Bassi, Busso, Urzúa, and Vargas (2012), and Lustig, Lopez-Calva, and Ortiz-Juarez (2012).

into college degrees (equivalent to a U.S. bachelor's degree, with a duration of four years) and two-year programs that offer vocational or technical instruction in different fields. We refer to all the institutions in postsecondary education as colleges, regardless of the duration of the program.

During high school, students take classes in mathematics, language (Spanish), and foreign languages as part of the school curriculum. Around 95 percent of schools choose to teach English as a foreign language; therefore, we refer to English and foreign language interchangeably. During postsecondary education, the level and intensity of instruction in these areas depend on the student's major, but most institutions require a minimum of foreign language knowledge as a graduation requirement.

After graduation, students' abilities and qualifications are extremely valuable in finding a job. Moreover, a mismatch of occupations and skills is more likely to happen among individuals with lower levels of abilities. A survey implemented in 2013 shows that 67 percent of the firms have employed at least one graduate with less than two years of experience and 73 percent, among these firms, consider that knowledge or specific abilities constitute the main selection criteria for hiring a graduate.

Measures of Skills. Since 1980, all high school seniors in Colombia have been evaluated before graduation through a mandatory, high-stakes exit exam (Saber 11). The exam is a requirement for graduation, and its results are sometimes used by colleges to make decisions.[4] The test resembles the SAT in the United States. It evaluates students in several subjects that, for convenience, we divide into two broad areas: i) general skills, which includes mathematics, reading and language (in Spanish), and foreign language; and ii) subject skills, which includes biology, philosophy, physics, chemistry, and social sciences. The test is administered in two sessions, each of four hours and 30 minutes. It evaluates general knowledge of each subject with roughly 40 questions. The mathematics test measures basic knowledge in algebra, calculus, geometry, probability, and statistics; students must interpret infor-

[4]Some colleges establish minimum scores that students must achieve to be considered for admission, while others apply their own admission exams and take Saber 11 only as an enrollment requirement.

mation, design solutions to problems, follow procedures, and justify steps in problem-solving. The reading test evaluates the student's ability to understand, interpret, and analyze critically written texts. The language exam tests the ability of the student to communicate in Spanish. The English test evaluates reading, grammar, and vocabulary.

Since 2003, college students who completed at least 70 percent of their coursework have been also required to take a college exit exam (Saber Pro). This exam is a graduation requirement and resembles the GRE, both the general and subject tests. The exam provides a strong signal for students readiness to make the transition from college into the labor market. It is very common for schools to provide incentives (e.g., public recognition to the best-performing students within each school) to motivate students to aim for high scores because the results are used for calculating school quality indexes that are routinely published by the Ministry of Education.

The college exit exam has two main components; one is general and the other is specific to the field of study from which the student is graduating.[5] The general section includes tests in numeracy, writing, reading, English, and citizenship abilities. Students have four hours and 40 minutes to complete the test, which includes a total of 161 questions (35 in numeracy, 35 in reading, 35 in citizenship abilities, 55 in English, and one in writing). The numeracy section evaluates basic mathematics knowledge needed to analyze and solve problems using quantitative methods and procedures. The reading section examines the capacity to read analytically by understanding the text and identifying different perspectives and value judgments. The writing section evaluates the ability to communicate ideas of a particular given topic. The English section focuses on testing the ability to communicate effectively in English. The specific section evaluates basic knowledge in the student's field of study. There is a total of 40 specific exams, and students have 90 minutes to answer between 30 and 60 questions included in each of them. The questions, designed by experts in each of the different areas, follow specific

[5] Although the general part of the exam started in 2003, major-specific exams were introduced by a staggered roll out. By the second semester of 2011, all students were taking the general tests, while the specific exams have been applied since then to a large share of students.

standards that assure comparability of the exams. Economics majors, for instance, are advised to take an Economics Analysis test with micro, macro, and econometrics questions.

In addition, since 2005, the Ministry of Education implemented a follow-up survey to recent college graduates that included measures of non-cognitive skills, with the purpose of tracking and evaluating professional performance. The survey was tested from 2005 to 2008 using a sub-sample of recent graduates that voluntarily completed an online form. Starting in 2011, and until 2014, the survey was implemented in person at the moment of graduation, and over the telephone to collect information from older cohorts.[6] Respondents were asked to answer a 10 to 15 minutes interview that demanded information about five modules: i) family background, ii) level of skills, iii) life plan, iv) job situation, and v) identity of the institution they graduated from.

The survey includes questions regarding the use of non-cognitive skills acquired in college and applied to the current job. Former students were asked to rank (from one to four) the use of skills related to communication, persuasion, innovation, and readiness to learn, among others. We use this information to compute a factor analysis index that measures non-cognitive skills among the sub-sample of surveyed graduates.[7] This measure quantifies those non-cognitive skills that have been proven to affect job market performance Heckman et al. (2006). More information and details about the different components and the index are given in Appendix C.2 on measures of skills.

Postsecondary Degrees. Similar to the experience in other Latin American countries, postsecondary education in Colombia has expanded dramatically in the last 10 years Busso, Cristia, Hincapie, Julian, and Ripani (2017). Postsecondary education in Colombia is delivered in the form of two-year and

[6]Respondents were contacted one, three, and five years after graduation by telephone.

[7]Appendix Table C.3 presents the loadings, corresponding to the factor with the largest eigenvalue, for each question used to compute the non-cognitive or non-cognitive measure used herein. We use a different set of non-cognitive questions to build a second measure, which is used later as an instrumental variable. Appendix Table C.4 presents the results from the factor model applied to compute this latter measure.

four-year programs. Two-year programs are mostly technically oriented and resemble vocational programs in the United States. Four-year programs, by contrast, are academically oriented and are equivalent to a bachelor's degree in the United States. The Colombian postsecondary education also includes public and private institutions; 30 percent of the postsecondary institutions in the country are public, while the other 70 percent are private. We classify academic programs into four categories: i) two-year public, ii) two-year private, iii) four-year public, and iv) four-year private. In addition, our data also allow us to identify individuals who have pursued graduate studies.

The degrees that students receive vary, not only in terms of the length of study required, but also in terms of quality, reputation, and tuition cost, among other dimensions.[8] In Colombia, as in the case of most countries in Latin America, each program (major) in each college makes its own admissions decisions. This decision is decentralized and puts a heavy weight on the high school exit exam. We compute a measure of reputation by program-college following MacLeod et al. (2017): the reputation measure is calculated as the average of the high school exit exam among students admitted in each program-college combination. The measure relies on the assumption that better regarded programs-colleges (i.e., with better reputations) can be more selective and will admit a set of applicants with higher high school exit exams.

Data and sample. The database used in this paper corresponds to a unique data that we assembled in collaboration with the Colombian Ministry of Education. We merged five different administrative datasets:

1. We use test score measures from the college exit exam from 2011 to 2015. The general components are compulsory for each student, but the specific tests are not. The decision on which field-specific tests to take is made by the college and not by the student. So, for instance, students graduating from economics may take the economics-specific test but also the business-specific test if their respective college indicates so. Some institutions do not require any specific exam, so a considerable number of students do not take field-specific tests. For this reason, we

[8] We lack information on the cost of tuition for each program in each university that would allow us to net out the cost of attending each program in calculating returns.

only observe field-specific test scores for a subgroup of students.

2. We merge data on the high school exit exam from 1996 to 2013. These data include information on the multiple test scores and, in addition, they include information on the student's municipality, school, and some individual characteristics, such as gender and age.

3. We merge records of all students enrolled in college between 1998 and 2016. These records include yearly information on graduation and enrollment of all students in postsecondary education for all universities and programs, and have detailed information on the program of study, as well as socioeconomic information about the student and her family.

4. We merge longitudinal yearly earnings records for workers who graduated from college. These data include individual earnings from 2011 to 2016 of all the workers who finished any college program after 2001, and who contributed to the Social Security System. Four-digit industry codes, the municipality where the contribution was paid, and establishment identifiers are also included.[9]

5. We merge data on the graduate follow-up survey implemented by the Ministry of Education. We linked the 2013 and 2014 cross-sections to obtain a random sample of graduates from cohorts 2011 to 2013. These data correspond to individuals surveyed in 2013 and 2014, who graduated in or before 2013. Therefore, the sample of individuals we observe with non-cognitive measures represents a subgroup of people with more tenure, compared to the average individual we observed in the full sample.[10]

The final sample includes 363,464 individuals who took the college exit exam between 2011 and 2015, graduated after 2011, and worked formally between 2012 and 2016. The high school and the college exit exams are not fully comparable across their different editions. Hence, all test scores are standardized to have mean zero and standard deviation one within each test's edition. We use three different samples: i) test-takers from 2011 to 2015 who

[9]Workers who do not contribute to health or pensions are not included in the data.

[10]In Appendix Table C.5 we show that individuals in this survey are older and have more tenure.

are working formally ($N = 363,330$), ii) test-takers with major-specific scores available who are working formally ($N = 155,939$), and iii) test-takers who were included in the graduate follow-up survey, for whom we have measures of non-cognitive ability, and who are working formally ($N = 2,401$).[11]

Descriptive Statistics Individuals included in the main sample are on average 27 years old. Sixty percent are female, 73 percent live in urban areas, and 45 percent belong to low-income households.[12] The majority (82 percent) are graduates of four-year programs, and the main fields of study are science, technology, engineering, and mathematics, hereafter STEM, (26 percent), business and economics (31 percent), social sciences and humanities (16 percent), and health and education (22 percent). Six out of 10 work for the service sector, and 28 percent are employed by large firms.[13]

Appendix Table C.6 presents a correlation matrix of all the different test score measures in our sample. We present a panel for each sample. Several highlights are in order. First, all correlations are positive and large which, as suggested by Rindermann (2007), indicates that these measures are capturing in part a broader trait such as cognitive ability. There is even a positive correlation between cognitive and non-cognitive (captured by non-cognitive) skills. Second, the largest correlations are found within exams (shown in bold type). This is suggestive of skill complementarities. Third, the correlations within subjects and across time (shown in non-bold type) are large and positive which suggests the existence of self-productivity Cunha and Heckman (2007); Cunha, Heckman, and Lochner (2006). Fourth, the magnitudes of the correlations are very stable across the three panels. Fifth, the largest

[11] Appendix C.3 describes in detail the steps followed to build the data and each sample. Initially, it is worth noting that the larger sample of students contains complete information on wages, socioeconomic background, and high school. The same happens to the sample of students with major-specific scores, which is then a perfect subset of the first sample. However, some observations (134 students) within the survey sample do not belong to the larger sample because it was not possible to identify the high school from which they graduated. We decided to keep these individuals to maintain as many observations as we can in the smaller sample.

[12] We identify low-income households using the Colombian housing stratification system. For purposes of targeting social assistance, all houses in the country are assigned to an economic stratum from one to six, depending on the neighborhood and house. We defined low-income households as those living in the first two strata.

[13] In Appendix Table C.5 we present a set of statistics that describe our sample.

correlation (0.94) is between the average high school exit exam result and what we call the subject exam result, which corresponds to the average of the high school exit exam excluding the numeracy, language, and English components. This high correlation allows us to replace the average result in the high school exit exam with the "subject" exam in our empirical strategy.

3.3 Returns to Degrees and Skills

The following equation describes our expanded Mincer earnings equation which we use to estimate jointly returns to degrees and skills:

$$\log(W_{ifct}) = P_{ifct}\beta_1 + T_{ifc\tau}\beta_2 + \beta_0\theta_i + \alpha_1 CQ_{ct} + X_i\gamma + \mu_f + \mu_t + \mu_\tau + \mu_s + \varepsilon_{ipct}. \tag{3.1}$$

W_{ifct} is the wage of individual i who graduated from field of study f at college c in year t. P_{ifct} is a vector that includes three indicator variables that take the value of one if the individual has a postsecondary degree corresponding to: i) a two-year private program, ii) a four-year public program, or iii) a four-year private program. The omitted category is two-year public programs. $T_{ipc\tau}$ is a vector of college exit tests scores that includes measures of numeracy, literacy, and English. Individual i's test scores correspond to results observed in edition τ of the college exit exam. Depending on the sample, vector $T_{ipc\tau}$ also includes measures of major-specific or non-cognitive skills. θ_i is a measure of the initial level of ability. We include the measure for college reputation (CQ_{ct}). X_i is a vector of individual characteristics that include gender, age, age-squared, mother's education, an indicator variable for graduate studies, and socioeconomic stratum. μ_f, μ_t, μ_τ, and μ_s are field of study, cohort, test edition, and high school fixed effects.

To reduce ability bias concerns, we include a measure of initial ability (θ_i) which is measured by the subject skills components of the high school exit exam.[14] This measure is highly correlated (0.94) with the average of all the

[14]We reserve the general high school skills components of the exam (mathematics, language, and foreign language) to be used as instruments later in the analysis. Results, however, are similar when including these general high school skills in the measure of initial ability.

tests in the high school exit exam.[15]

Equation 3.1 resembles an estimation of a *value-added* measure in which the effect on students' knowledge is conditioned on initial knowledge.[16] In our case, however, we estimate the economic return to skills enhanced during college and to types of degrees, conditioning on the abilities that each student had at the time of starting college. An advantage of that specification is that it eliminates observed and unobserved confounding elements about the history of parental and school inputs, and, therefore, reduces remarkably the likelihood of suffering of omitted variable bias Rivkin, Hanushek, and Kain (2005). The inclusion of θ_i in equation (3.1) together with the full set of skills measures and fixed effects attempts to address possible concerns regarding ability bias of $\hat{\beta}$. We interpret $\hat{\beta}_1$ as the return to skills acquired or enhanced during college –conditional on skills prior to college. $\hat{\beta}_2$ should be interpreted as the economic return to type of degrees conditioning on the skills a student has before and after graduation. Note that estimating these jointly isolates the returns of attending a specific program independently of college quality (which is controlled by the measures of skills and the measure of college reputation).

If our set of control variables is not rich enough to ensure our estimators are unbiased, our resulting estimates may still be informative about the relative returns if the bias is the same for all coefficients in the vectors $\hat{\beta}_1$ and $\hat{\beta}_2$. In other words, similar to methodology underpinning some of the previous related literature Hanushek et al. (2015); Lindqvist and Vestman (2011), the difference between coefficients will eliminate the bias, and we will still be able to correctly rank the returns to skills and types of degrees.[17]

Table 3.1 presents ordinary least squares (OLS) estimates of equation (3.1).

[15] See Appendix Table C.6 for more information about the correlation among the measures of skills.

[16] A considerable amount of papers have used value-added measures to estimate the return to better teachers. A discussion of this model is presented in Hanushek and Rivkin (2010).

[17] Assume a true model of the form: $W = \alpha + \beta_1 T_1 + \beta_2 T_2 + \theta + u$. If the estimated model is $W = \alpha + \beta_1 T_1 + \beta_2 T_2 + \varepsilon$, where $\varepsilon = \theta + u$, then the probability limits of the difference between the OLS estimators will be: $\text{Plim}(\hat{\beta}_1 - \hat{\beta}_2) = (\beta_1 - \beta_2) + (\hat{\beta}_{\theta T_1} - \hat{\beta}_{\theta T_2})$, where $\hat{\beta}_{\theta T_1}$ is the coefficient of a regression of θ and T_1. Our assumption states that $\hat{\beta}_{\theta T_1} \sim \hat{\beta}_{\theta T_2}$.

Given that we observe wages for multiple periods, we use the current observed wage as dependent variable.[18] Column (1) presents the estimation in the full sample. Columns (2) and (3) restrict the sample to those individuals with field-specific and non-cognitive skills, respectively.[19] Columns (4) to (6) follow the same format, but additionally include the measure of college reputation; these are our preferred estimates.

Returns to different types of degrees are fairly heterogeneous across programs. Using the point estimates in column (5) and adjusting by the years it takes to graduate from each program, we find that, relative to two-year public degrees, an additional year of education increases wages by 4.2 percent for two-year private, 3.9 percent for four-year public, and 5.9 percent for four-year private degrees. The reputation of the program also carries a wage premium of three percent. However, we observe that education does not capture the full extent of human capital. The returns to skills conditional on education are meaningful and comparable to the returns to education. Among the returns to skills, numeracy seems to have the largest return (up to 3.7 percent), except in the survey sample. Literacy, foreign language, non-cognitive, and field-specific skills have a similar return of, roughly, two percent. These results are fairly stable across columns, which vary both the specification and the sample used in the estimation.

3.4 Robustness and Measurement Error

Sensitivity of the Mincer Earnings Equation We analyze the sensitivity of the point estimates of the expanded Mincer equation by estimating alternative specifications, and present the results in Table 3.2. Column (1) presents estimates of the returns to degrees unconditionally of the returns to skills and field of study, whereas column (2) includes back field of study fixed effects aiming to control for potential pre-graduation sorting into fields. Columns

[18] In Appendix Table C.7 we present results with alternative measures such as the average wage observed for each individual, the first observed wage after graduation, and total annual earnings. The results are essentially the same.

[19] Columns (3) and (6) use the information of the follow-up survey of college graduates. For more details about this dataset visit the webpage `https://ole.mineducacion.gov.co/portal/A-quienes-aportamos/Investigadores/`

Table 3.1: Returns to Skills and Types of Degrees

	Dependent Variable: log(Current Wage)					
	(1)	(2)	(3)	(4)	(5)	(6)
Two-Years Private	0.086	0.088	0.109	0.080	0.084	0.128
	[0.018]	[0.015]	[0.039]	[0.014]	[0.012]	[0.049]
Four-Years Public	0.210	0.194	0.236	0.174	0.156	0.197
	[0.016]	[0.020]	[0.044]	[0.011]	[0.013]	[0.042]
Four-Years Private	0.268	0.257	0.229	0.249	0.236	0.212
	[0.025]	[0.026]	[0.037]	[0.020]	[0.021]	[0.036]
Graduate Studies	0.158	0.183	0.048	0.156	0.181	0.045
	[0.017]	[0.019]	[0.063]	[0.016]	[0.019]	[0.060]
College Reputation				0.032	0.031	0.061
				[0.004]	[0.005]	[0.023]
Literacy	0.028	0.022	0.018	0.027	0.020	0.017
	[0.001]	[0.002]	[0.007]	[0.001]	[0.002]	[0.007]
Numeracy	0.037	0.028	0.024	0.035	0.025	0.019
	[0.002]	[0.003]	[0.009]	[0.002]	[0.003]	[0.010]
Foreign language	0.020	0.020	0.036	0.016	0.016	0.026
	[0.006]	[0.006]	[0.007]	[0.006]	[0.006]	[0.006]
Field Specific		0.024			0.022	
		[0.004]			[0.004]	
Non-Congnitive			0.023			0.023
			[0.005]			[0.006]
Sample:	Full	Specific	Survey	Full	Specific	Survey
Observations	363,330	155,939	2,401	363,330	155,939	2,401
R-squared	0.192	0.218	0.239	0.194	0.219	0.247
Controls:						
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Ability	Yes	Yes	Yes	Yes	Yes	Yes
Field of study	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Ordinary least squares estimations. The dependent variable is the natural log of the last observed wage from 2011 to 2016 for each individual. Literacy, numeracy, and foreign language scores are computed from the college exit exam and standardized within each sample using the average and standard deviation of the corresponding test edition. The field specific scores are standardized with respect to the average not only of the test edition but also considering the groups of related majors. The omitted category of postsecondary degrees is 2-3 Years Public, which identifies individuals who graduated from technical or vocational tertiary programs taught by public universities. The college reputation variable is computed following MacLeod et al. (2017) and then standardized to have mean zero and standard deviation one within each sample. Individual control variables include gender, age, age squared, socioeconomic stratum, mother's education level, a dummy for graduate studies, a proxy of initial abilities, high school fixed effects, cohort fixed effects, and test edition fixed effects. Initial abilities are proxied using the standardized average of biology, physics, chemistry, philosophy, and social sciences scores computed from the high school exit exam. Field of Study refers to fixed effects of related majors: accounting, agricultural sciences, architecture, arts, business and related fields, economics, education, engineering, health, humanities, journalism, medicine, natural and exact sciences, nursing, law, psychology, social sciences, and sports. Standard errors are clustered by municipality and in brackets.

(3) to (5) present the returns to skills across different samples, but unconditionally of the types of degrees and field of study. In columns (8) to (10) we include field of study fixed effects, again aiming to control for potential pre-graduation sorting. Comparing these estimates with those in Table 3.1 give informative evidence on the degree to which the returns to degrees com-

plement the returns to skills.

Table 3.2: Returns to Different Types of Postsecondary Degrees

	Dependent Variable: log(Current Wage)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Two-Years Private	0.046	0.078						
	[0.012]	[0.014]						
Four-Years Public	0.127	0.175						
	[0.017]	[0.011]						
Four-Years Private	0.235	0.243						
	[0.011]	[0.020]						
Graduate Studies	0.163	0.160						
	[0.015]	[0.017]						
College Reputation	0.042	0.042						
	[0.005]	[0.006]						
Literacy			0.023	0.022	0.012	0.027	0.022	0.024
			[0.002]	[0.002]	[0.007]	[0.001]	[0.002]	[0.007]
Numeracy			0.055	0.058	0.044	0.034	0.029	0.025
			[0.005]	[0.004]	[0.012]	[0.002]	[0.003]	[0.010]
Foreign language			0.017	0.017	0.024	0.028	0.024	0.042
			[0.003]	[0.003]	[0.008]	[0.006]	[0.006]	[0.007]
Field Specific				-0.003			0.021	
				[0.003]			[0.004]	
Non-Congnitive					0.030			0.023
					[0.006]			[0.005]
Sample:	Full	Full	Full	Specific	Survey	Full	Specific	Survey
Observations	363,330	363,330	363,330	155,939	2,401	363,330	155,939	2,401
R-squared	0.130	0.188	0.118	0.143	0.137	0.177	0.207	0.224
Controls:								
Individual	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Ability	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Field of Study		Yes				Yes	Yes	Yes

Notes. Ordinary least squares estimations. The dependent variable is the log of the last observed wage for each individual from 2011 to 2016. 2-3 Years Private is a dummy variable for individuals graduated from technical or vocational tertiary programs taught by private universities, 4 > Years Public includes graduates from bachelor programs taught by public universities, and 4 > Years Private includes graduates from bachelor programs taught by private universities. The omitted category of postsecondary degrees is 2-3 Years Public, which identifies individuals graduated from technical or vocational tertiary programs taught by public universities. Columns (4) and (8) include measure of skills –literacy, numeracy and foreign language– computed from the college exit exam. Individual and field of study controls as described in the notes of Table 3.1. Standard errors clustered at the municipality level and in brackets.

The results in Column (1) are quite different from the returns to types of degrees in column (4) of Table 3.1. This difference can be attributed to the exclusion of the field of study fixed effects or the measures of skills. Including back the fixed effects in column (4), we recover point estimates very close to

those in Table 3.1. This implies that including the measures of skills does not affect the coefficients on the types of degrees.

Similar results are obtained in columns (3) to (5) when we estimate returns to skills unconditionally from the returns to degrees and fields of study. We do see that excluding these controls slightly affects the magnitudes, especially in the return to numeracy skills. However, controlling for potential pre-graduate sorting into fields recovers point estimates similar to those obtained in Table 3.1. These results as a whole imply that the measures of skills and the measures of types of degrees capture different information when estimating economic returns. In fact, it poses strong evidence suggesting that the returns to human capital cannot be exclusively attributed either to returns to education or returns to skills, but rather that a combination of both is important.

Skills Collinearity. The results in Table 3.1 include simultaneously multiple test score measures that are highly correlated. As previously discussed in Section 3.2, there is a high degree of skill complementarity: the level of skills of any given individual in a given subject is positively correlated with the skills of that individual in other subjects. This could be, for instance, because having better mathematics skills can allow a student to acquire science skills faster, or because having higher literacy skills, by better understanding the underlying structure of language, can affect the capacity of an individual to learn a foreign language. The simultaneous estimation of equation (3.1) will therefore yield a lower bound of the effect of skills on wages because the coefficient of one test score will bias the coefficient of the other test score. The separate inclusion of the test score measures, however, will estimate the upper bound of that effect (see Lindqvist and Vestman (2011)).

Table 3.3 shows OLS estimates of equation (3.1) including every measure of skills simultaneously and separately (to save space, these point estimates are presented stacked).[20] In columns (1) to (3) we present the same results as in Table 3.1, for comparison. Columns (4) to (6) show the point estimates of three and four separate regressions that use equation (3.1), but include

[20]Appendix Table C.9 presents the same results in Table 3.3, but including the point estimates for each type of degrees.

each measure of skills separately.

Using the full sample of workers, we again find that numeracy has the highest return (up to 4.4 percent), whereas literacy and foreign language have returns of 3.8 and 3 percent, respectively. Once again, we see that the return to numeracy decreases the most when considering the specific sample, although it remains the largest. Specific skills again show a point estimate among the highest (3.8 percent) when estimated in the same restricted sample. The coefficients on numeracy and literacy decrease, again suggesting that individuals who score high in generic skills also score high in specific skills. Regarding the estimations using the survey sample, we observe that the estimated returns to cognitive skills increase (compared to column (3)), while the return to non-cognitive skills remains stable (2.3 percent).

Table 3.3: Extensions: Estimates' Ranges and Measurement Error

	Dependent Variable: log(Current Wage)										
	OLS						IV: Measurement Error Correction				
	Simultaneous			Stacked			Simultaneous		Stacked		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Literacy	0.027	0.020	0.017	0.038	0.036	0.026	0.075	-0.004	0.105	0.106	0.112
	[0.001]	[0.002]	[0.007]	[0.002]	[0.002]	[0.007]	[0.011]	[0.057]	[0.012]	[0.013]	[0.046]
Numeracy	0.035	0.025	0.019	0.044	0.041	0.027	0.045	0.154	0.087	0.083	0.140
	[0.002]	[0.003]	[0.010]	[0.003]	[0.003]	[0.010]	[0.006]	[0.046]	[0.008]	[0.010]	[0.028]
Foreign Language	0.016	0.016	0.026	0.030	0.031	0.035	-0.002	0.002	0.040	0.041	0.053
	[0.006]	[0.006]	[0.006]	[0.006]	[0.006]	[0.007]	[0.004]	[0.027]	[0.006]	[0.005]	[0.022]
Field Specific		0.022			0.038					0.123	
		[0.004]			[0.003]					[0.017]	
Socioemotional			0.023			0.023		0.026			0.022
			[0.006]			[0.006]		[0.007]			[0.006]
Sample:	Full	Specific	Survey	Full	Specific	Survey	Full	Survey	Full	Specific	Survey
Observations	363,330	155,939	2,401	363,330	155,939	2,401	363,330	2,401	363,330	155,939	2,401
Controls:											
Individual & Field	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Types of Degrees	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable and measures of skills are as described in the notes of Table 3.1. Estimations within the full sample use the same specification as column (4) of Table 3.1. Likewise, estimations within the specific and survey samples use, respectively, the specifications of columns (3) and (5) of Table 3.1. Stacked results refer to separate regressions for each ability measure. In columns (7) to (9), cognitive skills were instrumented using each time numeracy, literacy and foreign language standardized scores computed from the high school exit exam. Individual and types of degrees area as described in the notes of Table 3.1. Standard errors clustered by municipality and in brackets.

Measurement Error Correction. Test scores are just a proxy variable of the true level of an individual's skills. In other words, even though the exit exams are official and well established it is possible that test scores measure skills with error, biasing the estimated coefficients towards zero (attenuation bias). To account for this possibility, we assume a classical measurement

error setting and instrument the college exit test scores with the high school exit test scores. We additionally control for initial ability. The identification of this model relies on the assumption that the level of a given skill in high school only affects wages through the skills level in college.

Columns (7) to (11) of Table 3.3 show the IV estimates. In column (7) we include all three measures of generic skills simultaneously and instrument them with the mathematics, language, and English result in the high school exit exam. The point estimates in the IV estimation increase considerably for literacy and numeracy, and they are similar in magnitude to those found by Lindqvist and Vestman (2011). The coefficient on literacy increases more than the others, and it becomes larger than the coefficient on numeracy. Notice as well that the returns to foreign language decrease and are negative, possibly because it estimates a lower bound when estimated jointly with the other measures of skills. In column (8) we restrict the sample to individuals from the follow-up survey of graduates. This column includes our three generic cognitive abilities and the measure of non-cognitive skills, which we instrument with the mentioned scores from the high school exit exam and a score computed from another set of non-cognitive related questions contained in the survey sample.[21] We observe that the returns to non-cognitive skills are similar in magnitude to those found using OLS, while there is a remarkable change for the returns to cognitive skills. Observe that literacy and foreign language returns are close to zero, and the returns to numeracy are up to 15 percent.

Column (9) includes each test score measure estimated separately but over-identified using the scores in mathematics, language, and English in the high school exam jointly as instruments.[22] We find that all coefficients increase and are much larger than those estimated separately by OLS (in column (4) of Table 3.3) or estimated jointly (in column (7)). The point estimate on foreign language (4 percent) is now positive. Column (10) deals with the measurement error for workers who took the major specific test. Each point

[21] Appendix Table C.4 shows the results from the factor model used to compute the instrument for the measure of non-cognitive skills.

[22] We also performed estimations using one instrument at a time. The results do not change.

estimate corresponds to a separate regression. The results show that, as in the OLS case, the returns to literacy and foreign language do not change largely in this sample (compared to columns (9)), whereas the returns to numeracy decrease. All the coefficients increase compared to column (5), but the increase in specific skills and literacy is remarkable changing from 3.8 to 12.3 percent and from 3.6 to 10.6 percent, respectively.

The last column in Table 3.3 shows the results, adjusting for measurement error, for workers contained in the follow-up survey. Estimations are obtained from separate regressions for each measure of skills. We over-identified each measure using mathematics, language, and English from the high school exam, as well as a non-cognitive score computed from the questions contained in the survey of graduates. Literacy and foreign language returns increase (in comparison with column (8)) and are up to 11.2 percent and 5.3 percent, respectively. Returns to numeracy are around 14 percent, and returns to non-cognitive skills stay close to 2 percent.[23]

Taken together, these estimates indicate that numeracy skills have a return that ranges between 1.7 percent and 14 percent, literacy a return between 1.4 percent and 11.2 percent, foreign language skills between 1 percent and 12.6 percent, specific skills from 2.1 percent to 12.3 percent, and non-cognitive skills return around 1.2 percent and 2.6 percent. These ranges include upper (regressions with each test score alone) and lower bounds (regressions with all the measures simultaneously) accounting and not accounting for measurement error.

3.5 Beyond Average Returns

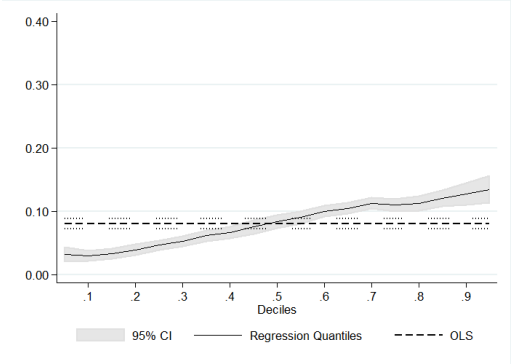
We now focus on the heterogeneity of returns to degrees and skills. First, we explore heterogeneity across the wage distribution, then we present returns by tenure and, finally, we estimate the returns among fields of study, economic sectors, gender, and firm size.

[23] We additionally estimate a latent variable model that implements alternative estimation methods to deal with attenuation bias. The results are presented in Appendix C.4.

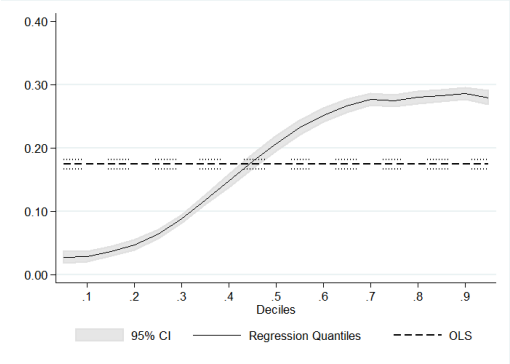
Returns Across Wage Distribution. We estimate conditional regression quantiles for each type of postsecondary degree and present the results in Figure 3.1. OLS point estimates are plotted to allow comparisons. We use the specification in column (4) of Table 3.1. Figure 3.1a shows the results for two-year private, figure 3.1b for four-year public, and 3.1c for four-year private.

Figure 3.1: Returns to Types of Degrees at Conditional Quantiles of the Wage Distribution

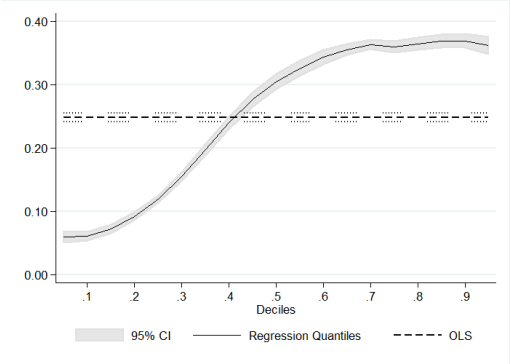
(a) Two-Year Private



(b) Four-Year Public



(c) Four-Year Private



Notes. The solid lines represent the regression quantiles using Koenker and Bassett (1978) estimator. The dashed lines correspond to an OLS specification. Quantile and OLS estimations for 2-3 years private, 4-year public and 4-year private programs used specification (4) in Table 3.1. Figures 3.1a, 3.1b, and 3.1c used the full sample of students who graduated from 2011 to 2016. OLS standard errors are clustered at the municipality level. Regression quantiles standard errors computed using 20 bootstrap replications. Confidence intervals of 95% are presented for all estimates.

The results suggest an important degree of heterogeneity in the returns to postsecondary degrees. The three graphs show strictly increasing returns

with wages, indicating that postsecondary degrees matter more among higher wage quantiles. Note that the lowest point estimate among four-year private degrees is around 0.8, which is comparable to the point estimates on returns to skills. It means that taking individuals from a two-year public program and placing them in a four-year private, keeping abilities constant, will increase the wage by at least 8 percent even if they get a job that pays in the first decile of the wage distribution. A similar increase in wage would be achieved if those same persons stayed in the two-year public program but increased their numeracy skills by one standard deviation.

In Figure 3.2 we present the results of conditional regression quantiles for each measure of skills.[24] We use again the main specification as in column (4) of Table 3.1 and contrast the quantile point estimates with the OLS.[25] Figures 3.2a, 3.2b, and 3.2c use the full sample of students who took the college exit exam between 2011 and 2016. Figure 3.2d uses the reduced sample of students with information about field-specific tests scores. Figure 3.2e shows the results within the sample of surveyed workers with measures of non-cognitive skills.

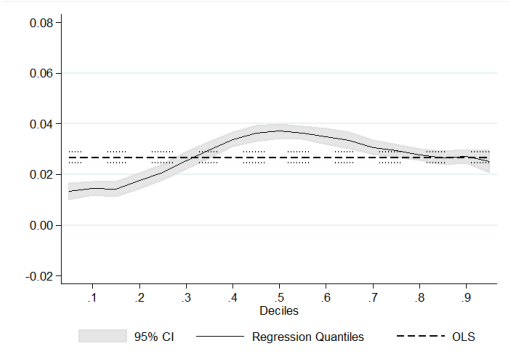
The returns to all the measures of skills are mostly positive along all deciles. They are lowest among people in the lowest percentiles and increase monotonically until roughly the 40th percentile. The returns to foreign language skills, however, are strictly increasing in the whole distribution, ranging from zero to 0.2. The returns to numeracy remain stagnant beyond the 40th percentile, although these are the highest returns we observe. The returns to literacy and major-specific skills decrease slightly beyond the 50th percentile, but that decrease is not statistically significant. The returns to non-cognitive skills are not precise because of the small sample size. However, they increase until the 20th percentile, remain steady until the 80th percentile and increase again for the top percentiles. One implication of these results is that inequality in the distribution of skills can explain part of the observed wage inequality in Colombia.

[24] We also estimate unconditional regression quantiles as in Firpo, Fortin, and Lemieux (2009). Results are very similar and available upon request.

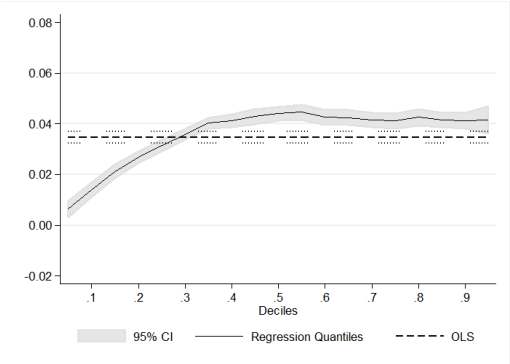
[25] For the case of specific and non-cognitive skills we use specifications (5) and (6) of Table 3.1, estimating the models in the corresponding restricted samples.

Figure 3.2: Returns to Skills at Conditional Quantiles of the Wage Distribution

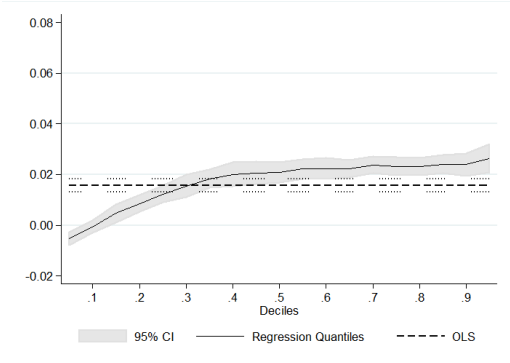
(a) Literacy



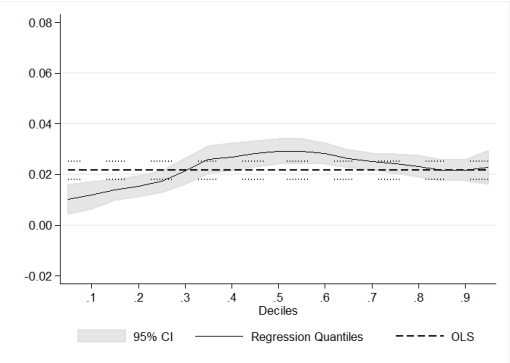
(b) Numeracy



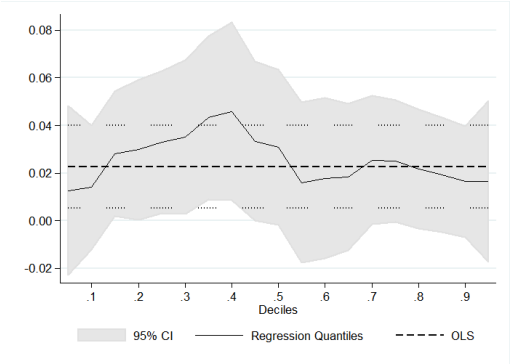
(c) Foreign Language



(d) Field Specific



(e) Non-Cognitive



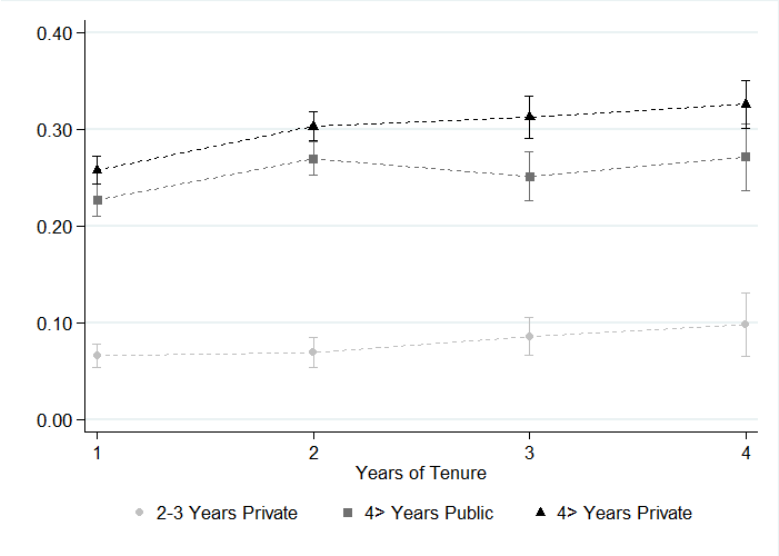
Notes. The solid lines represent the regression quantiles using Koenker and Bassett (1978) estimator. The dashed lines correspond to an OLS specification. Quantile and OLS estimations for numeracy, literacy and foreign language used specification (6) in Table 3.1. Figures 3.2a, 3.2b, and 3.2c used the full sample of students who graduated from 2011 to 2016. Figures 3.2d and fig:qregses used, respectively, the sample of students with major-specific test scores available and the sample of individuals in graduate follow-up survey. OLS standard errors are clustered at the municipality level. Regression quantiles standard errors computed using 20 bootstrap replications. Confidence intervals of 95% are presented for all estimates.

Returns Across Tenure. In some economics models, wages increase over time as a function of the capability of the employer to observe workers' abilities (Farber & Gibbons, 1996). Skills that are easily observable by the employer (such as foreign language) are predicted to have higher returns in the first year on the job, but then stay constant. The returns to less observable skills (such as numeracy or literacy), on the contrary, are expected to increase as the employer updates her beliefs about the worker.[26] Notice that the worker's type of postsecondary degree is very observable by employers and thus, following the previous argument, returns to degrees should not vary much with tenure.

We test that hypotheses between tenure and returns by exploiting the longitudinal feature of the data. Estimations use the same specification as column (4) in Table 3.1, but among workers with one, two, three, and four years of tenure. Figures 3.3 and 3.4 present the results. In Figure 3.3 we plot the returns to postsecondary degrees across years of tenure. Slight increases in the returns are observed, but the confidence intervals suggest that these slight increases are not in fact significant. We interpret this as evidence in favor of Farber and Gibbons (1996), suggesting that the returns to education degrees should not increase with tenure if they are observable for employers.

[26] We assume that foreign language skills are more observable in a face-to-face interview (which is very common in a hiring process), whereas literacy and numeracy skills are more difficult to observe.

Figure 3.3: Returns to Degrees by Years of Tenure



Notes. The plotted circles, squares and triangles represent the point estimates of 2-3 year private programs, 4-year public programs and 4-year private programs, respectively. Separate regressions were run among workers with different years of tenure to the estimate simultaneously the returns to different types of degrees. Estimations used the same specification as in column (4) of Table 3.1. The dependent variable is the log(Wage) for each year of tenure. Confidence levels of 95% are presented for all point estimates.

Returns across tenure for each measure of skills are presented in Figure 3.4. We additionally include the results among students with field-specific scores for comparability. Non-cognitive skills are not included because of the small sample size of workers who have available scores for this measure. The returns using the complete sample of students are plotted with circles, and the corresponding confidence intervals are represented with a dark area. These results show that numeracy and literacy are increasing with tenure. However, the returns to numeracy increase much more steeply, growing from 1.5 percent to 3.6 percent. The returns to foreign language, on the contrary, do not increase as much; they remain relatively steady across years of tenure. We interpret this as evidence in favor of the Farber and Gibbons (1996) model.

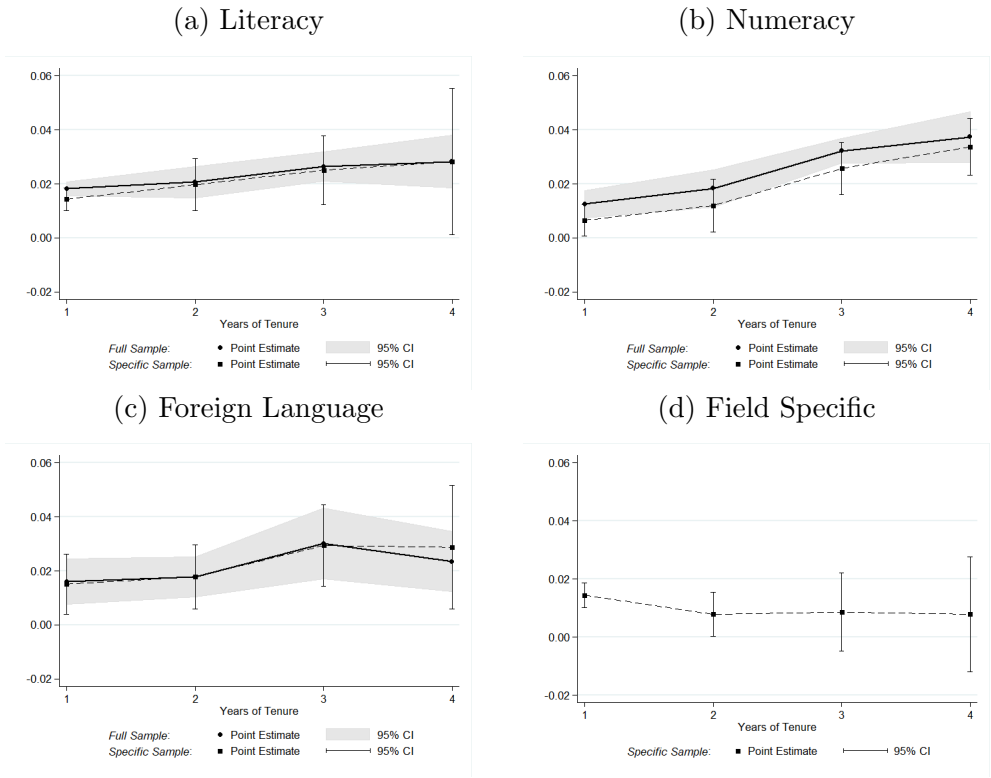
We find similar patterns if we restrict the sample to students with available field-specific tests scores. Estimations within this sample use specification (5) in Table 3.1. Returns are plotted using squares, with whiskers representing confidence intervals. Using this sample, we are able to estimate returns to

major-specific skills and observe that these do not change across years of tenure. We explain this by considering that during hiring processes employers may apply tests that allow them to know their applicants' level of specific abilities. On the other hand, applicants could also reveal their specific skills during interviews in order to increase their probability of being hired. Thus, specific skills could be more observable for employers than other skills, at least in the labor market for college graduates.

Returns to Specialization: Field of Study and Economic Sector. Returns to degrees and skills can in part reflect specialization. For instance, individuals with better mathematics skills can choose careers that place greater emphasis on those abilities. Then, they can find jobs that value their skills, receiving higher payments for greater levels. We explore whether the magnitude of returns varies with the person's specialization (either in field of study or in economic sector). Estimations of returns across different areas of study use specifications (4) and (5) in Table 3.1 without including field of study fixed effects. We exclude the estimation of returns to non-cognitive skills because of low sample sizes.

In Table 3.4 we present the returns to degrees and skills across groups of similar postsecondary programs or study areas. We estimate the returns for graduates of STEM, business and economics, social sciences and humanities, and health and education.

Figure 3.4: Returns to Skills by Years of Tenure



Notes. The plotted circles and squares represent the point estimates for each measure of skills using, respectively, specifications (6) and (8) of 3.1. Separate regressions were run among workers with different years of tenure to estimate simultaneously numeracy, literacy, foreign language and major-specific skills. The dependent variable is the log(Wage) for each year of tenure. Confidence levels of 95% are presented for all point estimates.

We find three main results. First, the positive gradient in the returns to degrees (by which four-year degrees pay more than two-year degrees and graduates from private school earn more than those of public schools) is observed in all fields of study. However, it is more pronounced in STEM, business and social sciences. Second, literacy and numeracy skills matter in all fields. In that sense, these basic skills are always remunerated, probably because they are highly transferable and ubiquitous across fields. Third, there is some evidence of positive returns to specialization in Table 3.4. Specific skills tend to be homogeneous across fields. Of course, different specific skills would be tested for each program. To shed some light on what those specific skills capture we estimate the model in the full sample without including specific skills (odd columns). Numeracy skills have the highest return in STEM, business and economics, but not in social sciences and humanities;

for these individuals, the largest return is to literacy skills. Workers with STEM degrees have the lowest return to literacy skills (2.3 percent). Graduates from health and education degrees have similar returns across all skills. Furthermore, we see that including field-specific skills decreases remarkably the returns to numeracy for most fields, suggesting a large complementarity between specific and numeracy skills.

Table 3.4: Heterogeneous Effects: Study Area

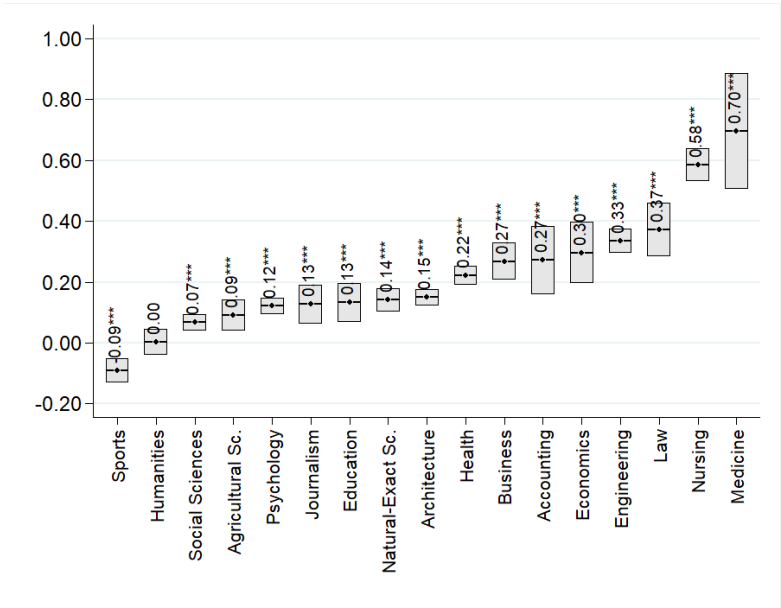
	Dependent Variable: log(Current Wage)									
	Study Area:									
	STEM		Business and Economics		Social Sciences and Humanities		Health and Education		Other	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Two-Year Private	0.062	0.052	0.029	0.031	-0.036	0.307	0.131	0.170	0.114	0.194
	[0.022]	[0.029]	[0.014]	[0.020]	[0.038]	[0.094]	[0.032]	[0.033]	[0.032]	[0.116]
Four-Year Public	0.234	0.218	0.091	0.078	0.359	0.522	0.228	0.205	0.140	0.173
	[0.024]	[0.025]	[0.026]	[0.028]	[0.028]	[0.068]	[0.025]	[0.021]	[0.029]	[0.086]
Four-Year Private	0.311	0.314	0.204	0.187	0.427	0.595	0.216	0.201	0.168	0.251
	[0.039]	[0.038]	[0.012]	[0.021]	[0.039]	[0.069]	[0.023]	[0.023]	[0.018]	[0.072]
Graduate Studies	0.085	0.115	0.177	0.238	0.163	0.186	0.142	0.144	0.042	-0.035
	[0.033]	[0.032]	[0.021]	[0.020]	[0.018]	[0.021]	[0.025]	[0.029]	[0.034]	[0.079]
College Reputation	0.024	0.023	0.052	0.051	0.044	0.049	0.012	0.009	0.017	0.020
	[0.004]	[0.003]	[0.013]	[0.013]	[0.003]	[0.007]	[0.007]	[0.007]	[0.007]	[0.013]
Literacy	0.023	0.020	0.033	0.024	0.032	0.033	0.026	0.015	0.013	0.024
	[0.002]	[0.002]	[0.002]	[0.004]	[0.004]	[0.004]	[0.003]	[0.004]	[0.005]	[0.013]
Numeracy	0.038	0.023	0.042	0.036	0.019	0.013	0.021	0.018	0.025	0.039
	[0.002]	[0.005]	[0.004]	[0.005]	[0.004]	[0.008]	[0.007]	[0.008]	[0.006]	[0.014]
Foreign language	0.010	0.012	0.020	0.019	0.012	0.003	0.023	0.020	0.004	-0.000
	[0.005]	[0.004]	[0.008]	[0.009]	[0.008]	[0.011]	[0.011]	[0.012]	[0.007]	[0.010]
Field Specific		0.024		0.024		0.022		0.022		-0.003
		[0.007]		[0.004]		[0.004]		[0.005]		[0.008]
Sample:	Full	Specific	Full	Specific	Full	Specific	Full	Specific	Full	Specific
Observations	95,270	40,668	112,184	42,572	56,568	23,697	78,526	43,690	20,782	5,312
R-squared	0.213	0.266	0.251	0.300	0.232	0.341	0.316	0.365	0.242	0.459
Controls:										
Individual	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Types of Degrees	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable and measures of skills are as described in the notes of Table 3.1. Estimations using the full sample follow the same specification as column (4) of Table 3.1, but within the set of individuals belonging to the population of interest –defined by the categories on top of the table. In the study area vector, the STEM samples in column (1) includes individuals graduated from engineering, mathematics and natural sciences. Column (2) includes individuals graduated from business and economics, column (3) from social sciences and humanities, column (4) from health and education sciences, and column (5) from agronomy and arts. Standard errors clustered at the municipality level are in brackets.

Using model 3.1 we can additionally estimate heterogeneity among fields of study. Figure 3.5 shows the field of study fixed effects estimated in Column

(4) of Table 3.1.[27] The returns to different fields of study vary dramatically. Sports have the lowest returns, while medicine is more than 70 percent higher. These results suggest that what a person studies is as important as the institution at which those studies take place.

Figure 3.5: Returns to Fields of Study



Notes. Every plotted dot is the point estimate for each category of the field of study fixed effects, taking Arts as the base for comparison. Confidence levels of 95% are represented by boxes for all point estimates. The estimation uses the same specification as in column (4) of Table 3.1. Standard errors were clustered at the municipality level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

In Table 3.5 we then present the returns among workers in different economic sectors. To save space we present only the results for the sample of individuals for whom we have a measure of field-specific skills. Results for the full sample are very similar.[28] Again, we find clear returns to specialization, in particular for workers in the trade and tourism industries, in which the highest return is to foreign language skills (6.9 percent and 7.5 percent, respectively). Field-specific skills have a sizable and precise return in manufacturing, trade, and services. Numeracy skills show the biggest return for manufacturing and services. We also observe some heterogeneity in the gradient of returns to degrees. In the tourism and in the trade sector

[27]The field of study fixed effects estimated using the specifications in the other columns of Table 3.1 are very similar to those reported in Figure 3.5.

[28]The results for the full sample are available from the authors upon request.

the differences between degrees is much more attenuated. Among the other economic sectors, the differences between types of degrees remain relatively constant.

Table 3.5: Heterogeneous Effects: Economic Activity

	Dependent Variable: log(Current Wage)								
	Economic Activity:					Gender:		Firm Size:	
	Manufacture	Trade	Services	Turism	Retail	Female	Male	Small	Large
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
2-3 Years Private	0.009	-0.021	0.118	0.077	0.135	0.087	0.078	0.070	0.070
	[0.037]	[0.046]	[0.012]	[0.052]	[0.043]	[0.013]	[0.016]	[0.015]	[0.013]
4¿ Years Public	0.187	0.126	0.192	0.056	0.182	0.154	0.175	0.191	0.257
	[0.023]	[0.054]	[0.017]	[0.069]	[0.046]	[0.011]	[0.021]	[0.020]	[0.019]
4¿ Years Private	0.243	0.189	0.275	0.222	0.299	0.237	0.245	0.257	0.326
	[0.022]	[0.069]	[0.018]	[0.058]	[0.038]	[0.016]	[0.032]	[0.016]	[0.020]
Graduate Studies	0.195	0.202	0.192	0.214	0.089	0.190	0.172	0.175	0.170
	[0.045]	[0.105]	[0.021]	[0.098]	[0.083]	[0.016]	[0.035]	[0.023]	[0.027]
College Reputation	0.037	0.040	0.032	0.066	0.048	0.037	0.022	0.040	0.034
	[0.018]	[0.016]	[0.004]	[0.013]	[0.017]	[0.005]	[0.004]	[0.005]	[0.007]
Literacy	0.031	0.011	0.017	0.017	0.039	0.020	0.020	0.018	0.019
	[0.008]	[0.016]	[0.002]	[0.012]	[0.012]	[0.003]	[0.002]	[0.002]	[0.003]
Numeracy	0.051	0.021	0.018	0.003	0.041	0.030	0.023	0.020	0.028
	[0.007]	[0.018]	[0.003]	[0.019]	[0.015]	[0.003]	[0.003]	[0.005]	[0.003]
Foreign language	0.027	0.069	0.010	0.075	0.048	0.020	0.011	0.030	0.011
	[0.010]	[0.016]	[0.005]	[0.019]	[0.021]	[0.006]	[0.006]	[0.007]	[0.008]
Field Specific	0.034	0.027	0.021	0.019	-0.011	0.019	0.027	0.025	0.021
	[0.007]	[0.014]	[0.005]	[0.022]	[0.010]	[0.004]	[0.004]	[0.004]	[0.003]
Sample:	Specific	Specific	Specific	Specific	Specific	Specific	Specific	Specific	Specific
Observations	9,610	4,907	86,051	1,410	5,300	98,182	57,757	39,626	82,124
R-squared	0.528	0.642	0.274	0.175	0.633	0.246	0.247	0.387	0.297
Controls:									
Individual	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Types of Degrees	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable and measures of skills are as described in the notes of Table 3.1. Estimations using the full sample follow the same specification as column (5) of Table 3.1, but within the set of individuals belonging to the population of interest –defined by the categories on top of the table. Economic activity categories were defined by grouping four-digit industry codes. Column (6) and (7) use the subsamples of women and men, respectively. columns (8) and (9) include individuals working for small and large firms, respectively. Individual, college and occupation controls as described in the notes of Table 3.1. Standard errors clustered at the municipality level and in brackets.

Returns by Gender and Firm Size. We also explore heterogeneity by gender and firm size. Columns (6) and (7) of Table 3.5 present the returns by gender and suggest that female workers have, for numeracy and foreign languages measures, slightly higher returns than male workers. However, males have higher returns to specific skills than females (2.7 percent vs. 1.9 percent). We do not observe heterogeneity by gender in terms of returns to degrees.

Columns (8) and (9) of the same table explore heterogeneity among firm

size. [29] We find that numeracy skills and returns to postsecondary degrees are higher among people working in large firms compared to those in smaller firms.

3.6 Conclusion

Understanding the reasons that make some workers earn higher wages than others is a key question in labor economics. In this paper, we investigate how skills and postsecondary degrees relate to labor market outcomes later in life. Our paper differs from the previous literature in that we are able to jointly measure the importance of a broad set of skills (literacy, numeracy, foreign language, field-specific and non-cognitive) and different types of postsecondary degrees (by length, quality and field of study).

Exams required for graduation from high school and college in Colombia evaluate students in multiple areas including mathematics, language (Spanish) and foreign language (English). A follow-up survey then assesses the non-cognitive abilities of former graduates. We combine a vast vector of test scores from these exams with college enrollment and graduation data and with Social Security records to create a data set that follows individuals from high school to college, and into the first years in the labor market. This uniquely rich dataset allows us to unbundle the returns to postsecondary education: we jointly estimate returns to different skills and to different types of degrees, accounting for the length, quality and field in which those degrees were granted.

We view the results presented in this paper as a building block toward better understanding which skills and degrees are valued in today's labor market. Our evidence confirms that the return to basic skills (like mathematics, literacy, foreign language, and non-cognitive) are sizable even for people who graduated in essentially the same program. On average, a one

[29] The firm size is computed using the number of workers observed by establishment in the earnings records. These records only include formal workers who have been enrolled in a postsecondary education program; they could be active students, dropouts or have a college degree. Although this is not a perfect measure of firm size, it still proxies fairly well for firm size.

standard deviation increase in numeracy skills has a return between 1.7 percent and 14 percent, in literacy skills between 1.4 percent and 11.2 percent, and in foreign language between 1 percent and 12.3 percent. Specific skills are also important, ranging between 2.1 percent and 12.3 percent. Returns to non-cognitive skills range from 1.2 to 2.6 percent. This is equivalent to half to one year extra of education in a 4-year private university. We also find that the type of postsecondary degrees individuals obtain are associated with very different wage returns. First, the returns to college reputation are as high as the returns to skills. Second, the annual returns to a four-year degree public program is 3.9 percent higher than returns to two-year public (and 5.9 higher in a private school). Third, what field people study can be as important as where they earn those degrees: students of engineering and medicine expect 33 to 70 percent higher wages than students of arts or education.

The results in this paper can be useful for a number of related literatures. There is a large literature in economics evaluating the effect of education interventions aimed at improving learning (most frequently of numeracy and literacy skills), including i) school choice (e.g., competition, vouchers), ii) human resources policies (e.g., teacher pay, incentives, and training), iii) school and classroom management policies (e.g., class sizes and student tracking), and iv) school resources (e.g., spending, computers, remedial teaching, student incentives). Many of these studies lack cost-benefit analysis, in part because it is difficult to monetize the benefits. Our paper provides wage returns that could be used for that purpose.

Finally, our results also provide some insights into a number of policy debates. First, wage gains associated with admission to some schools and fields can be sizable and, in fact, explain a substantial part of the variation of the wage variance. This suggests that interventions aimed at helping low-income and qualified students gain admission to certain fields and to four-year colleges can improve welfare (Hoxby & Turner, 2013). Second, returns to skills are positive and large, both on average and for most wage quantiles. This suggests that, in the presence of resources constraints, policies that aim at improving the quality of education of low-income individuals can be expected to reduce wage inequality.

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APPENDIX A

APPENDIX CHAPTER 1

A.1 Mathematical Appendix

Workers

Consider a continuum of skilled (S) and unskilled (U) workers indexed by i endowed with a unit of time, and a continuum of firms indexed by j. Workers choose between consumption and leisure, and have preferences over the firms as in Card et al. (2018). In this model, workers have heterogeneous preferences for firms who offer differentiated work environments. The utility function for each skill group takes the form of:

$$U(C_{ij}, L_{ij}) = \frac{C_{ij}^{1-\frac{1}{\nu}}}{1-\frac{1}{\nu}} - \frac{L_{ij}^{1+\frac{1}{\gamma_{i,j}}}}{1+\frac{1}{\gamma_{i,j}}},$$

where C_{ij} corresponds to aggregate consumption and L_{ij} to hours worked. Note that preferences depend on the individual, $i \in \{S, U\}$, and the firm, j, where the individual works. We define the wage elasticity of labor supply, γ_{ij}, as specific to every skill group i who works in a firm j. Individuals have preferences over firms such that the wage elasticity of supply is firm specific.

Aggregate consumption is a composite of national (N) and foreign (F) goods:

$$C_i = \left[(C_N^i)^\rho + (C_F^i)^\rho\right]^{\frac{1}{\rho}},$$

where $0 < \rho < 1$.[1] To determine the consumption of both types of goods,

[1] This implies that the elasticity of substitution, $\sigma = \frac{1}{1-\rho}$, is positive and bigger than one. If $\rho > 1$ then $\sigma < 0$. If $\rho < 0$ then $\sigma \in \{0, 1\}$.

agents solve:

$$\min_{C_N^i, C_F^i} P_N C_N^i + P_F C_F^i \text{ s.t. } \left[(C_N^i)^\rho + (C_F^i)^\rho\right]^{\frac{1}{\rho}} = C_i,$$

where $P_F = P_F^0(1+\tau)$ and τ is an ad-valorem tariff charged to foreign products. The solution to the worker's problem yields consumption levels of national and foreign goods for skilled and unskilled workers of the form:

$$C_k^i = \frac{P_k^{-\sigma}}{P^{1-\sigma}} C_i \text{ for } i \in \{S, U\}, \text{ and } k \in \{N, F\},$$

where $P = \left[P_N^{1-\sigma} + [P_F^0(1+\tau)]^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$ is the CES price index. National and foreign goods are substitutes because $\rho < 1$.[2]

The allocation of the worker's unit of time between consumption and leisure determines labor supply. Formally, agents choose:

$$\max_{C_{ij}, L_{ij}} \frac{C_{ij}^{1-\frac{1}{\nu}}}{1-\frac{1}{\nu}} - \frac{L_{ij}^{1+\frac{1}{\gamma_{ij}}}}{1+\frac{1}{\gamma_{ij}}} \text{ s.t. } PC \leq W_{ij} L_{ij},$$

where the amount of leisure corresponds to $N_{ij} = 1 - L_{ij}$, and it varies by skill group and firm. Labor supply is thus firm- and worker- specific:

$$\frac{W_{ij}}{P} = C_i^{\frac{1}{\nu}} L_{ij}^{\frac{1}{\gamma_{ij}}},$$

where γ_{ij} corresponds to the wage elasticity and ν to the price elasticity of demand.

Firms

Consider a continuum of J national firms. Each firm $j \in J$ produces output combining labor and capital goods. Labor can be skilled, L_S, or unskilled, L_U, whereas capital goods can be national, X_N, or foreign, X_F. The produc-

[2]The price elasticity of consumption of national goods with respect to the price of foreign goods (or the the tariff τ) is positive. This implies that a price decrease of foreign goods (or a decrease in tariffs) decreases the demand of national goods.

tion function is formally given by:

$$Y_j(L_j, X_j) = \varphi_j f\left(L_j(L_{Sj}, L_{Uj}), X_j(X_{Nj}, X_{Fj})\right)$$

where φ_j is the productivity associated to firm j, and $f(\cdot)$ is the production function of firm j that combines labor and capital goods to produce output.[3]

We assume a monopsonic setup in which firms have market power when hiring workers, but are price takers in the capital and output markets. Thus, firms offer a wage bundle $\{W_{Sj}, W_{Uj}\}$ that determines labor demand, and choose an optimal amount of national and foreign capital goods subject to the labor supply schedules of workers:

$$\begin{aligned} \min_{W_{Uj}, W_{Sj}, X_{Nj}, X_{Fj}} \quad & W_{Uj}L_{Uj} + W_{Sj}L_{Sj} + X_{Nj}Q_{Nj} + X_{Fj}Q_{Fj}(1+\tau) \\ \text{s.t. } & Y_j(L_j, X_j) \geq Y \\ & L_{ij} = \left[W_{ij}\left(PC_i^{\frac{1}{\nu}}\right)^{-1}\right]^{\gamma_{ij}}, \end{aligned}$$

where τ corresponds to the ad-valorem tariff charged to foreign goods.

Firm- and Individual- Wage Premiums

The wage setting condition for each skill group gives the following expression:

$$W_{ij}\left(\frac{1+\gamma_{L_{ij}}}{\gamma_{L_{ij}}}\right) = \lambda\varphi_j f_L(L_j, X_j)L'(L_i),$$

where $f_L(L_j, X_j)$ is the marginal product of labor which is specific for each firm, $L'(L_i)$ is the marginal product of skill type i, and λ is the lagrange multiplier that is constant across all workers. Note that we substitute $L'_j(L_{ij}) = L'(L_i)$ because the marginal product of skill type i is the same across all firms.

[3]Fieler et al. (2018) present a solution for this model assuming different functional forms of the production function. Their main results use a nested CES production function.

Reordering terms and taking logarithms yields:

$$\ln W_{ij} = \ln \lambda + \underbrace{\ln(\varphi_j f_L(L_j, X_j))}_{FWP} + \underbrace{\ln(L'(L_i))}_{WWP} - \ln(M_{ij}(\gamma_{Sj}))$$

where $M_{ij}(\gamma_{ij}) = \frac{1+\gamma_{ij}}{\gamma_{ij}}$ is a markdown that decreases wages from the competitive equilibrium because of the firm's market power. Thus, wages can be decomposed into four separate components: 1) A value, $\ln \lambda$, common to all workers; 2) A component common to all workers in firm j ($\ln(\varphi_j f_L(L_j, X_j))$, which we call the "firm-specific wage premium" (FWP); 3) an individual component determined by the marginal productivity of skill type i ($\ln(L'_j(L_{ij}))$, , which we call the worker-specific wage premium (WWP); and 4) a component that quantifies the match between the worker and the firm ($\ln(m_{ij}(\gamma_{ij}))$ that is equivalent to the wage markdown and is a function of the wage elasticity of supply of workers to firm j.

Note that the FWP depends on the firm's inherent productivity and on the marginal product of labor in that specific firm. Thus, following Card et al. (2018) we consider the FWP as a close proxy for the firm's productivity level.[4] The WWP, on the contrary, is the marginal product worker type i and is irrespective of the firm. Therefore, workers' wage is highly mediated by the firm where they work and their own productivity, which is a measure of skill.

In addition, note that the equilibrium wages are affected by tariff changes in two ways: 1) substituting with foreign capital goods; and 2) inducing firms to close (or enter) and worker reallocation. Workers can receive wage increases or decreases by moving to more or less productive firms, respectively. If skilled workers (or any particular group) are more likely to reallocate into more productive firms then we can expect wage inequality to rise.

[4]Card et al. (2018) present a discussion on how firm's productivity relates to the FWP and wages.

Firms' Productivity Level

Melitz (2003) suggested that international trade induces less productive firms to exit the markets and more productive firms to enter. In the setting of this paper, we observe similar patterns by analyzing the minimum productivity level required to make a firm profitable.

Assume that firms have a level of productivity, $\varphi_j \sim G$, that is known before entering the market. A firm will only decide to enter if their own productivity level ,φ_j, is not below a threshold level φ_j^* defined by:

$$\varphi_j^*(\tau) \equiv \frac{W_{Uj}L_{Uj} + W_{Sj}L_{Sj} + X_N Q_N + X_F Q_F(1+\tau)}{P_N(\tau) f(L_j, X_j)} \tag{A.1}$$

If a firm's productivity level is not below φ_j^* then the firm will enter the market. It can also be the case that a firm that is already in the market decides to exit if φ_j^* increases above its own productivity φ_j.

A change in tariffs affects the threshold in two ways. First, it decreases the price of foreign goods used as inputs in production. This implies a reduction in the input costs of firms which will enable less productive firms to enter the market by decreasing φ_j^*. This reduction decreases the numerator of (A.1). Second, the reduction in the price of foreign goods also affects the demand for local products. In fact, the price elasticity of consumption of local goods ($C_N = C_N^U + C_N^S$) with respect to foreign goods' prices is positive. Therefore, a decrease in tariffs reduces the demand of national goods and therefore the price P_N. Formally $P_N'(\tau) > 0$, which implies that a reduction in tariffs reduces P_N, increases φ_j^*, and induces less productive firms to exit the market.

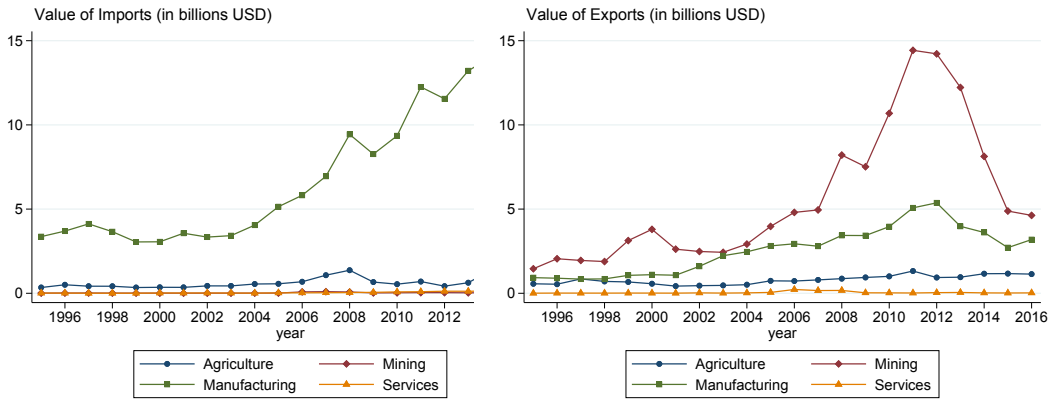
The change in the threshold with respect to a change in tariffs, $\frac{\partial \varphi_j^*(\tau)}{\partial \tau}$, depends on how these two forces interact. Formally, a change in tariffs motivates less productive firms to exit the market (i.e. $\frac{\partial \varphi_j^*(\tau)}{\partial \tau} < 0$) if $P_N < (1+\tau)P_N'(\tau)$. In words, firm exit depends on the degree in which national prices adjust to changes in tariffs, on the magnitude of the tariffs, and on the magnitude of the national prices. Therefore, under certain conditions an increase in imports penetration can motivate less productive firms to exit the market. Workers who are able sort to more productive firms/industries will

have wage increases, while workers who are less mobile will experience wage losses and potentially unemployment.

A.2 Appendix Figures

Appendix Figure A.1: Colombian Imports and Exports by Industry

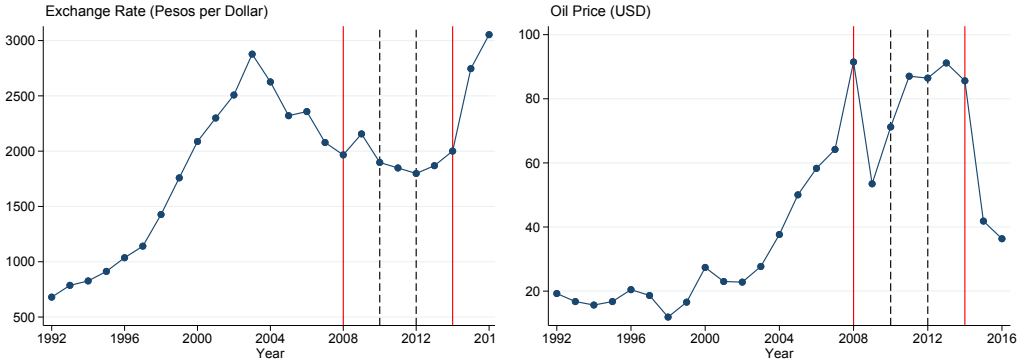
(a) Imports (b) Exports



Notes: These graphs present the value of imports and exports in USD millions by two-digit industry codes. These industries correspond to agriculture, manufacturing, mining, and services. The left panel presents imports, whereas the right presents exports.

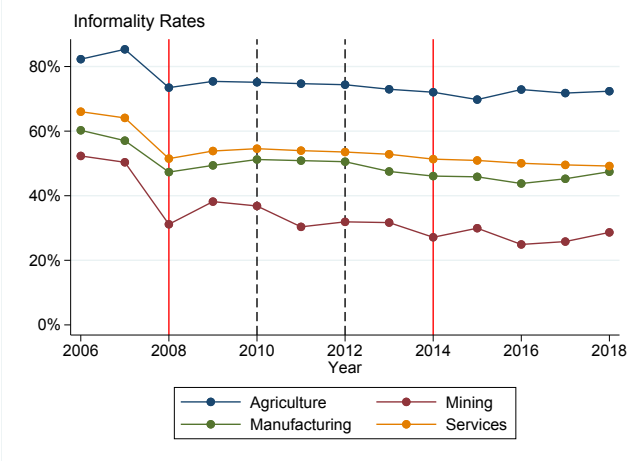
Appendix Figure A.2: Macroeconomic Environment

(a) Exchange Rates (b) Oil Prices



Notes: These graphs describe the macroeconomic environment around the implementation of the free-trade agreement. Panel A.2a presents the evolution of the exchange rate of U.S. dollars to Colombian pesos. Panel A.2b presents the evolution of the price of oil (in dollars). The vertical solid lines depict the period analyzed in this paper. The vertical dashed lines correspond to the year when Colombia (2010) and the United States (2012) reduced tariffs.

Appendix Figure A.3: Colombian Informality Rates



Notes: These graph presents the evolution of the informality rate in Colombia defined as persons who do not contribute to pension nor health. The vertical solid lines depict the period analyzed in this paper. The vertical dashed lines correspond to the year when Colombia (2010) and the United States (2012) reduced tariffs.

A.3 Appendix Tables

Appendix Table A.1: Descriptive Statistics Across Samples

	Count (1)	Mean (2)	S.D. (3)	Min. (4)	Max. (5)
A) Workers					
1(Male)	5,870,178	0.61	0.49	0.00	1.00
Age	5,870,178	41.52	9.36	26.00	63.00
1(Works at Endline)	5,870,178	0.77	0.42	0.00	1.00
1(Same 2-digit Ind. at Endline)	4,504,017	0.24	0.43	0.00	1.00
1(Same firm at Endline)	4,504,017	0.26	0.44	0.00	1.00
1(Same State)	4,504,017	0.60	0.49	0.00	1.00
log(Wage)	5,499,549	15.69	1.17	0.25	22.11
log(Daily Wage)	5,482,481	10.57	0.69	0.00	18.84
1(State that Trades)	5,870,178	0.74	0.44	0.00	1.00
1(Col Reduction)	5,870,178	0.06	0.23	0.00	1.00
1(USA Reduction)	5,870,178	0.01	0.11	0.00	1.00
1(Col Reduction)*1(State that Trades)	5,870,178	0.05	0.21	0.00	1.00
log(Output Import Competition)	5,870,178	0.59	2.45	0.00	19.66
B) Firms					
1(Open at Endline)	165,724	0.76	0.43	0.00	1.00
log(Days Worked)	165,724	7.40	1.87	0.22	15.20
log(Daily Wage)	165,693	10.48	0.55	0.04	15.92
Firm Wage Premium	152,793	-0.04	0.27	-10.75	6.96
1(State that Trades)	165,724	0.66	0.47	0.00	1.00
1(Col Reduction)	165,724	0.08	0.26	0.00	1.00
1(USA Reduction)	165,724	0.02	0.12	0.00	1.00
1(Col Reduction)*1(State that Trades)	165,724	0.05	0.23	0.00	1.00
log(Exports per Worker)	165,724	0.39	1.81	0.00	16.24
log(Output Import Competition)	165,724	0.53	2.31	0.00	19.66
C) Industry-State with at least one employee		.			
Employment	142,212	516.82	3,927.06	0.00	298,019.00
Wage with zeros (USD)	142,212	303.19	266.62	0.00	13,244.75
Number of Firms	142,212	20.90	264.90	0.00	34,465.00
Firm Size	142,212	23.46	114.31	0.00	16,671.00
Imports Per Worker (in millions)	142,212	0.03	2.36	0.00	384.50
Exports Per Worker (in millions)	142,212	0.01	1.09	0.00	179.68
1(COL reduction)	142,212	0.12	0.32	0.00	1.00
1(USA reduction)	142,212	0.02	0.14	0.00	1.00
1(States Trades with US)	142,212	0.30	0.46	0.00	1.00
1(COL reduction)*$Post$*1(Customs)	142,212	0.02	0.15	0.00	1.00
1(USA reduction)*$Post$*1(Customs)	142,212	0.00	0.06	0.00	1.00

Note: This table presents descriptive statistics of the three samples used. Panel A) describes the sample of incumbent workers observed in 2010 and tracked until 2014. Panel B) describe the sample of incumbent firms observed in 2010 and tracked until 2014. Panel C) describes the sample of industries-states-years from 2008 to 2014. Values of this sample were replaced to zero if that observation was missing for that year but we observed it in any other year. 1() stands for a dummy variable that takes the value of one if the condition inside parentheses is met.

Appendix Table A.2: First Stage at the Worker and Firm Level

	(1)	(2)	(3)	(4)	(5)
A) Worker Level					
1(COL reduction)1(Trade)	1.284**	1.277**	1.297**	1.725***	1.731***
	(0.589)	(0.589)	(0.589)	(0.367)	(0.361)
Log(Exports per Worker)	1.047***	1.047***	1.047***	0.676***	0.671***
	(0.073)	(0.073)	(0.073)	(0.042)	(0.042)
Observations	6,615,624	6,615,624	6,615,624	6,615,624	6,615,624
F-stat	4.751	4.696	4.696	22.038	22.985
B) Firm Level					
1(COL reduction)1(Trade)	4.065***	4.066***	4.157***	2.728***	2.734***
	(0.352)	(0.352)	(0.357)	(0.259)	(0.254)
Log(Exports per Worker)	0.615***	0.616***	0.611***	0.196***	0.191***
	(0.088)	(0.088)	(0.087)	(0.043)	(0.043)
Observations	165,724	165,724	165,724	165,724	165,724
F-stat	133.367	133.321	135.282	110.66	116.189
Demographics		Yes	Yes	Yes	Yes
Industry FE				Yes	Yes
State FE			Yes		Yes

Note: This table presents the first stage of equation 1.7. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.3: Effects of Import Competition on Indicator for whether firm open by Quartiles of Firms-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	0.000	0.001	-0.001	-0.003
	(0.005)	(0.004)	(0.004)	(0.004)
B) 2012				
log(IC)	-0.020*	-0.011	-0.007	-0.009
	(0.011)	(0.007)	(0.007)	(0.011)
C) 2013				
log(IC)	-0.026***	-0.017**	0.003	-0.016*
	(0.010)	(0.008)	(0.008)	(0.010)
D) 2014				
log(IC)	-0.027**	-0.010	-0.004	-0.009
	(0.011)	(0.008)	(0.008)	(0.012)
Observations	38,958	39,003	38,953	38,956
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether firm open separately by quartiles of the firms-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.4: Effects of Import Competition on firm size by Quartiles of Firms-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	-0.093*	0.031	-0.024	-0.034
	(0.053)	(0.045)	(0.055)	(0.057)
B) 2012				
log(IC)	-0.118*	-0.017	-0.007	-0.034
	(0.071)	(0.046)	(0.062)	(0.075)
C) 2013				
log(IC)	-0.102*	0.005	-0.012	-0.033
	(0.053)	(0.044)	(0.056)	(0.051)
D) 2014				
log(IC)	-0.099*	0.022	-0.014	-0.034
	(0.055)	(0.050)	(0.059)	(0.056)
Observations	37,505	37,512	37,858	38,013
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on firm size separately by quartiles of the firms-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.5: Effects of Import Competition on Indicator for whether the individual works by Quartiles of Firms-Specific Wage Premiums

	Quartile 1 (1)	Quartile 2 (2)	Quartile 3 (3)	Quartile 4 (4)
A) 2011				
log(IC)	0.009*	0.001	0.004	0.004
	(0.005)	(0.004)	(0.003)	(0.005)
B) 2012				
log(IC)	0.011	0.002	-0.000	-0.003
	(0.007)	(0.005)	(0.003)	(0.005)
C) 2013				
log(IC)	0.007	-0.002	0.000	-0.003
	(0.005)	(0.005)	(0.004)	(0.005)
D) 2014				
log(IC)	0.004	-0.001	-0.000	-0.001
	(0.004)	(0.005)	(0.004)	(0.005)
Observations	1,411,549	1,412,053	1,411,162	1,411,607
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether the individual works separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.6: Effects of Import Competition on Worker log Daily Wage by Quartiles of Firms-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	-0.032*	-0.002	-0.004	-0.004
	(0.017)	(0.008)	(0.005)	(0.008)
B) 2012				
log(IC)	-0.056**	0.016**	0.000	0.000
	(0.023)	(0.008)	(0.006)	(0.011)
C) 2013				
log(IC)	-0.045***	0.009	0.002	-0.001
	(0.017)	(0.007)	(0.005)	(0.008)
D) 2014				
log(IC)	-0.028**	0.013*	-0.000	0.003
	(0.013)	(0.008)	(0.006)	(0.008)
Observations	1,202,821	1,237,696	1,240,444	1,266,747
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on worker log daily wage separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.7: Effects of Import Competition on Indicator for whether working in unaffected industry by Quartiles of Firms-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	0.033**	0.018	0.012	0.017
	(0.016)	(0.015)	(0.011)	(0.012)
B) 2012				
log(IC)	0.038*	0.035*	0.006	0.007
	(0.021)	(0.018)	(0.009)	(0.013)
C) 2013				
log(IC)	0.016	0.033*	0.016	-0.007
	(0.014)	(0.019)	(0.015)	(0.012)
D) 2014				
log(IC)	0.016	0.028	-0.012	-0.004
	(0.014)	(0.021)	(0.016)	(0.014)
Observations	1,164,541	1,193,932	1,203,767	1,234,526
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether working in unaffected industry separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.8: Effects of Import Competition on Indicator for whether working in unaffected state by Quartiles of Firms-Specific Wage Premiums

	Quartile 1 (1)	Quartile 2 (2)	Quartile 3 (3)	Quartile 4 (4)
A) 2011				
log(IC)	0.005*	0.003	0.006***	0.005
	(0.003)	(0.002)	(0.002)	(0.004)
B) 2012				
log(IC)	0.008	0.005*	0.009***	0.010**
	(0.005)	(0.003)	(0.002)	(0.005)
C) 2013				
log(IC)	0.006	0.005*	0.010***	0.008**
	(0.004)	(0.003)	(0.003)	(0.003)
D) 2014				
log(IC)	0.007*	0.004	0.010***	0.005
	(0.004)	(0.003)	(0.003)	(0.004)
Observations	1,209,538	1,240,741	1,245,414	1,273,099
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether working in unaffected state separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.9: Effects of Import Competition on Indicator for whether the individual works by Quartiles of Workers-Specific Wage Premiums

	Quartile 1 (1)	Quartile 2 (2)	Quartile 3 (3)	Quartile 4 (4)
A) 2011				
log(IC)	-0.000	0.005	0.004	0.003
	(0.003)	(0.004)	(0.003)	(0.002)
B) 2012				
log(IC)	-0.008**	0.003	0.005	0.001
	(0.004)	(0.004)	(0.004)	(0.003)
C) 2013				
log(IC)	-0.011**	0.003	0.004	0.001
	(0.004)	(0.004)	(0.004)	(0.003)
D) 2014				
log(IC)	-0.006	0.002	0.005	0.000
	(0.004)	(0.004)	(0.004)	(0.003)
Observations	1,411,610	1,411,950	1,411,189	1,411,620
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether the individual works separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.10: Effects of Import Competition on Worker log Daily Wage by Quartiles of Workers-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	0.002	-0.010*	-0.011*	-0.001
	(0.011)	(0.005)	(0.006)	(0.005)
B) 2012				
log(IC)	0.003	-0.002	-0.009	-0.005
	(0.014)	(0.005)	(0.007)	(0.005)
C) 2013				
log(IC)	0.005	-0.002	-0.004	-0.006
	(0.012)	(0.005)	(0.006)	(0.006)
D) 2014				
log(IC)	0.002	0.000	-0.004	-0.001
	(0.012)	(0.005)	(0.007)	(0.007)
Observations	1,231,395	1,205,653	1,203,824	1,306,830
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on worker log daily wage separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.11: Effects of Import Competition on Indicator for whether working in unaffected industry by Quartiles of Workers-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	-0.000	0.013	0.046***	0.007
	(0.011)	(0.009)	(0.017)	(0.007)
B) 2012				
log(IC)	-0.009	0.020*	0.057***	0.011
	(0.010)	(0.011)	(0.019)	(0.007)
C) 2013				
log(IC)	-0.028*	0.014	0.054***	0.008
	(0.014)	(0.014)	(0.019)	(0.010)
D) 2014				
log(IC)	-0.005	-0.008	0.040**	0.004
	(0.013)	(0.013)	(0.018)	(0.012)
Observations	1,199,398	1,159,688	1,158,841	1,278,832
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether working in unaffected industry separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.12: Effects of Import Competition on Indicator for whether working in unaffected state by Quartiles of Workers-Specific Wage Premiums

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
	(1)	(2)	(3)	(4)
A) 2011				
log(IC)	-0.000	0.009***	0.006**	0.006***
	(0.002)	(0.002)	(0.002)	(0.002)
B) 2012				
log(IC)	0.001	0.014***	0.008***	0.009***
	(0.003)	(0.004)	(0.003)	(0.003)
C) 2013				
log(IC)	0.001	0.014***	0.008***	0.009***
	(0.003)	(0.003)	(0.003)	(0.003)
D) 2014				
log(IC)	0.001	0.014***	0.008**	0.005*
	(0.003)	(0.003)	(0.004)	(0.003)
Observations	1,235,568	1,209,926	1,209,570	1,313,721
Demographics	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
log(Exports)	Yes	Yes	Yes	Yes

Note: This table presents the estimation of Equation 1.7 on indicator for whether working in unaffected state separately by quartiles of the workers-specific wage premiums and year. Panel A) presents results for 2011, panel B) for 2012, panel C) for 2013, and panel D) for 2014. Column (1) plots estimates among quartile 1, column (2) for quartile 2, column (3) for quartile 3, and column (4) for quartile 4. All specifications control for age, age-squared, gender, wages in 2010, and exports per workers. State and industry fixed effects included. Standard errors clustered at the industry–state level. *** p¡0.01, ** p¡0.05, * p¡0.1

Appendix Table A.13: Most and Less Imported Products

Before free-trade agreement (2008-2010)		After free-trade agreement (2011-2014)	
Product	Value of Imports (USD)	Product	Value of Imports (USD)
More Imported Products			
Manufacture of refined petroleum products	3,454,817,062	Manufacture of refined petroleum products	18,043,048,647
Manufacture of basic chemicals	2,986,627,388	Manufacture of basic chemicals	4,618,210,761
Manufacture of aircraft and spacecraft	2,671,057,996	Manufacture of aircraft and spacecraft	3,904,964,753
Growing of cereals and other crops n.e.c.	2,434,170,631	Growing of cereals and other crops n.e.c.	2,918,795,023
Manufacture of machinery for mining, quarrying and construction	1,674,661,590	Manufacture of plastics in primary forms and of synthetic rubber	2,424,816,198
Manufacture of plastics in primary forms and of synthetic rubber	1,319,655,422	Manufacture of motor vehicles	2,011,364,592
Manufacture of pumps, compressors, taps and valves	1,046,711,803	Manufacture of machinery for mining, quarrying and construction	1,984,965,474
Manufacture of motor vehicles	1,045,354,478	Manufacture of other chemical products n.e.c.	1,657,742,734
Manufacture of other chemical products n.e.c.	901,847,834	Manufacture of pharmaceuticals, medicinal, and botanical products	1,604,872,773
Less Imported Products			
Manufacture of pharmaceuticals, medicinal, botanical	864,893,890	Manufacture of pumps, compressors, taps and valves	1,578,531,260
Dramatic arts, music and other arts activities	485,316	Motion picture and video production and distribution	810,482
Manufacture of structural non-refractory clay and ceramic products	351,996	Manufacture of television and radio transmitters	748,602
Cutting, shaping and finishing of stone	197,031	Cutting, shaping and finishing of stone	567,766
Photographic activities	191,111	Fishing, operation of fish hatcheries and fish farms	521,867
Manufacture of tobacco products	134,139	Manufacture of gas; distribution of gaseous fuels through mains	485,745
Manufacture of coke oven products	54,830	Manufacture of wooden containers	110,183
Manufacture of wooden containers	36,301	Photographic activities	79,673
Hairdressing and other beauty treatment	33,667	Manufacture of coke oven products	57,461
Dressing and dyeing of fur; manufacture of articles of fur	24,040	Architectural and engineering activities	38,926
Manufacture of gas; distribution of gaseous fuels through mains	10,515	Dressing and dyeing of fur; manufacture of articles of fur	29,811
Architectural and engineering activities	9,797	Hairdressing and other beauty treatment	4,214

Note: This table describes the most and less imported products in Colombia from the United States, before and after the free-trade agreement. The values are presented in USD.

APPENDIX B

APPENDIX CHAPTER 2

Appendix Table B.1: School Shooting Episodes

	Year	School	City	State	Victims	Fatalities
1	1998	Westside School	Jonesboro	Arkansas	15	5
2	1998	Thurston High School	Springfield	Oregon	29	4
3	1999	Columbine High School	Littleton	Colorado	37	13
4	1999	Heritage High School	Conyers	Georgia	6	0
5	1999	Fort Gibson Middle School	Fort Gibson	Oklahoma	4	0
6	2001	Santana High School	San Diego	California	15	2
7	2006	Orange High School	Hillsborough	North Carolina	3	1
8	2006	West Nickel Mines Amish School	Lancaster	Pennsylvania	10	5
9	2007	Springwater Trail High School	Gresham	Oregon	10	0
10	2007	Success Tech Academy	Cleveland	Ohio	4	1
11	2007	South Middle School Football Game	Saginaw	Michigan	4	0
12	2012	Chardon High School	Chardon	Ohio	6	3
13	2012	Sandy Hook Elementary School	Newtown	Connecticut	29	27
14	2013	Sparks Middle School	Sparks	Nevada	3	1
15	2014	Marysville-Pilchuck High School	Marysville	Washington	5	4

Appendix Table B.2: Taking out one episode at a time

	(1) Chardon	(2) Cleveland	(3) Columbine	(4) Conyers	(5) Fort Gibson	(6) Gresham	(7) Hillsborough	(8) Jonesboro
1(Within Affected SD)	-0.120***	-0.040**	-0.124***	-0.118***	-0.113***	-0.114***	-0.113***	-0.111***
*1(After Shooting)	(0.017)	(0.016)	(0.018)	(0.017)	(0.017)	(0.017)	(0.017)	(0.016)
Observations	787,851	728,053	612,455	676,761	791,852	784,542	750,846	797,611
R-squared	0.710	0.712	0.711	0.720	0.703	0.707	0.710	0.706

	(9) Lancaster	(10) Marysville	(11) Newton	(12) Saginaw	(13) San Diego	(14) Sparks	(15) Springfield
1(Within Affected SD)	-0.112***	-0.121***	-0.111***	-0.071***	-0.124***	-0.142***	-0.128***
*1(After Shooting)	(0.016)	(0.018)	(0.016)	(0.011)	(0.018)	(0.019)	(0.018)
Observations	788,216	762,995	788,427	788,506	604,951	724,930	778,558
R-squared	0.707	0.706	0.703	0.699	0.673	0.703	0.721
			Yes	Yes	Yes	Yes	Yes
Year*Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted Regression	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: All columns use a three-year window around the episode. For each column we dropped observations for one of the episodes and estimate the model in the remaining ones. All the regressions are weighted by one over the total number of transactions per episode. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Appendix Table B.3: Effect of School Shootings around Boundary with Different Radius

	(1) 0.3 Mi.	(2) 0.5 Mi.	(3) 1 Mi.	(4) 2 Mi.	(5) 3 Mi.
1(Within Affected SD)	0.054	0.050	0.044	0.025	0.021
	(0.046)	(0.040)	(0.041)	(0.036)	(0.034)
1(Within Affected SD)	-0.104***	-0.114***	-0.125***	-0.126***	-0.112***
*1(After Shooting)	(0.037)	(0.032)	(0.023)	(0.016)	(0.013)
Constant	352.736***	334.771***	298.816***	191.569***	-26.281
	(62.475)	(51.393)	(40.065)	(33.737)	(49.847)
Observations	11,855	19,661	39,481	70,548	98,493
R-squared	0.733	0.739	0.726	0.735	0.744
Census Tract FE	Yes	Yes	Yes	Yes	Yes
Year*Month FE	Yes	Yes	Yes	Yes	Yes
Episode Specific Trend	Yes	Yes	Yes	Yes	Yes
Property Characteristics	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the log of the deflated price per property. All the columns vary the radius around the school district boundary. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not is an apartment, a dummy for whether or not is a condo, a dummy if the property has a fireplace, a dummy if the house is constructed with brick, and the number of bedrooms. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

Appendix Table B.4: Average Effect of School Shootings on Non-Residential Properties

	Dependent Variable : ln(Price)			
	(1)	(2)	(3)	(4)
1(Within Affected SD)*1(After Shooting)	0.011	0.005	0.033	0.091*
	(0.048)	(0.043)	(0.042)	(0.047)
Constant	7.054***	2.052***	201.955	-99.178
	(0.438)	(0.344)	(366.194)	(435.684)
Observations	13,151	13,151	13,151	13,151
R-squared	0.509	0.621	0.623	0.633
Year*Month FE	Yes	Yes	Yes	Yes
Census Tract FE	Yes	Yes	Yes	Yes
Property Characteristics		Yes	Yes	Yes
Episode Specific Trend			Yes	Yes
Weighted Regression				Yes

Note: The dependent variable is the log of the deflated price per property. Estimations include all non-residential properties (i.e. industrial and commercial properties) within the school district. Property characteristics include the log of area of land, the log of area of building, a dummy for whether or not the property is an apartment. The regression in column 4 weighs the observations by the inverse of the total number of properties sold within each episode. Standard errors clustered at the census tract level. *** p<0.01, ** p<0.05, * p<0.1.

APPENDIX C

APPENDIX CHAPTER 3

C.1 Appendix: Additional Tables

Appendix Table C.1: Estimates of the Effect of Cognitive Skills from Previous Literature

Skills	Reference	Country/ City	Population in Estimating Sample	Identification Strategy	Estimation	Dependent Variable (log)	Range[a] Lowest Estimate	Range[a] Highest Estimate
Panel A: Numeracy								
	(Levine & Zimmerman, 1995)	USA	1980 HS graduates	Selection on Observables	OLS	1990 & 1986 weekly wage	0.028	0.030[b]
	(Murnane et al., 1995)	USA	1972 and 1980 HS graduates	Selection on Observables	OLS	1978 & 1986 hourly wage	0.026	0.069[c]
	(Tyler, 2004)	Florida (USA)	HS dropouts	Selection on Observables	OLS	1995-1999 quarterly earnings	0.063	0.074
	(Song et al., 2008)	USA	College graduates	Selection on Observables	IV	1993 earnings	0.181	0.210[d]
	(Joensen & Nielsen, 2009)	Denmark	1986 and 1987 HS graduates	Quasi-experiment	IV	1999-2002 annual earnings	0.12	0.32[e]
	(Hanushek et al., 2015)	Several	Adults aged 20 to 50 from 23 OECD countries	Selection on Observables	OLS	2011-2012 hourly earnings	0.079	0.178[f]
Panel B: Literacy								
	(Ishikawa & Ryan, 2002)	USA	Adults above 16	Selection on Observables	OLS	weekly wages	0.001	0.008[g]
	(Fasih et al., 2013)	Several	Males aged 22 to 65 from 20 countries, mostly OECD	Selection on Observables	OLS	hourly wage	0.021	0.210[h]
	(Hanushek et al., 2015)	Several	Adults aged 20 to 50 from 23 OECD countries	Selection on Observables	OLS	1993 earnings	0.068	0.171
	(Sanders, 2016)	USA	Populations represented in 5 longitudinal surveys	Selection on Observables	OLS	real wages	-0.056	-0.024

[a] Estimates points correspond to standardized test scores, unless another interpretation is suggested. [b] Estimations correspond to the number of mathematics classes taken during high school. [c] Point estimates are given originally for levels of a mathematics score. Since a one standard deviation is 6.25, then coefficients are translated into this scale. Lower and upper bounds correspond, respectively, to 1972 and 1980 high school graduates. [d] The mathematics score is estimated in levels. [e] Numeracy is a dummy valued 1 if individuals took a high-level mathematics course during high school. Reported bounds correspond to the pilot school sample. [f] Numeracy is also estimated using literacy as instrument and the coefficient found is 0.201. [g] Lower and Upper bounds correspond to the point estimates for Black men's and Hispanic men's samples, which respectively are the lowest and highest point estimates. Literacy was estimated in levels. [h] Lower and Upper bounds correspond to the point estimates for Denmark and Bermuda, which are respectively the lowest and largest estimates found. See the paper for more details.

Appendix Table C.2: Estimates of the Effect of Cognitive Skills from Previous Literature (Cont.)

Skills	Reference	Country/ City	Population in Estimating Sample	Identification Strategy	Estimation	Dependent Variable (log)	Range	
							Lowest Estimate	Highest Estimate
Panel C: Foreign Language								
	(Bleakley & Chin, 2004)	USA	1960-1974 Young immigrants	Quasi-experiment	IV	1990 annual wage	0.222	0.334[i]
	(Saiz & Zoido, 2005)	USA	College graduates	Selection on Observables	OLS	1997 hourly wage	0.025	0.028[j]
	(Christofides & Swidinsky, 2010)	Quebec (CA)	Fulltime native workers aged 15 to 64	Selection on Observables	OLS	2000 earnings	0.109	0.139[k]
	(Azam et al., 2013)	India	Male workers aged 18 to 65	Selection on Observables	OLS	2005 earnings	0.345	0.603[l]
	(Guo & Sun, 2014)	China	College graduates	Selection on Observables	OLS	2010 monthly wage	0.033	0.131[m]
	(Budría & Swedberg, 2015)	Spain	Male immigrants aged 18 to 65	Quasi-experiment	IV	2006-2007 hourly wages	0.049	0.204[n]
	(Di Paolo & Tansel, 2015)	Turkey	Male workers	Selection on Observables	OLS	2007 wage	0.107	0.072[o]
	(Stöhr, 2015)	Germany	Fulltime workers	Selection on Observables	OLS	2005-2006 gross monthly wage	0.033	0.093[p]

[i] The independent variable takes 1 as value if individual speaks English very well. [j] IV, Panel and PSM estimations are also considered. For instance, PSM point estimates ranged from 0.020 to 0.021. [k] Point estimates correspond to a subsample of only men. The independent variable takes 1 as value if individual uses English in his/her workplace. Estimations are also carried for women and ranged from 0.068 to 0.076. [l] Point estimates for a dummy variable that takes 1 as value if the individual is fluent in English. Estimations for knowing little English can be seen in the paper. [m] The English score of CET-4 test is used to measure foreign language proficiency, check the paper for more details. [n] Lower bound corresponds to OLS estimation and Upper bound corresponds to IV estimation using simultaneously the following instruments: 1(arrived before 12), 1(has a child proficient in spanish) and 1(willingness to stay in Spain). [o] The independent variable takes 1 as value if the individual knows english. Other languages are estimated (French, German, Arabic and Russian), but those who know english account for 76%. [p] The independent variable takes 1 as value if the individual's occupation requires expertise in a foreign language.

Appendix Table C.3: Non-Cognitive Skills in the Survey of Graduates

Questionnaire item	**Factor Loadings**
Accept the difference and work under multicultural contexts	0.565
Learn and keep updated	0.593
Work under pressure	0.504
Adopt a coexistence culture	0.589
Work independently without permanent supervision	0.557
Convince and persuade others	0.532
Identify and use communication symbols (i.e. non-verbal or iconic language)	0.344
Abstraction capacity, analysis and synthesis	0.625
Identify, plan and solve problems	0.680

Notes. The answers for each item ranges from 1 to 4: (1) Very unsatisfied, (2) Unsatisfied, (3) Satisfied and (4) Very satisfied. The largest eigenvalue from a factor analysis model is 2.836, and the second largest is 0.109. For each item, the factor loadings in this table correspond to the factor with the largest eigenvalue. The scale reliability coefficient, α, is 0.80.

Appendix Table C.4: Non-Cognitive Skills Used As Instrumental Variable

Questionnaire item	**Factor Loadings**
To be creative and innovative	0.5836
Look for, analyze, manage and share information	0.6014
Understand the surrounding reality	0.6350
Assume responsibilities and make decisions	0.6959
Plan and use time effectively to achieve the goals	0.6294
Formulate and execute projects	0.5479
Work in teams to achieve common goals	0.6575
Consider values and professional ethics to perform tasks	0.6479
Adapt to changes	0.6486
Take risks	0.6712
Identify opportunities and resources	0.6660

Notes. The answers for each item ranges from 1 to 4: (1) Very unsatisfied, (2) Unsatisfied, (3) Satisfied and (4) Very satisfied. The largest eigenvalue from a factor analysis model is 4.453, and the second largest is 0.109. For each item, the factor loadings in this table correspond to the factor with the largest eigenvalue. The scale reliability coefficient, α, is 0.88.

Appendix Table C.5: Descriptive Statistics

	Estimation Sample						Mean Difference	
	Full (N = 363,330)		Specific (N = 155,939)		Survey (N = 2,401)			
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	*p*-value	
	(1)		(2)		(3)		(1) - (2)	(1) - (3)
Panel A: Socioeconomic Statistics								
Age	26.93	3.44	26.86	3.5	28.74	4.57	0.00	0.00
Share female	0.61	0.49	0.63	0.48	0.59	0.49	0.00	0.08
Share living in big urban areas	0.73	0.45	0.71	0.45	0.80	0.40	0.00	0.00
Share living in low income households	0.45	0.50	0.43	0.50	0.46	0.50	0.00	0.40
Share graduate students	0.02	0.14	0.03	0.17	0.04	0.19	0.00	0.00
Panel B: Education Statistics								
Share graduated from:								
STEM	0.26	0.44	0.26	0.44	0.25	0.44	0.29	0.36
Business and Economics	0.31	0.46	0.27	0.45	0.27	0.44	0.00	0.00
Social Sc. and Humanities	0.16	0.36	0.15	0.36	0.12	0.33	0.00	0.00
Health and Education	0.22	0.41	0.28	0.45	0.22	0.41	0.00	0.66
Share postsecondary degrees:								
Two-Year Private	0.09	0.29	0.04	0.20	0.40	0.49	0.00	0.00
Two-Year Public	0.10	0.30	0.05	0.21	0.10	0.3	0.00	0.64
Four-Year Private	0.50	0.50	0.56	0.50	0.32	0.47	0.00	0.00
Four-Year Public	0.32	0.47	0.36	0.48	0.18	0.38	0.00	0.00
Panel C: Occupation Statistics								
Share working in:								
Manufacture	0.07	0.26	0.07	0.26	0.08	0.27	0.00	0.60
Commerce	0.03	0.18	0.03	0.17	0.04	0.19	0.00	0.27
Services	0.54	0.50	0.55	0.50	0.60	0.49	0.00	0.00
Turism	0.01	0.10	0.01	0.09	0.01	0.11	0.00	0.83
Retail	0.04	0.20	0.04	0.19	0.04	0.20	0.00	0.95
Share working as:								
Public Employee	0.03	0.18	0.04	0.21	0.04	0.19	0.00	0.12
Independent Workers	0.07	0.25	0.07	0.26	0.08	0.27	0.09	0.03
Share working in firms that are:								
$[1-10]$	0.15	0.36	0.14	0.35	0.12	0.32	0.00	0.00
$[11-50]$	0.18	0.39	0.18	0.39	0.16	0.37	0.97	0.02
$[51-100]$	0.08	0.28	0.08	0.28	0.09	0.29	0.32	0.13
$[101-500]$	0.21	0.40	0.21	0.41	0.21	0.41	0.15	0.50
$[501-1000]$	0.10	0.30	0.10	0.30	0.09	0.29	0.02	0.26
[1001+]	0.28	0.45	0.28	0.45	0.33	0.47	0.01	0.00
Panel D: Labor Statistics								
Current Wage	16.86	11.24	17.48	11.59	19.75	10.79	0.00	0.00
First Wage	13.42	8.48	14.07	8.88	12.85	7.72	0.00	0.00
Average Wage	15.17	8.86	15.8	9.21	16.19	7.92	0.00	0.00
Current Tenure	1.77	1.02	1.78	1.11	2.28	1.3	0.00	0.00

Notes. Descriptive statistics of students who took the college exit exam from 2011 to 2015 for whom data were matched to earnings and college records. Big urban area refers to the largest 13 cities in Colombia. Low-income households refer to individuals in the first two income strata designated by place of residence. Wages are presented in nominal 2016 USD currency (1 USD = 3,050.98 COP).

Appendix Table C.6: Correlation Matrix across Test Scores

	High School Exit Exams					College Exit Exams		
	Average	Subject	Literacy	Numeracy	Foreign	Literacy	Numeracy	Foreign
Panel A: Full Sample (N = 363,330)								
High School Exit Exams:								
Subject	**0.941***							
Literacy	**0.708***	**0.595***						
Numeracy	**0.671***	**0.546***	**0.394***					
Foreign Language	**0.726***	**0.571***	**0.460***	**0.428***				
College Exit Exams:								
Literacy	0.537*	0.497*	0.438*	0.314*	0.390*			
Numeracy	0.584*	0.553*	0.394*	0.487*	0.401*	**0.449***		
Foreign Language	0.611*	0.515*	0.414*	0.393*	0.678*	**0.458***	**0.464***	
Panel B: Specific Sample (N = 155,939)								
High School Exit Exams:								
Subject	**0.941***							
Literacy	**0.705***	**0.593***						
Numeracy	**0.676***	**0.553***	**0.396***					
Foreign Language	**0.727***	**0.573***	**0.459***	**0.431***				
College Exit Exams:								
Literacy	0.553*	0.513*	0.446*	0.331*	0.401*			
Numeracy	0.611*	0.579*	0.409*	0.509*	0.424*	**0.470***		
Foreign Language	0.623*	0.526*	0.419*	0.405*	0.687*	**0.472***	**0.490***	
Major-Specific	0.520*	0.496*	0.394*	0.344*	0.345*	**0.519***	**0.511***	**0.413***
Panel C: Survey Sample (N = 2,401)								
High School Exit Exams:								
Subject	**0.933***							
Literacy	**0.708***	**0.586***						
Numeracy	**0.620***	**0.483***	**0.369***					
Foreign Language	**0.668***	**0.508***	**0.420***	**0.347***				
College Exit Exams:								
Literacy	0.505*	0.461*	0.430*	0.278*	0.335*			
Numeracy	0.513*	0.483*	0.373*	0.417*	0.321*	**0.395***		
Foreign Language	0.597*	0.500*	0.434*	0.352*	0.636*	**0.463***	**0.400***	
Socioemotional:								
Factor Score	0.054*	0.053*	0.038†	0.047*	0.039†	0.026	0.025	0.058*

Notes. Pairwise correlations are estimated using the Pearson's formula. For both the college exit exam (*Saber Pro*) and the high school exit exam (*Saber 11*), individuals' scores are standardized with respect to the corresponding average in each test edition. The specific scores from the college exit exam are standardized with respect to the average of the test edition and the corresponding group of related programs. The subject score from the high school exit exam is computed as the standardized average of biology, philosophy, physics, chemistry, and social science tests. The non-cognitive scores were computed as the predictions from a factor model considering categorical answers to nine questions (see Appendix Table C.3). † $p<0.1$, * $p<0.05$.

Appendix Table C.7: Other Outcomes

	Dependent Variable: log(First Wage After Graduation)			*Dependent Variable*: log(Avg. Wage Since Graduation)			*Dependent Variable*: log(Current Earnings)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Two-Year Private	0.068	0.061	0.144	0.078	0.077	0.137	0.080	0.084	0.130
	[0.009]	[0.010]	[0.026]	[0.011]	[0.009]	[0.036]	[0.014]	[0.013]	[0.050]
Four-Year Public	0.149	0.156	0.287	0.174	0.171	0.263	0.177	0.159	0.199
	[0.010]	[0.013]	[0.029]	[0.010]	[0.012]	[0.029]	[0.011]	[0.014]	[0.041]
Four-Year Private	0.211	0.221	0.216	0.244	0.244	0.225	0.251	0.239	0.218
	[0.017]	[0.019]	[0.027]	[0.018]	[0.018]	[0.025]	[0.020]	[0.022]	[0.035]
Literacy	0.016	0.010	0.008	0.023	0.016	0.017	0.027	0.020	0.015
	[0.001]	[0.002]	[0.009]	[0.001]	[0.002]	[0.006]	[0.001]	[0.002]	[0.008]
Numeracy	0.025	0.018	0.032	0.031	0.023	0.020	0.035	0.026	0.022
	[0.001]	[0.002]	[0.011]	[0.001]	[0.002]	[0.010]	[0.002]	[0.003]	[0.010]
Foreign language	0.011	0.013	0.047	0.015	0.016	0.032	0.015	0.015	0.025
	[0.003]	[0.003]	[0.008]	[0.004]	[0.004]	[0.006]	[0.005]	[0.005]	[0.007]
Field Specific		0.016			0.020			0.022	
		[0.003]			[0.004]			[0.004]	
Socioemotional			0.010			0.021			0.021
			[0.011]			[0.008]			[0.006]
College Reputation	0.026	0.029	0.030	0.031	0.032	0.050	0.032	0.031	0.059
	[0.005]	[0.005]	[0.017]	[0.004]	[0.005]	[0.020]	[0.004]	[0.005]	[0.023]
Sample:	Full	Specific	Survey	Full	Specific	Survey	Full	Specific	Survey
Observations	363,330	155,939	2,401	363,330	155,939	2,401	363,330	155,939	2,401
R-squared	0.196	0.237	0.244	0.233	0.264	0.324	0.195	0.220	0.244
Controls:									
Individual	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Field of Study	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Estimations in columns (1), (4) and (7) follow the specification of column (4) in Table 3.1. Estimations in columns (2), (5) and (8) follow the specification of column (5) in Table 3.1. Estimations in columns (3), (6) and (9) follow the specification of column (6) in Table 3.1. The dependent variable for columns (1) to (3) is the log of the first wage observed after the individual's graduation year. The dependent variable for columns (4) to (6) is the log of the average wage from 2011 to 2016 for each individual, taking into account only wages after the year of graduation. The dependent variable for columns (7) to (9) is the log of the last observed annual earnings for each individual. Individual and field of study controls as described in the notes of Table 3.1. Standard errors are clustered at the municipality level and in brackets.

Appendix Table C.8: Extensions: Estimates' Ranges and Measurement Error

	Dependent Variable: log(Current Wage)										
	OLS						IV: Measurement Error Correction				
	Simultaneous			Stacked			Simultaneous		Stacked		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Literacy	0.026	0.020	0.015	0.036	0.034	0.022	0.075	0.010	0.099	0.100	0.110
	[0.001]	[0.002]	[0.008]	[0.001]	[0.002]	[0.009]	[0.010]	[0.059]	[0.012]	[0.014]	[0.046]
Numeracy	0.034	0.025	0.017	0.043	0.040	0.024	0.048	0.114	0.085	0.081	0.127
	[0.003]	[0.003]	[0.009]	[0.003]	[0.003]	[0.009]	[0.007]	[0.044]	[0.010]	[0.013]	[0.029]
Foreign Language	0.010	0.011	0.024	0.024	0.026	0.031	-0.009	0.021	0.034	0.036	0.060
	[0.005]	[0.005]	[0.013]	[0.006]	[0.006]	[0.014]	[0.004]	[0.028]	[0.006]	[0.006]	[0.022]
Field Specific		0.021			0.037					0.119	
		[0.004]			[0.002]					[0.019]	
Socioemotional			0.012			0.012		0.012			0.007
			[0.006]			[0.006]		[0.007]			[0.007]
Sample:	Full	Specific	Survey	Full	Specific	Survey	Full	Survey	Full	Specific	Survey
Observations	363,330	155,939	2,401	363,330	155,939	2,401	363,330	2,401	363,330	155,939	2,401
Controls:											
Individual & Field	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
College	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable and measures of skills are as described in the notes of Table 3.1. Estimations within the full sample use the same specification as column (6) of Table 3.1. Likewise, estimations within the specific and survey samples use respectively the specifications of columns (8) and (10) of Table 3.1. Stacked results refer to separate regressions for each ability measure. In columns (7) to (9), cognitive skills were instrumented, each time using numeracy, literacy and foreign language standardized scores computed from the high school exit exam. Individual and types of degrees area as described in the notes of Table 3.1. Standard errors clustered by municipality and in brackets.

Appendix Table C.9: Extensions: Estimates' Ranges and Measurement Error

	Dependent Variable: log(Current Wage)													
Panel A:	Ordinary Least Squares Estimations													
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Two-Year Private	0.080	0.084	0.128	0.081	0.078	0.078	0.086	0.083	0.083	0.084	0.129	0.129	0.126	0.132
	[0.014]	[0.012]	[0.049]	[0.014]	[0.014]	[0.014]	[0.012]	[0.012]	[0.012]	[0.012]	[0.053]	[0.052]	[0.051]	[0.052]
Four-Year Public	0.174	0.156	0.197	0.176	0.175	0.173	0.159	0.156	0.161	0.164	0.197	0.199	0.206	0.208
	[0.011]	[0.013]	[0.042]	[0.011]	[0.011]	[0.011]	[0.013]	[0.013]	[0.013]	[0.013]	[0.043]	[0.043]	[0.042]	[0.044]
Four-Year Private	0.249	0.236	0.212	0.247	0.249	0.240	0.235	0.234	0.232	0.240	0.215	0.221	0.211	0.222
	[0.020]	[0.021]	[0.036]	[0.020]	[0.020]	[0.019]	[0.021]	[0.021]	[0.021]	[0.021]	[0.039]	[0.038]	[0.037]	[0.038]
Literacy	0.027	0.020	0.017	0.038			0.036				0.026			
	[0.001]	[0.002]	[0.007]	[0.002]			[0.002]				[0.007]			
Numeracy	0.035	0.025	0.019		0.044			0.041				0.027		
	[0.002]	[0.003]	[0.010]		[0.003]			[0.003]				[0.010]		
Foreign language	0.016	0.016	0.026			0.030			0.031				0.035	
	[0.006]	[0.006]	[0.006]			[0.006]			[0.006]				[0.007]	
Field Specific		0.022								0.038				
		[0.004]								[0.003]				
Socioemotional			0.023											0.023
			[0.006]											[0.006]
Sample:	Full	Specific	Survey	Full	Full	Full	Specific	Specific	Specific	Specific	Survey	Survey	Survey	Survey
Observations	363,330	155,939	2,401	363,330	363,330	363,330	155,939	155,939	155,939	155,939	2,401	2,401	2,401	2,401
R-squared	0.194	0.219	0.247	0.191	0.192	0.189	0.216	0.217	0.215	0.217	0.243	0.243	0.243	0.243

Panel B:	IV: Measurement Error Correction												
	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)
Two-Year Private	0.083	0.128	0.085	0.079	0.078	0.090	0.083	0.083	0.083	0.126	0.126	0.124	0.132
	[0.015]	[0.045]	[0.016]	[0.014]	[0.014]	[0.013]	[0.011]	[0.013]	[0.012]	[0.054]	[0.047]	[0.050]	[0.052]
Four-Year Public	0.176	0.169	0.177	0.174	0.173	0.153	0.148	0.160	0.168	0.165	0.170	0.206	0.208
	[0.012]	[0.045]	[0.012]	[0.011]	[0.011]	[0.014]	[0.013]	[0.013]	[0.015]	[0.040]	[0.041]	[0.042]	[0.044]
Four-Year Private	0.257	0.225	0.255	0.254	0.238	0.235	0.232	0.231	0.252	0.197	0.222	0.206	0.222
	[0.022]	[0.034]	[0.022]	[0.021]	[0.020]	[0.022]	[0.021]	[0.021]	[0.023]	[0.042]	[0.035]	[0.036]	[0.038]
Literacy	0.075	-0.004	0.105			0.106				0.112			
	[0.011]	[0.057]	[0.012]			[0.013]				[0.046]			
Numeracy	0.045	0.154		0.087			0.083				0.140		
	[0.006]	[0.046]		[0.008]			[0.010]				[0.028]		
Foreign language	-0.002	0.002			0.040			0.041				0.053	
	[0.004]	[0.027]			[0.006]			[0.005]				[0.022]	
Field Specific									0.123				
									[0.017]				
Socioemotional		0.026											0.022
		[0.007]											[0.006]
Sample:	Full	Survey	Full	Full	Full	Specific	Specific	Specific	Specific	Survey	Survey	Survey	Survey
Observations	363,330	2,401	363,330	363,330	363,330	155,939	155,939	155,939	155,939	2,401	2,401	2,401	2,401
R-squared	0.042	0.046	0.034	0.042	0.043	0.028	0.036	0.037	0.023	0.071	0.054	0.097	0.097
Controls:													
Individual & Field	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Types of Degrees	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*Notes.*The dependent variable and measures of skills are as described in the notes of Table 3.1. Estimations within the full sample use the same specification as column (1) of Table 3.1. Likewise, estimations within the specific and survey samples use, respectively, the specifications of columns (3) and (5) of Table 3.1. Measures of numeracy, literacy and foreign language computed from the high school exit exam are used as instruments within the full and specific samples. We additionally include an alternative measure of non-cognitive skill as instrument within the survey sample. Individual and types of degrees area as described in the notes of Table 3.1. Standard errors clustered by municipality and in brackets.

C.2 Appendix: Skills Measures

In this appendix, we describe the cognitive abilities evaluated in the college exit exam and the high school exit exam. We also describe the non-cognitive abilities collected in the follow-up survey of graduates, and how we compute the different measures of skills used in the paper.

The abilities tested in the college exit exam (Saber Pro) are divided into two sections. The first section has five mandatory general tests and consists of 160 multiple choice questions and one open question, lasting a maximum of four hours and 40 minutes. The main objective of this section is to evaluate common abilities that students from the wide range of fields should be able to apply in non-specialized tasks. On the other hand, some students also take a second section with specific tests. This section is only available if the student's college previously decided which specific tests will be applied to their undergraduate programs. There are 40 specific tests and combinations of theses suggested for each field of study or group of related programs.[1] Following those combinations, colleges can require up to three specific tests. The maximum time allowed for students taking one specific test is one hour and 30 minutes, while students taking two or three specific tests have a maximum of four hours and 30 minutes. Appendix Table C.10 presents more details on the abilities evaluated and a sample question for each general test. It also shows one question for the Economic Analysis test, to present an example for one of the specific tests evaluated in the second part of the college exit exam.

We now describe how the different measures of skills were computed. Let η_{ti} be student i's scores for η_t, which is test η applied in time t. Let μ_{η_t} be the mean of η_t, this is:

$$\mu_{\eta_t} = \frac{1}{|I_t|} \sum_{i \in I_t} \eta_{ti}$$

where I_t is the set of students who took η_t. Thus, the standardized score of η_{ti} with respect to μ_{η_t} is $\tau_{ti} = \frac{(\eta_{ti} - \mu_{\eta_t})}{\sigma_{\eta_t}}$, where the standard deviation of η_t

[1] Throughout the years there have been changes in some of these specific evaluations and, thus, our data contains a total of 87 specific tests. For instance, some specific tests were divided into others, and some have disappeared.

is defined as:

$$\sigma_{\eta_t} = \sqrt{\frac{1}{|I_t| - 1} \sum_{i \in I_t} (\eta_{ti} - \mu_{\eta_t})^2}.$$

Now let $\mu^s_{\eta_t}$ be the mean of η_t, but computed within a sample $s \subseteq |I_t|$. If $i \in s$, then $\tau^s_{t\,i}$ is the standardized score of η_{ti} within sample s.

Taking the previous framework into account, we use the test scores from the Saber Pro exam to compute literacy, numeracy and foreign language measures as the standardized scores within the four samples mentioned in Appendix C.3. Hence, for each individual in a sample, the numeracy ability level is the standardized score of the quantitative reasoning test with respect to the mean in the time period in which the test was taken, considering only individuals within the sample. Likewise, the foreign language ability level is computed as the standardized score of the English proficiency test within each sample. On the other hand, to define the literacy measure we first compute the average score of written communication and critical reading tests, and then standardized the resulting vector in the same way as we did for numeracy and foreign language. To compute the specific skills measure, we first average all field-specific test scores available for each student and then proceed to standardize the resulting vector, this time considering both the sample and group of related majors.

Appendix Table C.10: Description of Tests and Abilities Evaluated in the Saber Pro Exam

Test	Evaluated Abilities	Sample Question
Section 1:		
Critical Reading (35 Questions)	Abilities that allow individuals to understand, interpret and analyze texts found in both, common life and non-specialized academic scenarios.	The text's author states that "seeking justice is the eternal seeking of human happiness". This statement: A) implies that "every human pursues hapiness", B) does not imply that "seeking justice is seeking happiness"
Quantitative Reasoning (35 Questions)	Mathematic abilities that every citizen should have, independently of their profession or occupacion, such as: algebra, calculus, geometry, statistics, interpretation of numeric information, use of mathematics to formulate and execute plans, and use of mathematics to solve problems.	Four utility companies estimated their daily efficiency to solve customer complaints: Electricity: 2 out of 3 solved complaints per day. Aqueduct: 5 out of 6. Telephone: 9 out of 10. The average efficiency for one of these companies is : A) 72%, B) 79%
Citizenship Abilities (35 Questions)	Knowledge and abilities to understand the social environment and its issues, as well as abilities to analyze positions taken by different parties involved in a conflicting situation.	To reduce traffic jams within a city, a major decided to restrict the free circulation of vehicles using the last digit of license plates. The offer of public transporation in the city is limited and has low quality. What undesired effects may cause to the mentioned policy for citizens using the public transportation?
Written Communication (1 Open Question)	Abilities to communicate ideas in writing, regarding a given topic. Students are asked to produce a text in response to a non-specialized problem.	Some consider that national and international sport competitions are used for political and commercial means. Do you agree or disagree with this opinion? Discuss.
English Proficiency (55 Questions)	Communication abilities in English throughout reading, grammar and vocabulary tasks.	The Ozone Layer is a "blanket" (1) _____ earth. It protects (2) _____ from the sun's UV rays. Fill the blanks: (1) A) around, B) through; (2) A) our, B) us.
Section 2:		
Specific Tests (30 - 60 Questions)	Abilities that different postsecondary programs must provide to its students. These abilities have been defined between the Ministry of Education, the academic and professional community, and the industry.	(Economic Analysis Test:) Consider a linear model $y = X\beta + \varepsilon$, where ε is an error term. Assuming that $E(X'\varepsilon) = 0$, then: A) OLS are consistent, and 2SLS are consistent and efficient. B) OLS are inconsistent and 2SLS are consistent and efficient. (OLS: Ordinary Least Squares; 2SLS: Two-Stage Least Squares)

Notes. Information adapted by the authors from Icfes (2017).

To compute the non-cognitive skills measure, we rely on a module of abilities contained in the follow-up survey of graduates. The questionnaire for this module emphasizes, previous to ask any question, that an acquired ability allows individuals to use their knowledge to solve problems in different contexts, and to perform efficiently in their social, academic and working activities. The items we use from the module of abilities are presented in Appendix Table C.3. Surveyed individuals are asked to pinpoint the satisfaction level, regarding the academic and labor impact of each item or ability acquired during their undergraduate studies. Additionally, we present the loadings from a factor analysis model performed using the nine items in this table. Using these results we predict a vector of non-cognitive scores, which is then standardized to have mean zero and standard deviation one.

Following MacLeod et al. (2017), we used the administrative records of undergraduate students to build a measure of college reputation or quality. This measure is defined as the mean admission score of graduates, then for college c, in time t, the reputation measure is:

$$R_{ct} = \frac{1}{|G_{ct}|} \sum_{i \in G_{ct}} \tilde{\eta}_i$$

where G_{ct} is the set of students graduating from college c in time t, and $\tilde{\eta}_i$ is the percentile score of individual $i \in G_{ct}$ in the high school exit exam. We then standardized R_{ct} to have mean zero and standard deviation one.

C.3 Appendix: Data Construction

In this appendix, we explain in detail the construction of samples used in the paper. The starting point is to append the individual records of Saber Pro exams applied between the second semester of 2011 and 2015. This is our population.[2] The Colombian Institute for the Evaluation of Education (ICFES, in Spanish) applies Saber Pro twice per year to undergraduate students who have passed at least three-fourths of their coursework. For thesis students, the exam constitutes a graduation requirement. Our data contain information for around 1,448,000 students who have been tested in critical reading, written communication, English proficiency, and quantitative reasoning. A test in citizenship abilities has been also evaluated since 2012, and almost 90% of students have available scores for this test.

For almost 42% of the population, the data include scores from different field-specific tests. The specific exam taken by each student depends exclusively on the college decision, which has to be supported by ICFES. For each program, ICFES suggests a list of potential specific tests per program. Then, each college decides which specific test(s) are their students taking. Students can take up to three specific tests. For more details on measures of skills, and the test scores used to compute them, see Appendix C.2.

[2]It is worth noting that the decision to choose the mentioned period of time was driven by the fact that students attending certain undergraduate programs were exempted from taking Saber Pro during previous editions of the exam.

The second step in our data construction is to link the data from Saber Pro with college and labor market records from different sources. We merge individuals using the national identification number. For the subset of individuals that we are unable to merge by these means, we conduct a record linkage procedure using names and dates of birth.[3] Appendix Table C.11 shows the merging results by technique and a measure of reliability for observations merged using record linkage.

We merge the population of students in Saber Pro with three other datasets. First, we merge it with the corresponding results in the high school exit examination, known as Saber 11. We merge 83% of observations with a file containing high school test-takers from 1996 to 2013. Saber 11 data include socioeconomic variables such as gender, household stratum and the high school from which the student graduated.[4] Most importantly, these data allow us to obtain a proxy of initial abilities, since individuals taking Saber 11 were evaluated in mathematics, physics, chemistry, biology, language, philosophy, geography, history, social sciences and foreign language.[5]

Second, we merge individuals in Saber Pro with a file containing records of individuals enrolled in college between 1998 and 2016. This dataset is known as Spadies (in Spanish "Sistema para la prevención de la deserción de la Educación Superior"). The information in this file is collected each semester by the Ministry of Education directly from all colleges. Eighty-two percent of students in Saber Pro were linked to this administrative data, which allowed us to compute a measure of college quality proposed by MacLeod et al.

[3]In Colombia, individuals change their identification number when they turn 18. A large portion of individuals graduate from high school before this age and, thus, using the identification number is not enough to merge information from postsecondary education, or labor market information, with high school information. We use a crosswalk of national identification numbers between youth IDs (before they turn 18) and adult IDs (after they turn 18) provided by the Colombian registry's office (in Spanish "Registraduría General de la Nación"). We minimize the number of fuzzy matches using this crosswalk. For the remaining sample, we rely on other merging techniques to link our available datasets.

[4]In the text we explain that each household in Colombia is given a stratum number depending on the neighborhood in which it is located.

[5]Before the second semester of 2006, students taking Saber 11 were allowed to choose among three foreign language tests: English, French or German. But, despite having these options, 99% of students chose English as their test. Because of this reason, nowadays the only foreign language tested is English.

(2017). By these means, we obtain information regarding college attendance, students' major, and access to college loans.

Third, we merge records on monthly wages of college graduates. These records are gathered by the Colombian social security administration office.[6] We merge 64% of Saber Pro students with these records. We use a sample of graduates who received their degrees between 2012 and 2016, and whose monthly wage is observed between the second and third quarters after their corresponding semester of graduation. The merged data provided us with an observation of the monthly wage after graduation for each year between 2012 and 2016. The data also allow us to have information on the city or municipality where individuals work, four-digit industrial codes, and the tax-identification of employers or firms. Notice that some individuals, among Saber Pro test-takers, may have received their diplomas after the period of time for our sample of graduates, or may have dropped out of college. In these two cases, we are not able to link or merge them into our sample, explaining our merging results.

Appendix Table C.11: Results by Merging Technique with Saber Pro Dataset

Dataset	Identity Card	Record Linkage	
High School Exit Exam (Saber 11)	281,861	905,748	(0.98)
Undergraduate Enrollees (Spadies)	1,027,802	160,435	(0.97)
Graduates Social Security Records (OLE)	922,684	12,844	(0.96)

Notes. The numbers in this table correspond to individuals merged with a total of 1,448,395 Saber Pro test-takers. Record linkage merging method used names and birthdates to pair individuals. Average Jaccard index of similarity between names in parenthesis. A score of 1 is assigned to names with perfect similarity, a score of 0 is assigned to names without similarity. We required birth dates to be equal in order to merge individuals.

After assembling the mentioned datasets with the Saber Pro population, we obtain a subset of 743,542 students who belong to the intersection of the three merging processes. However, we also make some selection of individuals in order to have more homogeneous samples. For instance, some individuals may have taken the college exit exam on different occasions. These could be individuals attending different undergraduate programs. For these, we keep the information from the first observed test scores, considering that those

[6] We were given access to these data by the Labor Observatory for Education (OLE, in Spanish), which is part of the Ministry of Education.

should reflect their level of skills at the time when they entered the labor market as graduates. It is also possible that some students took the high school exit exam more than once. In that case, we leave the information from the first observed results, keeping in mind that some individuals could have prepared after graduating from high school to apply for funding or a scholarship. It is also possible that students took the test more than once in order to get the minimum scores required to be admitted to some undergraduate programs. Also, we drop individuals who are under 19 or above the retirement age in Colombia (i.e. 57 years old for women and 62 for men). Individuals who graduated from other undergraduate programs before 2012 are dropped as well.

We exclude observations with missing values in any of the independent variables commonly used in our regressions, remaining with a pool of 437,673 test-takers. We link them to individuals in a follow-up survey of graduates that contains information on non-cognitive abilities. The survey takes a stratified probabilistic sample from students who received their diplomas during the past year, three years ago and five years ago. Using the 2013 and 2014 surveys, we obtain information for a random sample of individuals whobelong to cohorts of graduates from 2011 to 2013.

Notice that for the described sample, we observe values for gender, age, socioeconomic stratum, and also for literacy, numeracy and foreign language skills from both the college and high school exit exams. However, we may not have information about their high school, their specific skills, their non-cognitive skills or current wage. Thus, from this pool or *complete sample* ($N = 437,673$), we obtained three subsets: i) a *full sample* with formal workers for whom we observe high school and current wage ($N = 363,330$); ii) a *specific sample* with formal workers for whom we observe high school, current wage, and a measure of specific skill (N = 155,939); and iii) a *survey sample* with formal workers for whom we observe a measure of non-cognitive skills and current wage, but for whom we may not observe their high school of graduation (N = 2,401).

C.4 Appendix: Alternative Robustness on Returns to Skills

Following Heckman et al. (2006) and Acosta, Muller, and Sarzosa (2015), we additionally address the potential attenuation bias by estimating the returns using a latent skills model. This model identifies the underlying latent parameters of a set of skills and estimates the associated economic returns. Assume the following model in which wages (W) are a function of latent skills (Θ) and other observables (X_W). The reduced equation is given by:

$$W = \alpha^W \Theta + X_W \beta^W + e^W \tag{D.1}$$

While the latent variable is unobservable, there are measures available in the data (i.e., test scores) that represent realizations of latent skills:

$$T = \alpha^T \Theta + X_T \beta^T + e^T \tag{D.2}$$

where T is an $L \times 1$ vector of test scores, and L corresponds to the number of associated test scores per latent skill. The identification assumption states that $e^T \perp e^W$ conditional on (Θ, X_W), and that $L \geq 3$.[7]. Parameters $\alpha^W, \alpha^T, \beta^W$ and β^T, from equations (D.1) and (D.2), can be jointly estimated by maximizing the following likelihood function:

$$\mathcal{L} = \prod_{i=1}^{N} \int \Big\{ f_{e^W}(X_W, W, \zeta^\Theta) \times f_{e^{T_1}}(X_{T_1}, T_1, \zeta^\Theta) \times ... \times f_{e^{T_L}}(X_{T_L}, T_L, \zeta^\Theta) \Big\} dF_\Theta(\zeta^\Theta) \tag{D.3}$$

The results of this model (α^W) should be interpreted as the economic return of a latent skill allowing for the presence of unobserved heterogeneity (Sarzosa & Urzúa, 2016; Carneiro, Hansen, & Heckman, 2003). Since the return should be read in levels of the latent skill, we use the estimated distribution of the skill, $\widehat{F_\Theta}(\zeta^\Theta)$, to rescale α^W so it can be read in terms of standard deviations. This way we can compare the results from this model with those previously obtained.

[7] More details about these identification assumptions are given in Sarzosa and Urzúa (2016).

Identification requires at least three test scores for each latent skill. Thus, to estimate the effect of latent literacy skills we use writing and reading scores from the college exit exam and the language (Spanish) test scores from the high school exit exam. To identify the latent effect of numeracy we use the numeracy measures from the college exams and the standardized mathematics and physics scores from the high school exit exams. For foreign language we use the test scores from the college and high school exit exams, and the social science score from the high school exit exam, since it is strongly correlated with the results on foreign language (0.46). Among the survey sample we replace the social sciences scores with self-assessed information in the graduates' follow-up survey about respondents' abilities in listening, speaking, reading and writing in a foreign language.[8] For major-specific skills we use different scores from the set of field-specific tests in the college exit exam, and a combination of scores from the high school exit exams. [9] For non-cognitive skills we divided the nine questions used to build the non-cognitive measure into three dimensions. We therefore exploit the full extent of the data and the large number of available tests scores to estimate the returns to latent literacy, numeracy, major-specific, and non-cognitive skills.

This model is computationally demanding, and estimation times increase dramatically with sample size, controls, and the number of factors or latent skills to estimate (Sarzosa & Urzúa, 2016). Therefore, we respectively took two random samples of 36,403 and 15,585 individuals from both the complete set of test takers working formally and the set of formal workers with specific scores available.[10] Notice as well that we use all the individuals in the survey

[8]The follow-up survey includes a module in which respondents assess their own ability in listening, speaking, reading and writing in a foreign language. This information is only available for the sub-sample that was surveyed.

[9]Given the constraint on the number of test scores, we do this only for individuals who have taken only one or two specific tests and exclude those who took three. For students in health and agronomy, we use biology and chemistry test scores. For education, social sciences, humanities, and economics and business, we use history and geography tests scores. For students in arts, we use philosophy and history test scores. For engineering, mathematics and natural sciences we use chemistry and biology.

[10]We conducted a stratified probabilistic sampling within both sets of information: i) the set of workers who took the college exit exam between 2011 and 2015, and ii) the workers for whom specific test scores are available. Strata were defined using the nine test editions of the college exit exam, 18 groups of related majors, six cohorts of graduates, four groups or quartiles defined using the average score in the college exit exam, and quartiles defined by the current wage of individuals. Within each stratum, we applied a simple

sample to estimate the model.

Appendix Table C.12 presents the estimation of the latent skill model for each sample and, for comparison, it also presents OLS estimations. Again, simultaneous and stacked results could be interpreted as lower and upper bounds. Within the survey sample, the results suggest that latent numeracy skills have the largest returns (from 4.8 percent to 7.2 percent), followed by latent literacy skills (5.1 percent to 6 percent), foreign language skills (up to 1.8 percent) and, finally, non-cognitive skills (around 2 percent). On the other hand, using the random sample representing our complete set of students, we see that numeracy returns up to 6.4 percent, literacy latent skills returns from 3.7 percent to 7.1 percent, and foreign language returns up to 12 percent. The results from the sample representing the students with field-specific scores show that latent specific skills returns up to 6.7 percent.

random sampling methodology to select 10 percent of test-takers. Expansion factors were computed as $\pi_{kh} = \frac{n_h}{N_h}$, with n_h as the number of sampled individuals in stratum h, and N_h as the original number of individuals in h.

Appendix Table C.12: Structural Estimations: Measurement Error Correction

	Dependent Variable: log(Current Wage)									
	OLS in Random Sample				Unobserved Heterogeneity					
	Simultaneous		Stacked		Simultaneous			Stacked		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Literacy	0.029	0.013	0.040	0.030	0.037	0.035	0.051	0.071	0.070	0.060
	[0.002]	[0.007]	[0.003]	[0.006]	[0.006]	[0.009]	[0.012]	[0.004]	[0.006]	[0.011]
Numeracy	0.032	0.033	0.042	0.047	-0.002	0.014	0.048	0.064	0.073	0.072
	[0.004]	[0.007]	[0.005]	[0.007]	[0.007]	[0.010]	[0.020]	[0.004]	[0.006]	[0.012]
English	0.018	0.016	0.031	0.031	0.096	0.126	0.011	0.049	0.058	0.018
	[0.007]	[0.009]	[0.008]	[0.008]	[0.011]	[0.016]	[0.011]	[0.003]	[0.005]	[0.011]
Field Specific		0.023		0.040		0.031			0.067	
		[0.005]		[0.005]		[0.007]			[0.005]	
Socioemotional							0.023			0.022
							[0.012]			[0.011]
Sample	Full	Specific	Full	Specific	Full	Specific	Survey	Full	Specific	Survey
Observations	36,403	15,585	36,403	15,585	36,403	15,585	2,401	36,403	15,585	2,401
Controls:										
Individual & Field	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Types of Degrees	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable is as described in the notes of Table 3.1. Columns (1) and (3) use the same specifications as columns (1) and (4) of Table 3.3, but within a stratified random sample. Columns (2) and (4) use the specifications of columns (2) and (5) of Table 3.3, within another random sample. Expansion factors for each random sample were used to estimate columns (1) to (4). The point estimates from the unobserved heterogeneity model are originally computed in levels of each latent skill. Using the standard deviation of these skills, we rescale the coefficients and the standard errors applying the "delta method." From columns (5) to (10), to estimate literacy we used i) writing scores from the college exit exam, ii) reading scores from the college exam, and iii) language (Spanish) scores from the high school exit exam. Numeracy used i) quantitative reasoning scores from the college exam, ii) mathematics scores from the high school exam, and iii) physics scores from the high school exam. Foreign language used i) foreign language scores from the college exam, ii) foreign language scores from high school exam, and iii) a factor score predicted using self-reported ability in a foreign language for columns (7) and (10). For columns (5), (6), (8) and (9) we used the social science scores from the high school exam as the third score for foreign language. For non-cognitive estimations, we divided the nine categorical questions of Appendix Table C.3 into three scores, each time computed using a factor analysis model. Major-specific skills used the specific test scores from the college exit exam. When a student has only taken one or two specific tests in the college exam, we used the set of test scores from the high school exam, depending on the study area of the student (see footnote 26). Stacked results refer to separate regressions for each ability measure. Field of study and type of degrees controls as described in the notes of Table 3.1, however individual controls in columns (5) to (10) do not include high school fixed effects and the initial ability proxy.